

1. Public API / ABI

Ada (spec)

```
driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 4)
[package] package DS_Buddy_Math with
driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 17)
[function] function Power_Of_Two (Exponent : Block_Size) return Phys_Addr with
driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 23)
[function] function Is_Aligned (Addr : Phys_Addr; Size_Exp : Block_Size) return
Boolean with
driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 32)
[function] function Get_Buddy_Address (
driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 47)
[function] function Get_Left_Child_Address (
driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 55)
[function] function Get_Right_Child_Address (
driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads (line 69)
[function] function Safe_Calculate_DMA_End (
kernel_Workspace/kernel/src/policy/ada/policy.ads (line 3)
[package] package Policy
kernel_Workspace/kernel/src/policy/ada/policy.ads (line 21)
[function] function Evaluate (Ring : Ring_Id;
kernel_Workspace/kernel/src/policy/ada/policy_c.ads (line 4)
[package] package Policy_C is
kernel_Workspace/kernel/src/policy/ada/policy_c.ads (line 6)
[function] function policy_evaluate (ring : Unsigned_8;
```

Assembly

```
comgrub/src/boot.asm (line 30)
[label] global _start
kernel_Workspace/kernel/src/cpu/trampoline.s (line 5)
[label] .global trampoline_start
kernel_Workspace/kernel/src/cpu/trampoline.s (line 6)
[label] .global trampoline_end
kernel_Workspace/kernel/src/cpu/trampoline.s (line 7)
[label] .global trampoline_cr3
kernel_Workspace/kernel/src/cpu/trampoline.s (line 8)
[label] .global trampoline_entry
kernel_Workspace/kernel/src/cpu/trampoline.s (line 9)
[label] .global trampoline_stack
kernel_Workspace/kernel/src/cpu/trampoline.s (line 10)
[label] .global trampoline_arg
kernel_Workspace/kernel/src/cpu/trampoline.s (line 100)
[label] .global trampoline_end
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 3)
[label] .global tlb_asm_invlpg
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 9)
[label] .global tlb_asm_read_cr3
```

```

kernel_workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 15)
[label] .global tlb_asm_write_cr3
kernel_workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 21)
[label] .global tlb_asm_read_cr4
kernel_workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 27)
[label] .global tlb_asm_write_cr4
kernel_workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 34)
[label] .global tlb_asm_invpcid
kernel_workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 40)
[label] .global tlb_asm_wbinvd
kernel_workspace/kernel/src/mm/arch/x86_64/tlb_asm.s (line 46)
[label] .global tlb_asm_clflush
kernel_workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.s (line 102)
[label] .global tr_init
kernel_workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.s (line 111)
[label] .global tr_restore_ctx
kernel_workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.s (line 9)
[label] .global trap_entry
kernel_workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.s (line 98)
[label] .global tr_init
kernel_workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.s (line 114)
[label] .global tr_restore_ctx
kernel_workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 3)
[label] .global tr_gdt
kernel_workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 12)
[label] .global tr_tss_desc
kernel_workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 21)
[label] .global tr_tss
kernel_workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 27)
[label] .global tr_init
kernel_workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 54)
[label] .global isr_hypercall
kernel_workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 94)
[label] .global isr_default
kernel_workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.s (line 99)
[label] .global tr_restore_ctx
triang-lang/out/x86_64/prog.asm (line 7)
[label] global main

```

C

```

comlimine/src/main.c (line 6)
[declaration] extern void kernel_main(uint64_t magic, void *info);
comlimine/src/main.c (line 28)
[function] void _start(void) {
comlimine/src/main.c (line 35)
[declaration] kernel_main(0, (void *)framebuffer.response);
drivers/vgpu/vgpu_driver.c (line 15)
[declaration] ds_poll();

```

```

drivers/vgpu/vgpu_driver.c (line 17)
[declaration] ds_take(id, &r);
drivers/vgpu/vgpu_driver.c (line 23)
[declaration] ds_poll();
drivers/vgpu/vgpu_driver.c (line 26)
[function] if (ds_take(id, &r) && r.status == 0) {
drivers/vgpu/vgpu_driver.c (line 35)
[declaration] pci_void(PORT_WRITE, VBE_INDEX, idx, 16);
drivers/vgpu/vgpu_driver.c (line 36)
[declaration] pci_void(PORT_WRITE, VBE_DATA, val, 16);
drivers/vgpu/vgpu_driver.c (line 42)
[function] if (g_ready) {
drivers/vgpu/vgpu_driver.c (line 47)
[declaration] ds_poll();
drivers/vgpu/vgpu_driver.c (line 51)
[function] if (!ds_take(id, &r) || r.status != 0) {
drivers/vgpu/vgpu_driver.c (line 57)
[function] if (r.arg1 != (0x1234 | (0x1111 << 16))) {
drivers/vgpu/vgpu_driver.c (line 63)
[function] if (fb_phys == 0) {
drivers/vgpu/vgpu_driver.c (line 67)
[declaration] pci_void(PCI_ENABLE, bdf, 0, 0);
drivers/vgpu/vgpu_driver.c (line 69)
[declaration] vbe_write(4, 0);
drivers/vgpu/vgpu_driver.c (line 70)
[declaration] vbe_write(0, 0xB0C4);
drivers/vgpu/vgpu_driver.c (line 71)
[declaration] vbe_write(1, w);
drivers/vgpu/vgpu_driver.c (line 72)
[declaration] vbe_write(2, h);
drivers/vgpu/vgpu_driver.c (line 73)
[declaration] vbe_write(3, 32);
drivers/vgpu/vgpu_driver.c (line 74)
[declaration] vbe_write(4, 0x01 | 0x40 | 0x80);
drivers/vgpu/vgpu_driver.c (line 78)
[declaration] ds_poll();
drivers/vgpu/vgpu_driver.c (line 82)
[function] if (!ds_take(mid, &mr) || mr.status != 0) {
drivers/vgpu/vgpu_driver.c (line 106)
[function] if (!g_ready || x >= g_info.width || y >= g_info.height) {
drivers/vgpu/vgpu_driver.c (line 115)
[function] if (!g_ready) {
drivers/vgpu/vgpu_driver.c (line 133)
[declaration] ds_poll();
drivers/vgpu/vgpu_driver.c (line 137)
[function] if (!ds_take(aid, &ar) || ar.status == 0 ? ar.arg0 == 0 : 1) {
drivers/vgpu/vgpu_driver.c (line 144)
[declaration] ds_poll();

```

```

drivers/vgpu/vgpu_driver.c (line 148)
[function] if (!ds_take(pid, &pr) || pr.status != 0) {
drivers/vgpu/vgpu_driver.c (line 153)
[function] if (!g_surf[i].used) {
drivers/vgpu/vgpu_driver.c (line 170)
[function] if (id < 0 || id >= VGPU_MAX_SURFACES || !g_surf[id].used) {
drivers/vgpu/vgpu_driver.c (line 181)
[function] if ((c & 0xFF000000) != 0) {
drivers/vgpu/vgpu_driver.c (line 192)
[function] if (ring == (void *)0) {
drivers/vgpu/vgpu_driver.c (line 199)
[function] if (s->id == 0 || s->done != 0) {
drivers/vgpu/vgpu_driver.c (line 203)
[function] switch (s->op) {
drivers/vgpu/vgpu_driver.c (line 215)
[function] if (sid < 0) {
drivers/vgpu/vgpu_driver.c (line 248)
[declaration] ds_init(params_va);
drivers/vgpu/vgpu_driver.c (line 250)
[function] if (!initd) {
drivers/vgpu/vgpu_driver.c (line 251)
[function] if (vgpu_init(1024, 768)) {
drivers/vgpu/vgpu_driver.c (line 253)
[declaration] ds_poll();
drivers/vgpu/vgpu_driver.c (line 257)
[function] if (ds_take(rid, &rr) && rr.status == 0) {
drivers/vgpu/vgpu_driver.c (line 261)
[declaration] ds_poll();
drivers/vgpu/vgpu_driver.c (line 265)
[function] if (ds_take(pid, &pr) && pr.status == 0) {
drivers/vgpu/vgpu_driver.c (line 266)
[declaration] ds_call(SVC_VGPU, VGPU_REGISTER, pr.arg0, 0, 0);
drivers/vgpu/vgpu_driver.c (line 267)
[declaration] ds_poll();
drivers/vgpu/vgpu_driver.c (line 270)
[declaration] ds_take(0, &zr);
drivers/vgpu/vgpu_driver.c (line 274)
[declaration] vgpu_clear(0xFF101020);
drivers/vgpu/vgpu_driver.c (line 280)
[declaration] vgpu_process_ring(g_ring);
drivers/vgpu/vgpu_driver.c (line 282)
[declaration] ds_yield();
kernel_Workspace/kernel/src/battery/init.c (line 8)
[function] if (is_ready) {
kernel_Workspace/kernel/src/battery/operation.c (line 23)
[function] if (!model.present) {
kernel_Workspace/kernel/src/battery/operation.c (line 40)
[function] if (b != (void *)0 && b->status != (void *)0) {

```

```

kernel_Workspace/kernel/src/battery/operation.c (line 52)
[function] if (out == (void *)0) {
kernel_Workspace/kernel/src/battery/operation.c (line 63)
[function] if (!battery_status(&s)) {
kernel_Workspace/kernel/src/battery/operation.c (line 79)
[function] if (low_pct > 50) {
kernel_Workspace/kernel/src/battery/operation.c (line 103)
[function] if (!model.present || model.state == BATT_STATE_CHARGING) {
kernel_Workspace/kernel/src/battery/operation.c (line 109)
[function] if (tick_counter % 2000 == 0 && model.percent > 0) {
kernel_Workspace/kernel/src/battery/operation.c (line 112)
[function] if (model.percent <= low_threshold && !low_pending) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 14)
[function] if (ret_len != 0 && ret == NULL) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 19)
[function] if (!bt_event_poll(buf, EVT_MAX, &len)) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 23)
[function] if (len < 7 || buf[0] != HCI_EVT_PKT || buf[1] != EVT_CMD_COMPLETE) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 27)
[function] if (buf[4] != (uint8_t)(opcode & 0xFF) || buf[5] != (uint8_t)(opcode >>
8)) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 31)
[function] if (buf[6] != 0 || len < (uint8_t)(7 + ret_len)) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 47)
[function] if (bt_is_ready) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 51)
[declaration] bt_op_model_init();
kernel_Workspace/kernel/src/bluetooth/init.c (line 53)
[function] if (!bt_hci_cmd(HCI_RESET, NULL, 0)) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 58)
[function] if (!wait_cmd_complete(HCI_RESET, tmp, 0)) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 62)
[function] if (!bt_hci_cmd(HCI_READ_LOCAL_VERSION, NULL, 0)) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 66)
[function] if (!wait_cmd_complete(HCI_READ_LOCAL_VERSION, tmp, 1)) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 72)
[function] if (!bt_hci_cmd(HCI_READ_BD_ADDR, NULL, 0)) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 76)
[function] if (!wait_cmd_complete(HCI_READ_BD_ADDR, saved_addr, 6)) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 91)
[function] if (hci_ver != NULL) {
kernel_Workspace/kernel/src/bluetooth/init.c (line 95)
[function] if (bdaddr != NULL) {
kernel_Workspace/kernel/src/bluetooth/operation.c (line 18)
[function] if (data == NULL || len == 0 || len > EVT_MAX) {
kernel_Workspace/kernel/src/bluetooth/operation.c (line 23)
[function] if (next == evt_tail) {
kernel_Workspace/kernel/src/bluetooth/operation.c (line 49)
[function] if (len != 0 && params == NULL) {

```

```

kernel_Workspace/kernel/src/bluetooth/operation.c (line 53)
[function] switch (opcode) {
kernel_Workspace/kernel/src/bluetooth/operation.c (line 62)
[declaration] return evt_push(ev, 7);
kernel_Workspace/kernel/src/bluetooth/operation.c (line 80)
[declaration] return evt_push(ev, 15);
kernel_Workspace/kernel/src/bluetooth/operation.c (line 93)
[declaration] return evt_push(ev, 13);
kernel_Workspace/kernel/src/bluetooth/operation.c (line 103)
[declaration] return evt_push(ev, 7);
kernel_Workspace/kernel/src/bluetooth/operation.c (line 109)
[function] if (!op_ready) {
kernel_Workspace/kernel/src/bluetooth/operation.c (line 113)
[declaration] return bt_op_model_run_cmd(opcode, params, len);
kernel_Workspace/kernel/src/bluetooth/operation.c (line 118)
[function] if (buf == NULL || len == NULL || evt_head == evt_tail) {
kernel_Workspace/kernel/src/bluetooth/operation.c (line 123)
[function] if (n > cap) {
kernel_Workspace/kernel/src/bluetooth/operation.c (line 138)
[function] if (!op_ready || (len != 0 && data == NULL) || len > ACL_MAX) {
kernel_Workspace/kernel/src/bluetooth/operation.c (line 143)
[function] if (next == acl_tail) {
kernel_Workspace/kernel/src/bluetooth/operation.c (line 158)
[function] if (data == NULL || len == NULL || acl_head == acl_tail) {
kernel_Workspace/kernel/src/bluetooth/operation.c (line 163)
[function] if (n > cap) {
kernel_Workspace/kernel/src/camera/init.c (line 7)
[function] if (is_ready) {
kernel_Workspace/kernel/src/camera/operation.c (line 43)
[function] if (((x + y + f * 8) & 0xFF) < 16) {
kernel_Workspace/kernel/src/camera/operation.c (line 58)
[function] if (!streaming) {
kernel_Workspace/kernel/src/camera/operation.c (line 62)
[declaration] gen_frame();
kernel_Workspace/kernel/src/camera/operation.c (line 66)
[function] if (cap < n) {
kernel_Workspace/kernel/src/camera/operation.c (line 78)
[function] if (fid) {
kernel_Workspace/kernel/src/camera/operation.c (line 96)
[function] if (b != (void *)0 && b->frame != (void *)0) {
kernel_Workspace/kernel/src/camera/operation.c (line 108)
[function] if (out == (void *)0 || !camera_present()) {
kernel_Workspace/kernel/src/camera/operation.c (line 123)
[function] if (!camera_present()) {
kernel_Workspace/kernel/src/camera/operation.c (line 137)
[function] if (buf == (void *)0) {
kernel_Workspace/kernel/src/core-lang/ast.c (line 14)
[function] if (a->pos + aligned > a->cap) {

```

```

kernel_Workspace/kernel/src/core-lang/ast.c (line 29)
[function] if (n == (void *)0) {
kernel_Workspace/kernel/src/core-lang/bridge.c (line 3)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 4)
[declaration] extern int32_t k_input_key(void);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 5)
[declaration] extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 6)
[declaration] extern uint32_t k_getpid(void);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 7)
[declaration] extern uint64_t k_world_cr3(void);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 8)
[declaration] extern uint64_t k_kernel_cr3(void);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 9)
[declaration] extern int64_t k_spawn(const char *path, uint32_t parent, uint64_t
cr3);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 10)
[declaration] extern int32_t k_ipc_send(uint32_t dst, uint64_t a0, uint64_t a1);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 11)
[declaration] extern int32_t k_ipc_recv(uint64_t *a0, uint64_t *a1);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 12)
[declaration] extern uint64_t k_tick(void);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 16)
[declaration] extern bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t
*a2);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 26)
[declaration] kprintf("%llu\n", (unsigned long long)v);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 34)
[declaration] kprintf("%c", (char)v);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 45)
[function] if (g_ring == 0) {
kernel_Workspace/kernel/src/core-lang/bridge.c (line 49)
[function] while (s[i] && i < 127) {
kernel_Workspace/kernel/src/core-lang/bridge.c (line 56)
[declaration] extern bool k_user_cstr(uint64_t ptr, char *buf, uint32_t cap);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 58)
[function] if (!k_user_cstr(ptr, buf, sizeof(buf))) {
kernel_Workspace/kernel/src/core-lang/bridge.c (line 63)
[declaration] kprintf("%s\n", buf);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 81)
[declaration] return k_getpid();
kernel_Workspace/kernel/src/core-lang/bridge.c (line 88)
[declaration] return k_tick();
kernel_Workspace/kernel/src/core-lang/bridge.c (line 96)
[function] if (g_vm != (void *)0) {
kernel_Workspace/kernel/src/core-lang/bridge.c (line 108)
[function] if (g_ring >= 3) {

```

```

kernel_Workspace/kernel/src/core-lang/bridge.c (line 114)
[declaration] return (uint64_t)k_spawn((const char *)ptr, k_getpid(), cr3);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 121)
[declaration] return (uint64_t)(int64_t)k_ipc_send((uint32_t)dst, a0, a1);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 140)
[function] if (g_ring >= 3) {
kernel_Workspace/kernel/src/core-lang/bridge.c (line 153)
[function] if (g_ring >= 3) {
kernel_Workspace/kernel/src/core-lang/bridge.c (line 171)
[function] if (!battery_status_packed(&a0, &a1, &a2)) {
kernel_Workspace/kernel/src/core-lang/bridge.c (line 181)
[declaration] return cl_vm_register_extern(vm, name, fn);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 189)
[declaration] cl_bridge_add(vm, "put", br_put);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 190)
[declaration] cl_bridge_add(vm, "putc", br_putc);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 191)
[declaration] cl_bridge_add(vm, "log", br_log);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 192)
[declaration] cl_bridge_add(vm, "key", br_key);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 193)
[declaration] cl_bridge_add(vm, "tick", br_tick);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 194)
[declaration] cl_bridge_add(vm, "bat", br_bat);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 195)
[declaration] cl_bridge_add(vm, "exit", br_exit);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 197)
[function] if (ring <= 1) {
kernel_Workspace/kernel/src/core-lang/bridge.c (line 198)
[declaration] cl_bridge_add(vm, "getpid", br_getpid);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 199)
[declaration] cl_bridge_add(vm, "spawn", br_spawn);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 200)
[declaration] cl_bridge_add(vm, "ipc_send", br_ipc_send);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 201)
[declaration] cl_bridge_add(vm, "ipc_recv", br_ipc_recv);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 202)
[declaration] cl_bridge_add(vm, "fsread", br_fsread);
kernel_Workspace/kernel/src/core-lang/bridge.c (line 203)
[declaration] cl_bridge_add(vm, "vfill", br_vfill);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 22)
[function] if (P->code_len >= P->code_cap) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 45)
[function] if (P->gnames[i][k] != name[k]) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 63)
[function] if (g->locals[i].name[k] != name[k]) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 79)
[function] if (i >= 0) {

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 83)
[function] if (g->nlocals >= 32) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 112)
[function] if (neg) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 115)
[declaration] cl_sub(sub, z, P->consts[P->nconsts].b, w);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 119)
[declaration] cl_mask(P->consts[P->nconsts].b, w, bits);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 126)
[declaration] static void gen_expr(cg_t *g, ast_node_t *n, uint8_t bits);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 134)
[function] switch (n->kind) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 136)
[declaration] emit(g, OP_CONST, bits, (uint16_t)add_const(g, n->value, bits, false),
0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 142)
[function] if (li >= 0) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 143)
[declaration] emit(g, OP_LOAD, bits, g->locals[li].slot, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 147)
[function] if (gi >= 0) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 148)
[declaration] emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 150)
[declaration] emit(g, OP_CONST, bits, (uint16_t)add_const(g, 0, bits, false), 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 157)
[function] if (n->pool_kind == 0) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 158)
[declaration] emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 161)
[declaration] emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 162)
[declaration] emit(g, OP_NEG, bits, 0, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 163)
[declaration] emit(g, OP_GSTORE, bits, (uint16_t)gi, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 164)
[declaration] emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 167)
[declaration] emit(g, OP_SHARED, bits, (uint16_t)gi, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 168)
[declaration] emit(g, OP_GLOAD, bits, (uint16_t)gi, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 175)
[function] if (li >= 0) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 176)
[declaration] emit(g, OP_LOAD, bits, g->locals[li].slot, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 178)
[declaration] emit(g, OP_GLOAD, bits, (uint16_t)find_global(P, n->name), 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 184)
[declaration] gen_expr(g, n->a, bits);

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 186)
[function] if (n->op == '-') {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 187)
[declaration] emit(g, OP_NEG, bits, 0, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 189)
[declaration] emit(g, OP_NOT, bits, 0, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 196)
[function] switch (n->op) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 215)
[declaration] gen_expr(g, n->a, bits);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 216)
[declaration] gen_expr(g, n->b, bits);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 217)
[declaration] emit(g, op, bits, 0, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 222)
[declaration] gen_expr(g, n->b, bits);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 224)
[function] if (n->a->kind == ND_VAR) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 227)
[function] if (li >= 0) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 228)
[declaration] emit(g, OP_STORE, bits, g->locals[li].slot, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 230)
[declaration] emit(g, OP_GSTORE, bits, (uint16_t)find_global(P, n->a->name), 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 237)
[declaration] gen_expr(g, n->a, bits);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 241)
[function] if (n->a != (void *)0 && n->a->kind == ND_VAR) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 248)
[function] if (P->ext_names[i][k] != n->a->name[k]) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 254)
[function] if (same) { ei = (int)i; break; }
kernel_Workspace/kernel/src/core-lang/codegen.c (line 258)
[declaration] gen_expr(g, n->list[i], 64);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 261)
[function] if (ei >= 0) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 262)
[declaration] emit(g, OP_EXTERN, 64, (uint16_t)ei, (uint16_t)n->nlist, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 270)
[function] if (P->fn_names[i][k] != n->a->name[k]) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 276)
[function] if (same) { fi = (int)i; break; }
kernel_Workspace/kernel/src/core-lang/codegen.c (line 279)
[function] if (fi >= 0) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 280)
[declaration] emit(g, OP_CALL, bits, (uint16_t)fi, (uint16_t)n->nlist, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 296)
[function] switch (n->kind) {

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 300)
[declaration] gen_expr(g, n->a, n->type.bits);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 301)
[declaration] emit(g, OP_STORE, n->type.bits, slot, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 308)
[function] if (li >= 0) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 309)
[declaration] emit(g, OP_FREE, 0, 0, g->locals[li].slot, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 311)
[declaration] emit(g, OP_FREE, 0, 1, (uint16_t)find_global(P, n->name), 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 317)
[function] if (n->a != (void *)0) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 318)
[declaration] gen_expr(g, n->a, 1);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 322)
[declaration] gen_stmt(g, n->b);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 324)
[function] if (n->c != (void *)0) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 328)
[declaration] gen_stmt(g, n->c);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 334)
[declaration] gen_stmt(g, n->b);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 344)
[function] if (li2 >= 0 && g->locals[li2].slot != tslot) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 345)
[declaration] emit(g, OP_LOAD, 64, g->locals[li2].slot, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 346)
[declaration] emit(g, OP_STORE, 64, tslot, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 348)
[declaration] emit(g, OP_GLOAD, 64, (uint16_t)find_global(P, n->name), 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 349)
[declaration] emit(g, OP_STORE, 64, tslot, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 359)
[declaration] gen_stmt(g, n->list[i]);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 364)
[declaration] gen_expr(g, n, 64);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 365)
[declaration] emit(g, OP_POP, 0, 0, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 387)
[function] if (n->kind == ND_LET && P->nglobals < 64) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 403)
[declaration] cl_wbytes(n->type.bits), n->type.bits);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 436)
[function] if (P->fn_names[k][x] != n->name[x]) {
kernel_Workspace/kernel/src/core-lang/codegen.c (line 442)
[function] if (same) { fi = (int)k; break; }
kernel_Workspace/kernel/src/core-lang/codegen.c (line 450)
[declaration] add_local(&g, n->list[k]->name, n->list[k]->type.bits);

```

```

kernel_Workspace/kernel/src/core-lang/codegen.c (line 455)
[declaration] gen_stmt(&g, n->a);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 457)
[declaration] emit(&g, OP_RET, 0, 0, 0, 0);
kernel_Workspace/kernel/src/core-lang/codegen.c (line 465)
[function] if (m[k] != n->name[k]) { is_main = false; break; }
kernel_Workspace/kernel/src/core-lang/codegen.c (line 468)
[function] if (is_main) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 49)
[function] if (c == '\n') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 77)
[function] if (c == ' ' || c == '\t' || c == '\r') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 78)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 80)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 82)
[function] while (l->pos < l->len && peek(l) != '\n') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 83)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 86)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 87)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 91)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 94)
[function] if (l->pos < l->len) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 95)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 96)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 111)
[declaration] return is_id_start(c) || (c >= '0' && c <= '9');
kernel_Workspace/kernel/src/core-lang/lexer.c (line 133)
[function] if (peek(l) == '0' && (peek2(l) == 'x' || peek2(l) == 'X')) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 134)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 135)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 137)
[function] while (hex_val(peek(l)) >= 0) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 141)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 142)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 144)
[function] while (peek(l) == '0' || peek(l) == '1') {

```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 148)
[function] while (is_digit(peek(l))) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 163)
[function] while (is_id_char(peek(l))) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 164)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 175)
[function] if (kwlen == len) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 179)
[function] if (start[j] != k->kw[j]) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 185)
[function] if (same) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 186)
[declaration] emit(l, t, k->kind, start, len, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 193)
[function] if (len == 1 && start[0] == '_') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 194)
[declaration] emit(l, t, TK_UNDERSCORE, start, len, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 198)
[declaration] emit(l, t, TK_IDENT, start, len, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 204)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 208)
[function] while (l->pos < l->len && peek(l) != '"') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 209)
[function] if (peek(l) == '\\') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 210)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 213)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 218)
[function] if (peek(l) == '"') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 219)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 221)
[declaration] emit(l, t, TK_ERR, start, len, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 225)
[declaration] emit(l, t, TK_STR, start, len, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 231)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 235)
[function] if (peek(l) == '\\') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 236)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 240)
[function] switch (e) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 252)
[function] if (peek(l) == '\\') {

```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 253)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 255)
[declaration] emit(l, t, TK_ERR, l->src + l->pos, 1, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 259)
[declaration] emit(l, t, TK_CHAR, l->src + l->pos - 1, 1, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 265)
[declaration] skip_ws(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 270)
[function] if (l->pos >= l->len) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 271)
[declaration] emit(l, t, TK_EOF, l->src + l->pos, 0, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 277)
[function] if (is_digit(c)) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 278)
[declaration] lex_number(l, t, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 282)
[function] if (is_id_start(c)) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 283)
[declaration] lex_ident(l, t, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 287)
[function] if (c == '"') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 288)
[declaration] lex_string(l, t, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 292)
[function] if (c == '\\') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 293)
[declaration] lex_char(l, t, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 299)
[function] if (c == '$') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 300)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 302)
[function] if (peek(l) == '!') {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 303)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 304)
[declaration] emit(l, t, TK_DOLLAR_BANG, start, 2, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 306)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 307)
[declaration] emit(l, t, TK_DOLLAR_AT, start, 2, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 309)
[declaration] emit(l, t, TK_DOLLAR, start, 1, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 317)
[function] if (c == '=' && d == '=') { take(l); take(l); emit(l, t, TK_EQ, start, 2,
line, col); return true; }

```

```

kernel_Workspace/kernel/src/core-lang/lexer.c (line 318)
[function] if (c == '=' && d == '>') { take(l); take(l); emit(l, t, TK_ARROW, start,
2, line, col); return true; }
kernel_Workspace/kernel/src/core-lang/lexer.c (line 319)
[function] if (c == '=' && d == ':') { take(l); take(l); emit(l, t, TK_EQCOLON,
start, 2, line, col); return true; }
kernel_Workspace/kernel/src/core-lang/lexer.c (line 320)
[function] if (c == '!' && d == '=') { take(l); take(l); emit(l, t, TK_NEQ, start, 2,
line, col); return true; }
kernel_Workspace/kernel/src/core-lang/lexer.c (line 321)
[function] if (c == '<' && d == '=') { take(l); take(l); emit(l, t, TK_LEQ, start, 2,
line, col); return true; }
kernel_Workspace/kernel/src/core-lang/lexer.c (line 322)
[function] if (c == '>' && d == '=') { take(l); take(l); emit(l, t, TK_GEQ, start, 2,
line, col); return true; }
kernel_Workspace/kernel/src/core-lang/lexer.c (line 323)
[function] if (c == '<' && d == '<') { take(l); take(l); emit(l, t, TK_SHL, start, 2,
line, col); return true; }
kernel_Workspace/kernel/src/core-lang/lexer.c (line 324)
[function] if (c == '>' && d == '>') { take(l); take(l); emit(l, t, TK_SHR, start, 2,
line, col); return true; }
kernel_Workspace/kernel/src/core-lang/lexer.c (line 325)
[function] if (c == ':' && d == ':') { take(l); take(l); emit(l, t, TK_DCOLON, start,
2, line, col); return true; }
kernel_Workspace/kernel/src/core-lang/lexer.c (line 328)
[declaration] take(l);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 330)
[function] switch (c) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 355)
[declaration] emit(l, t, TK_ERR, start, 1, line, col);
kernel_Workspace/kernel/src/core-lang/lexer.c (line 374)
[function] if (n >= cap) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 380)
[function] if (!ok) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 388)
[function] if (k == TK_EOF) {
kernel_Workspace/kernel/src/core-lang/lexer.c (line 399)
[function] switch (kind) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 19)
[function] if (toks == (void *)0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 25)
[declaration] cl_lexer_init(&l, src, len);
kernel_Workspace/kernel/src/core-lang/loader.c (line 27)
[function] if (cl_lex_all(&l, toks, tcap, &n) != 0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 36)
[function] if (prog == (void *)0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 44)
[function] if (s == (void *)0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 50)
[declaration] cl_sema_run(s, prog);

```

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 52)
[function] if (cl_sema_diag_count(s) > 0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 62)
[function] if (P == (void *)0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 88)
[function] if (!w->ok || w->o >= w->cap) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 99)
[declaration] wr_u8(w, (uint8_t)(v >> (i * 8)));
kernel_Workspace/kernel/src/core-lang/loader.c (line 106)
[declaration] wr_u8(w, ((const uint8_t *)src)[i]);
kernel_Workspace/kernel/src/core-lang/loader.c (line 112)
[function] if (!r->ok || r->o >= r->len) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 142)
[declaration] wr_bytes(&w, "CLBC1", 5);
kernel_Workspace/kernel/src/core-lang/loader.c (line 144)
[declaration] wr_u32(&w, (uint32_t)P->code_len);
kernel_Workspace/kernel/src/core-lang/loader.c (line 149)
[declaration] wr_u8(&w, in->op);
kernel_Workspace/kernel/src/core-lang/loader.c (line 150)
[declaration] wr_u8(&w, in->bits);
kernel_Workspace/kernel/src/core-lang/loader.c (line 151)
[declaration] wr_u8(&w, (uint8_t)(in->a & 0xFF));
kernel_Workspace/kernel/src/core-lang/loader.c (line 152)
[declaration] wr_u8(&w, (uint8_t)(in->a >> 8));
kernel_Workspace/kernel/src/core-lang/loader.c (line 153)
[declaration] wr_u8(&w, (uint8_t)(in->b & 0xFF));
kernel_Workspace/kernel/src/core-lang/loader.c (line 154)
[declaration] wr_u8(&w, (uint8_t)(in->b >> 8));
kernel_Workspace/kernel/src/core-lang/loader.c (line 155)
[declaration] wr_u32(&w, in->c);
kernel_Workspace/kernel/src/core-lang/loader.c (line 158)
[declaration] wr_u32(&w, (uint32_t)P->nconsts);
kernel_Workspace/kernel/src/core-lang/loader.c (line 161)
[declaration] wr_bytes(&w, P->consts[i].b, 64);
kernel_Workspace/kernel/src/core-lang/loader.c (line 162)
[declaration] wr_u8(&w, P->const_bits[i]);
kernel_Workspace/kernel/src/core-lang/loader.c (line 165)
[declaration] wr_u32(&w, (uint32_t)P->nfns);
kernel_Workspace/kernel/src/core-lang/loader.c (line 168)
[declaration] wr_u32(&w, P->fns[i].start);
kernel_Workspace/kernel/src/core-lang/loader.c (line 169)
[declaration] wr_u8(&w, (uint8_t)P->fns[i].nslots);
kernel_Workspace/kernel/src/core-lang/loader.c (line 170)
[declaration] wr_u8(&w, (uint8_t)P->fns[i].nargs);
kernel_Workspace/kernel/src/core-lang/loader.c (line 171)
[declaration] wr_bytes(&w, P->fn_names[i], 64);
kernel_Workspace/kernel/src/core-lang/loader.c (line 174)
[declaration] wr_u32(&w, (uint32_t)P->nglobals);

```

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 177)
[declaration] wr_bytes(&w, P->gnames[i], 64);
kernel_Workspace/kernel/src/core-lang/loader.c (line 178)
[declaration] wr_u8(&w, P->gbits[i]);
kernel_Workspace/kernel/src/core-lang/loader.c (line 179)
[declaration] wr_u8(&w, P->gneg[i]);
kernel_Workspace/kernel/src/core-lang/loader.c (line 180)
[declaration] wr_bytes(&w, P->ginit[i].b, 64);
kernel_Workspace/kernel/src/core-lang/loader.c (line 183)
[declaration] wr_u32(&w, (uint32_t)P->nexts);
kernel_Workspace/kernel/src/core-lang/loader.c (line 186)
[declaration] wr_bytes(&w, P->ext_names[i], 64);
kernel_Workspace/kernel/src/core-lang/loader.c (line 189)
[declaration] wr_u32(&w, (uint32_t)P->main_fn);
kernel_Workspace/kernel/src/core-lang/loader.c (line 200)
[declaration] rd_bytes(&r, magic, 5);
kernel_Workspace/kernel/src/core-lang/loader.c (line 209)
[function] if (P == (void *)0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 218)
[function] if (P->code == (void *)0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 224)
[function] if (cl > P->code_cap) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 242)
[function] if (P->nconsts > 64) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 247)
[declaration] rd_bytes(&r, P->consts[i].b, 64);
kernel_Workspace/kernel/src/core-lang/loader.c (line 253)
[function] if (P->nfns > 16) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 261)
[declaration] rd_bytes(&r, P->fn_names[i], 64);
kernel_Workspace/kernel/src/core-lang/loader.c (line 266)
[function] if (P->nglobals > 64) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 271)
[declaration] rd_bytes(&r, P->gnames[i], 64);
kernel_Workspace/kernel/src/core-lang/loader.c (line 274)
[declaration] rd_bytes(&r, P->ginit[i].b, 64);
kernel_Workspace/kernel/src/core-lang/loader.c (line 279)
[function] if (P->nexts > 16) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 284)
[declaration] rd_bytes(&r, P->ext_names[i], 64);
kernel_Workspace/kernel/src/core-lang/loader.c (line 289)
[function] if (!r.ok) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 305)
[function] if (f == (void *)0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 311)
[declaration] fclose(f);
kernel_Workspace/kernel/src/core-lang/loader.c (line 316)
[declaration] extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);

```

```

kernel_Workspace/kernel/src/core-lang/loader.c (line 320)
[declaration] return (long)k_fs_read(path, buf, (uint32_t)cap);
kernel_Workspace/kernel/src/core-lang/loader.c (line 331)
[function] if (n <= 0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 337)
[function] if (m == (void *)0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 343)
[function] if (n > 5 && io[0] == 'C' && io[1] == 'L' && io[2] == 'B') {
kernel_Workspace/kernel/src/core-lang/loader.c (line 353)
[function] if (m->bc == (void *)0) {
kernel_Workspace/kernel/src/core-lang/loader.c (line 357)
[function] if (try_native) {
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 27)
[declaration] nb_u8(d, 0x48); nb_u8(d, 0xB8); nb_u64(d, v);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 37)
[declaration] nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a
* 8);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 38)
[declaration] push_rax(d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 43)
[declaration] pop_rax(d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 44)
[declaration] nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x83); nb_u32(d, (uint32_t)a
* 8);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 49)
[declaration] nb_u8(d, 0x48); nb_u8(d, 0x8B); nb_u8(d, 0x85); nb_u32(d, (uint32_t)a
* 8);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 50)
[declaration] push_rax(d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 55)
[declaration] pop_rax(d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 56)
[declaration] nb_u8(d, 0x48); nb_u8(d, 0x89); nb_u8(d, 0x85); nb_u32(d, (uint32_t)a
* 8);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 76)
[declaration] nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xFB);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 77)
[declaration] nb_u8(&d, 0x48); nb_u8(&d, 0x89); nb_u8(&d, 0xF5);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 78)
[declaration] nb_u8(&d, 0x49); nb_u8(&d, 0x89); nb_u8(&d, 0xD2);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 89)
[function] switch (in.op) {
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 97)
[declaration] mov_rax_u64(&d, v);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 98)
[declaration] push_rax(&d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 102)
[declaration] case OP_LOAD: load_slot(&d, in.a); break;
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 103)
[declaration] case OP_STORE: store_slot(&d, in.a); break;

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 104)
[declaration] case OP_GLOAD: load_glob(&d, in.a); break;
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 105)
[declaration] case OP_GSTORE: store_glob(&d, in.a); break;
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 108)
[declaration] pop_rax(&d); pop_rcx(&d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 109)
[declaration] nb_u8(&d, 0x48); nb_u8(&d, 0x01); nb_u8(&d, 0xC1);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 110)
[declaration] push_rcx(&d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 114)
[declaration] pop_rax(&d); pop_rcx(&d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 115)
[declaration] nb_u8(&d, 0x48); nb_u8(&d, 0x29); nb_u8(&d, 0xC1);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 116)
[declaration] push_rcx(&d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 120)
[declaration] pop_rax(&d); pop_rcx(&d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 121)
[declaration] nb_u8(&d, 0x48); nb_u8(&d, 0x0F); nb_u8(&d, 0xAF); nb_u8(&d, 0xC8);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 122)
[declaration] push_rcx(&d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 129)
[declaration] pop_rax(&d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 131)
[function] switch (i) {
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 132)
[declaration] case 0: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC7); break;
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 133)
[declaration] case 1: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC6); break;
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 134)
[declaration] case 2: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC2); break;
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 135)
[declaration] case 3: nb_u8(&d,0x48); nb_u8(&d,0x89); nb_u8(&d,0xC1); break;
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 136)
[declaration] case 4: nb_u8(&d,0x49); nb_u8(&d,0x89); nb_u8(&d,0xC0); break;
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 137)
[declaration] case 5: nb_u8(&d,0x49); nb_u8(&d,0x89); nb_u8(&d,0xC1); break;
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 142)
[declaration] nb_u8(&d, 0x49); nb_u8(&d, 0x8B); nb_u8(&d, 0x82);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 143)
[declaration] nb_u32(&d, (uint32_t)in.a * 8);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 144)
[declaration] nb_u8(&d, 0xFF); nb_u8(&d, 0xD0);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 145)
[declaration] push_rax(&d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 149)
[declaration] case OP_POP: pop_rax(&d); break;

```

```

kernel_Workspace/kernel/src/core-lang/native_x86.c (line 152)
[declaration] nb_u8(&d, 0xE9);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 156)
[declaration] nb_u32(&d, 0);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 160)
[declaration] pop_rax(&d);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 161)
[declaration] nb_u8(&d, 0x48); nb_u8(&d, 0x85); nb_u8(&d, 0xC0);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 162)
[declaration] nb_u8(&d, 0x0F); nb_u8(&d, 0x84);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 166)
[declaration] nb_u32(&d, 0);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 171)
[declaration] nb_u8(&d, 0x48); nb_u8(&d, 0x31); nb_u8(&d, 0xC0);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 172)
[declaration] nb_u8(&d, 0xC3);
kernel_Workspace/kernel/src/core-lang/native_x86.c (line 197)
[declaration] return sizeof(g_nb_mem);
kernel_Workspace/kernel/src/core-lang/parser.c (line 7)
[function] if (n >= cap) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 27)
[function] if (i >= p->n) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 38)
[function] if (p->pos + 1 < p->n) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 52)
[function] if (check(p, k)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 53)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 62)
[function] if (!p->failed) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 71)
[function] if (!check(p, k)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 72)
[declaration] perr(p, m);
kernel_Workspace/kernel/src/core-lang/parser.c (line 73)
[declaration] return peek(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 76)
[declaration] return advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 81)
[declaration] return cl_node(p->ar, kind, peek(p)->line);
kernel_Workspace/kernel/src/core-lang/parser.c (line 99)
[function] if (l->n < LIST_MAX) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 102)
[declaration] perr(p, "list too long");
kernel_Workspace/kernel/src/core-lang/parser.c (line 111)
[function] if (arr == (void *)0) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 112)
[declaration] perr(p, "arena full");

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 128)
[function] if (check(p, TK_TYPE)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 131)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 135)
[declaration] perr(p, "expected type (u4..u512/str)");
kernel_Workspace/kernel/src/core-lang/parser.c (line 145)
[declaration] expect(p, TK_LET, "let");
kernel_Workspace/kernel/src/core-lang/parser.c (line 147)
[declaration] parse_type(p, &n->type);
kernel_Workspace/kernel/src/core-lang/parser.c (line 149)
[declaration] copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "let name"));
kernel_Workspace/kernel/src/core-lang/parser.c (line 151)
[declaration] expect(p, TK_ASSIGN, "let =");
kernel_Workspace/kernel/src/core-lang/parser.c (line 158)
[function] if (match(p, TK_UNDERSCORE)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 159)
[declaration] expect(p, TK_ARROW, "_=>");
kernel_Workspace/kernel/src/core-lang/parser.c (line 174)
[declaration] expect(p, TK_STATIC, "static");
kernel_Workspace/kernel/src/core-lang/parser.c (line 175)
[declaration] expect(p, TK_DATA, "static data");
kernel_Workspace/kernel/src/core-lang/parser.c (line 176)
[declaration] expect(p, TK_LBRACK, "[");
kernel_Workspace/kernel/src/core-lang/parser.c (line 178)
[declaration] parse_type(p, &n->type);
kernel_Workspace/kernel/src/core-lang/parser.c (line 180)
[declaration] expect(p, TK_COLON, ":");
kernel_Workspace/kernel/src/core-lang/parser.c (line 188)
[declaration] expect(p, TK_RBRACK, "]");
kernel_Workspace/kernel/src/core-lang/parser.c (line 200)
[declaration] expect(p, TK_FN, "fn");
kernel_Workspace/kernel/src/core-lang/parser.c (line 202)
[declaration] copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "fn name"));
kernel_Workspace/kernel/src/core-lang/parser.c (line 204)
[declaration] expect(p, TK_LPAREN, "fn (");
kernel_Workspace/kernel/src/core-lang/parser.c (line 208)
[function] while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 211)
[declaration] parse_type(p, &arg->type);
kernel_Workspace/kernel/src/core-lang/parser.c (line 213)
[declaration] expect(p, TK_IDENT, "param name"));
kernel_Workspace/kernel/src/core-lang/parser.c (line 215)
[declaration] nl_push(p, &params, arg);
kernel_Workspace/kernel/src/core-lang/parser.c (line 217)
[function] if (!match(p, TK_COMMA)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 222)
[declaration] expect(p, TK_RPAREN, "fn )");

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 224)
[declaration] nl_commit(p, &params, n);
kernel_Workspace/kernel/src/core-lang/parser.c (line 235)
[declaration] expect(p, TK_ENUM, "enum");
kernel_Workspace/kernel/src/core-lang/parser.c (line 237)
[declaration] copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "enum name"));
kernel_Workspace/kernel/src/core-lang/parser.c (line 244)
[function] while (!check(p, TK_RBRACE) && !check(p, TK_EOF)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 248)
[declaration] expect(p, TK_IDENT, "enum item"));
kernel_Workspace/kernel/src/core-lang/parser.c (line 252)
[declaration] nl_push(p, &items, it);
kernel_Workspace/kernel/src/core-lang/parser.c (line 254)
[function] if (!match(p, TK_COMMA)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 259)
[declaration] expect(p, TK_RBRACE, "enum }");
kernel_Workspace/kernel/src/core-lang/parser.c (line 261)
[declaration] nl_commit(p, &items, n);
kernel_Workspace/kernel/src/core-lang/parser.c (line 270)
[declaration] expect(p, TK_EXTERN, "extern");
kernel_Workspace/kernel/src/core-lang/parser.c (line 281)
[function] while (!check(p, TK_RBRACE) && !check(p, TK_EOF)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 285)
[declaration] expect(p, TK_IDENT, "extern fn"));
kernel_Workspace/kernel/src/core-lang/parser.c (line 287)
[declaration] nl_push(p, &fns, f);
kernel_Workspace/kernel/src/core-lang/parser.c (line 289)
[function] if (!match(p, TK_COMMA)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 294)
[declaration] expect(p, TK_RBRACE, "extern }");
kernel_Workspace/kernel/src/core-lang/parser.c (line 296)
[declaration] nl_commit(p, &fns, n);
kernel_Workspace/kernel/src/core-lang/parser.c (line 305)
[declaration] expect(p, TK_DATAIMPL, "dataimpl");
kernel_Workspace/kernel/src/core-lang/parser.c (line 312)
[function] while (!check(p, TK_RBRACE) && !check(p, TK_EOF)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 313)
[function] if (match(p, TK_DATA)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 314)
[declaration] expect(p, TK_COLON, "data:");
kernel_Workspace/kernel/src/core-lang/parser.c (line 318)
[declaration] nl_push(p, &base, parse_expr(p));
kernel_Workspace/kernel/src/core-lang/parser.c (line 320)
[function] if (!match(p, TK_COMMA)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 325)
[declaration] expect(p, TK_RBRACE, "dataimpl }");
kernel_Workspace/kernel/src/core-lang/parser.c (line 327)
[declaration] expect(p, TK_ARROW, "=>");

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 330)
[declaration] nl_commit(p, &base, baseblk);
kernel_Workspace/kernel/src/core-lang/parser.c (line 348)
[function] while (!check(p, TK_RBACE) && !check(p, TK_EOF)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 351)
[function] switch (peek(p)->kind) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 358)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 361)
[declaration] expect(p, TK_IDENT, "set_free var");
kernel_Workspace/kernel/src/core-lang/parser.c (line 362)
[declaration] expect(p, TK_RBACE, "set_free }");
kernel_Workspace/kernel/src/core-lang/parser.c (line 371)
[function] if (s == (void *)1) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 372)
[function] if (check(p, TK_IF)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 384)
[declaration] nl_push(p, &stmts, s);
kernel_Workspace/kernel/src/core-lang/parser.c (line 387)
[declaration] expect(p, TK_RBACE, "}");
kernel_Workspace/kernel/src/core-lang/parser.c (line 389)
[declaration] nl_commit(p, &stmts, n);
kernel_Workspace/kernel/src/core-lang/parser.c (line 398)
[declaration] expect(p, TK_IF, "if");
kernel_Workspace/kernel/src/core-lang/parser.c (line 400)
[function] if (!check(p, TK_LBRACE)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 407)
[function] if (match(p, TK_ELSE)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 408)
[function] if (check(p, TK_IF) && peek_at(p, 1)->kind != TK_ELSE) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 409)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 413)
[function] if (!check(p, TK_LBRACE)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 423)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 424)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 435)
[declaration] expect(p, TK_WHEN, "when");
kernel_Workspace/kernel/src/core-lang/parser.c (line 437)
[declaration] copy_name(n->name, sizeof(n->name), expect(p, TK_IDENT, "when
target"));
kernel_Workspace/kernel/src/core-lang/parser.c (line 439)
[declaration] expect(p, TK_EQCOLON, "=:");
kernel_Workspace/kernel/src/core-lang/parser.c (line 444)
[function] while (!check(p, TK_RBACE) && !check(p, TK_EOF)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 450)
[declaration] nl_push(p, &cases, c);

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 453)
[declaration] expect(p, TK_RBRACE, "when ");
kernel_Workspace/kernel/src/core-lang/parser.c (line 455)
[declaration] nl_commit(p, &cases, n);
kernel_Workspace/kernel/src/core-lang/parser.c (line 466)
[function] if (check(p, TK_NUM)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 472)
[function] if (check(p, TK_STR)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 475)
[declaration] copy_name(n->name, sizeof(n->name), t);
kernel_Workspace/kernel/src/core-lang/parser.c (line 479)
[function] if (check(p, TK_CHAR)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 486)
[function] check(p, TK_DOLLAR_AT)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 494)
[declaration] expect(p, TK_IDENT, "$var");
kernel_Workspace/kernel/src/core-lang/parser.c (line 498)
[function] if (check(p, TK_CALL) && peek_at(p, 1)->kind == TK_LBRACK) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 500)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 501)
[declaration] expect(p, TK_LBRACK, "call[");
kernel_Workspace/kernel/src/core-lang/parser.c (line 503)
[declaration] expect(p, TK_IDENT, "call[x]");
kernel_Workspace/kernel/src/core-lang/parser.c (line 504)
[declaration] expect(p, TK_RBRACK, "]);
kernel_Workspace/kernel/src/core-lang/parser.c (line 508)
[function] if (match(p, TK_LPAREN)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 510)
[declaration] expect(p, TK_RPAREN, "));
kernel_Workspace/kernel/src/core-lang/parser.c (line 514)
[function] if (check(p, TK_IDENT)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 516)
[declaration] copy_name(n->name, sizeof(n->name), advance(p));
kernel_Workspace/kernel/src/core-lang/parser.c (line 520)
[declaration] perr(p, "unexpected token in expr");
kernel_Workspace/kernel/src/core-lang/parser.c (line 521)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 523)
[declaration] return new_node(p, ND_NUM);
kernel_Workspace/kernel/src/core-lang/parser.c (line 531)
[function] if (match(p, TK_DOT)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 536)
[declaration] expect(p, TK_IDENT, "member");
kernel_Workspace/kernel/src/core-lang/parser.c (line 539)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 547)
[function] if (n->kind == ND_VAR) {

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 554)
[declaration] expect(p, TK_IDENT, "scope fn");
kernel_Workspace/kernel/src/core-lang/parser.c (line 556)
[declaration] expect(p, TK_LPAREN, "scope (");
kernel_Workspace/kernel/src/core-lang/parser.c (line 560)
[function] while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 561)
[declaration] nl_push(p, &args, parse_expr(p));
kernel_Workspace/kernel/src/core-lang/parser.c (line 563)
[function] if (!match(p, TK_COMMA)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 568)
[declaration] expect(p, TK_RPAREN, "::~");
kernel_Workspace/kernel/src/core-lang/parser.c (line 570)
[declaration] nl_commit(p, &args, m);
kernel_Workspace/kernel/src/core-lang/parser.c (line 574)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 582)
[function] while (!check(p, TK_RPAREN) && !check(p, TK_EOF)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 583)
[declaration] nl_push(p, &args, parse_expr(p));
kernel_Workspace/kernel/src/core-lang/parser.c (line 585)
[function] if (!match(p, TK_COMMA)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 590)
[declaration] expect(p, TK_RPAREN, "call ");
kernel_Workspace/kernel/src/core-lang/parser.c (line 592)
[declaration] nl_commit(p, &args, m);
kernel_Workspace/kernel/src/core-lang/parser.c (line 606)
[function] check(p, TK_TILDE)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 615)
[declaration] return parse_postfix(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 626)
[function] switch (k) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 638)
[function] if (lv > level) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 642)
[declaration] advance(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 662)
[function] if (match(p, TK_ASSIGN)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 676)
[declaration] return parse_assign(p);
kernel_Workspace/kernel/src/core-lang/parser.c (line 694)
[function] while (!check(p, TK_EOF)) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 697)
[function] switch (peek(p)->kind) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 705)
[declaration] perr(p, "expected top-level decl");
kernel_Workspace/kernel/src/core-lang/parser.c (line 706)
[declaration] advance(p);

```

```

kernel_Workspace/kernel/src/core-lang/parser.c (line 710)
[declaration] nl_push(p, &tops, d);
kernel_Workspace/kernel/src/core-lang/parser.c (line 712)
[function] if (p->failed) {
kernel_Workspace/kernel/src/core-lang/parser.c (line 717)
[declaration] nl_commit(p, &tops, prog);
kernel_Workspace/kernel/src/core-lang/parser.c (line 719)
[function] if (p->failed) {
kernel_Workspace/kernel/src/core-lang/run.c (line 4)
[declaration] extern long cl_kernel_read(const char *path, uint8_t *buf, size_t
cap);
kernel_Workspace/kernel/src/core-lang/run.c (line 13)
[function] if (n <= 0) {
kernel_Workspace/kernel/src/core-lang/run.c (line 22)
[function] if (P == (void *)0) {
kernel_Workspace/kernel/src/core-lang/run.c (line 23)
[declaration] extern void kprintf(const char *, ...);
kernel_Workspace/kernel/src/core-lang/run.c (line 24)
[declaration] kprintf("core-lang: %s line %u: %s\n", path, el, em);
kernel_Workspace/kernel/src/core-lang/run.c (line 30)
[declaration] cl_vm_init(&vm, P);
kernel_Workspace/kernel/src/core-lang/run.c (line 31)
[declaration] cl_bridge_init(&vm, ring);
kernel_Workspace/kernel/src/core-lang/run.c (line 35)
[function] if (code != (void *)0) {
kernel_Workspace/kernel/src/core-lang/run.c (line 36)
[declaration] extern void cl_make_exec(void *p, size_t len);
kernel_Workspace/kernel/src/core-lang/run.c (line 37)
[declaration] cl_make_exec(code, 16384);
kernel_Workspace/kernel/src/core-lang/run.c (line 57)
[declaration] cl_vm_run(&vm);
kernel_Workspace/kernel/src/core-lang/sema.c (line 45)
[function] if (s == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 58)
[function] if (s->ndiags < 64) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 66)
[function] if (name != (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 77)
[function] if (s->top < 15) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 86)
[function] if (sym->kind == SYM_VAR && sym->live) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 92)
[function] if (s->top > 0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 104)
[function] if (name[i] != sym->name[i]) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 110)
[function] if (same) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 111)
[declaration] diag(s, line, "redeclaration in same scope", name, false);

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 118)
[function] if (sym == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 149)
[function] if (name[k] != sym->name[k]) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 155)
[function] if (same) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 167)
[function] if (t.bits == 0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 171)
[function] if (t.bits >= 64) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 175)
[function] if (negative) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 176)
[function] if (!t.neg) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 183)
[function] if (t.neg) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 197)
[declaration] static etype_t check_expr(sema_ctx_t *s, ast_node_t *n);
kernel_Workspace/kernel/src/core-lang/sema.c (line 202)
[function] if (s->sh_count < 2) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 216)
[function] if (s->sh_count == 1) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 220)
[function] if (s->sh_type[0].bits != s->sh_type[1].bits) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 233)
[function] if (n == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 237)
[function] switch (n->kind) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 255)
[function] if (n->op == 45 ) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 256)
[function] if (r.is_const) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 259)
[declaration] diag(s, n->line, "negation needs _=> range", (void *)0, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 268)
[function] if (sym == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 269)
[declaration] diag(s, n->line, "undeclared identifier", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 273)
[function] if (sym->kind == SYM_VAR || sym->kind == SYM_STATIC) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 274)
[function] if (!sym->live) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 275)
[declaration] diag(s, n->line, "use after set_free", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 282)
[declaration] diag(s, n->line, "not a value", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 291)
[function] if (sym == (void *)0) {

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 292)
[declaration] diag(s, n->line, "undeclared identifier", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 296)
[function] if (sym->depth != 0 || sym->kind != SYM_VAR) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 301)
[function] if (!sym->live) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 302)
[declaration] diag(s, n->line, "use after set_free", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 307)
[function] if (n->pool_kind == 1 && !sym->type.neg) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 308)
[declaration] diag(s, n->line, "$! needs neg range (=>-1)", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 311)
[function] if (n->pool_kind == 2) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 313)
[declaration] shared_note(s, n->name, sym->type, n->line);
kernel_Workspace/kernel/src/core-lang/sema.c (line 323)
[function] if (sym == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 324)
[declaration] diag(s, n->line, "call[] on undeclared", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 328)
[function] if ((sym->kind == SYM_VAR || sym->kind == SYM_STATIC) && !sym->live) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 329)
[declaration] diag(s, n->line, "use after set_free", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 340)
[function] if (sc == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 341)
[declaration] diag(s, n->line, "unknown scope", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 345)
[function] if (sc->kind == SYM_ENUM) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 348)
[function] if (sc->node != (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 355)
[function] if (it->name[k] != n->name2[k]) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 361)
[function] if (same) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 370)
[function] if (!found) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 371)
[declaration] diag(s, n->line, "no such enum item", n->name2, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 377)
[function] if (sc->kind == SYM_DATA) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 380)
[function] if (sc->node != (void *)0 && sc->node->a != (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 386)
[function] if (e->kind == ND_VAR) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 390)
[function] if (e->name[k] != n->name2[k]) {

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 396)
[function] if (same) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 404)
[function] if (!found) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 405)
[declaration] diag(s, n->line, "no such data entry", n->name2, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 411)
[declaration] diag(s, n->line, "scope is not enum/data", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 416)
[function] if (n->a == (void *)0 || n->a->kind != ND_VAR) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 417)
[declaration] diag(s, n->line, "bad call target", (void *)0, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 423)
[function] if (fn == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 424)
[declaration] diag(s, n->line, "call of undeclared", n->a->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 428)
[function] if (fn->kind == SYM_FN) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 429)
[function] if (fn->nparams != n->nlist) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 430)
[declaration] diag(s, n->line, "arity mismatch", fn->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 438)
[declaration] diag(s, n->line, "arg out of range", fn->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 446)
[function] if (fn->kind == SYM_EXTERN) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 448)
[declaration] check_expr(s, n->list[i]);
kernel_Workspace/kernel/src/core-lang/sema.c (line 455)
[declaration] diag(s, n->line, "not callable", n->a->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 467)
[function] if (l.t.bits != 0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 468)
[function] if (rhs.is_const) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 469)
[function] if (!value_fits(rhs.cval, rhs.cneg, l.t)) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 470)
[declaration] diag(s, n->line, "value out of range", (void *)0, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 473)
[function] if (rhs.t.bits > l.t.bits) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 474)
[declaration] diag(s, n->line, "narrowing assignment", (void *)0, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 493)
[function] if (cmp) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 498)
[function] if (a.t.bits == 0 && b.t.bits != 0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 499)
[function] if (a.is_const && !value_fits(a.cval, a.cneg, b.t)) {

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 500)
[declaration] diag(s, n->line, "value out of range", (void *)0, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 505)
[function] if (b.is_const && !value_fits(b.cval, b.cneg, a.t)) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 506)
[declaration] diag(s, n->line, "value out of range", (void *)0, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 524)
[declaration] static void check_block(sema_ctx_t *s, ast_node_t *n);
kernel_Workspace/kernel/src/core-lang/sema.c (line 530)
[function] switch (n->kind) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 536)
[declaration] diag(s, n->line, "literal out of type range", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 539)
[function] if (init.t.bits != 0 && init.t.bits > n->type.bits) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 540)
[declaration] diag(s, n->line, "narrowing assignment", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 545)
[function] if (sym != (void *)0 && n->type.neg) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 553)
[declaration] declare(s, "data", SYM_STATIC, n->type, n->line);
kernel_Workspace/kernel/src/core-lang/sema.c (line 559)
[function] if (sym == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 560)
[declaration] diag(s, n->line, "set_free on undeclared", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 562)
[declaration] diag(s, n->line, "double set_free", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 572)
[function] if (n->a != (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 573)
[declaration] check_expr(s, n->a);
kernel_Workspace/kernel/src/core-lang/sema.c (line 576)
[declaration] check_block(s, n->b);
kernel_Workspace/kernel/src/core-lang/sema.c (line 578)
[function] if (n->c != (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 579)
[function] if (n->c->kind == ND_BLOCK) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 580)
[declaration] check_block(s, n->c);
kernel_Workspace/kernel/src/core-lang/sema.c (line 582)
[declaration] check_stmt(s, n->c);
kernel_Workspace/kernel/src/core-lang/sema.c (line 591)
[function] if (tgt == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 592)
[declaration] diag(s, n->line, "when on undeclared", n->name, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 606)
[function] if (cn[k] != n->name[k]) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 612)
[function] if (!same) {

```

```

kernel_Workspace/kernel/src/core-lang/sema.c (line 613)
[declaration] diag(s, c->line, "case target mismatch", cn, false);
kernel_Workspace/kernel/src/core-lang/sema.c (line 624)
[declaration] check_block(s, c->b);
kernel_Workspace/kernel/src/core-lang/sema.c (line 631)
[declaration] check_block(s, n);
kernel_Workspace/kernel/src/core-lang/sema.c (line 635)
[declaration] check_expr(s, n);
kernel_Workspace/kernel/src/core-lang/sema.c (line 636)
[declaration] shared_validate(s);
kernel_Workspace/kernel/src/core-lang/sema.c (line 640)
[function] if (n->kind != ND_BLOCK) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 641)
[declaration] shared_validate(s);
kernel_Workspace/kernel/src/core-lang/sema.c (line 647)
[declaration] push_scope(s);
kernel_Workspace/kernel/src/core-lang/sema.c (line 650)
[declaration] check_stmt(s, n->list[i]);
kernel_Workspace/kernel/src/core-lang/sema.c (line 653)
[declaration] pop_scope(s);
kernel_Workspace/kernel/src/core-lang/sema.c (line 659)
[function] switch (n->kind) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 664)
[function] if (sym == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 677)
[declaration] declare(s, n->name, SYM_VAR, n->type, n->line);
kernel_Workspace/kernel/src/core-lang/sema.c (line 681)
[declaration] declare(s, "data", SYM_STATIC, n->type, n->line);
kernel_Workspace/kernel/src/core-lang/sema.c (line 688)
[function] if (sym != (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 698)
[declaration] declare(s, n->list[i]->name, SYM_EXTERN, t, n->line);
kernel_Workspace/kernel/src/core-lang/sema.c (line 707)
[function] if (sym != (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 721)
[function] if (prog == (void *)0) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 726)
[declaration] declare_top(s, prog->list[i]);
kernel_Workspace/kernel/src/core-lang/sema.c (line 732)
[function] if (n->kind != ND_FN) {
kernel_Workspace/kernel/src/core-lang/sema.c (line 736)
[declaration] push_scope(s);
kernel_Workspace/kernel/src/core-lang/sema.c (line 739)
[declaration] declare(s, n->list[k]->name, SYM_VAR, n->list[k]->type, n->line);
kernel_Workspace/kernel/src/core-lang/sema.c (line 742)
[declaration] check_block(s, n->a);
kernel_Workspace/kernel/src/core-lang/sema.c (line 744)
[declaration] pop_scope(s);

```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 19)
[declaration] for (int i = full + (rem ? 1 : 0); i < w; i++) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 23)
[function] if (rem) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 85)
[function] if (a[i] != b[i]) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 95)
[function] if (!sign) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 96)
[declaration] return cmp_u(a, b, w);
kernel_Workspace/kernel/src/core-lang/vm.c (line 113)
[declaration] return cmp_u(x, y, w);
kernel_Workspace/kernel/src/core-lang/vm.c (line 121)
[function] if (bit_get(a, i) && (int)(i + n) < w * 8) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 122)
[declaration] bit_set(t, i + (int)n);
kernel_Workspace/kernel/src/core-lang/vm.c (line 136)
[function] if (bit_get(a, i) && i >= (int)n) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 137)
[declaration] bit_set(t, i - (int)n);
kernel_Workspace/kernel/src/core-lang/vm.c (line 154)
[declaration] cl_shl(shl, rem, 1, w);
kernel_Workspace/kernel/src/core-lang/vm.c (line 160)
[function] if (bit_get(n, i)) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 164)
[function] if (cmp_u(rem, d, w) >= 0) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 166)
[declaration] cl_sub(sub, rem, d, w);
kernel_Workspace/kernel/src/core-lang/vm.c (line 172)
[declaration] bit_set(quo, i);
kernel_Workspace/kernel/src/core-lang/vm.c (line 207)
[function] if (vm->prog->ext_names[i][k] != name[k]) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 213)
[function] if (same) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 224)
[function] if (vm->sp < 256) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 233)
[function] if (vm->sp > 0) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 260)
[function] if (P->main_fn < 0) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 273)
[function] while (vm->depth > 0 && vm->err == CL_OK && !vm->halt_req) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 276)
[function] if (f->pc >= P->code_len) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 287)
[function] switch (in.op) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 292)
[declaration] cl_mask(r.b, w, in.bits);

```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 293)
[declaration] push(vm, &r);
kernel_Workspace/kernel/src/core-lang/vm.c (line 297)
[function] if (!f->slive[in.a]) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 301)
[declaration] push(vm, &f->slots[in.a]);
kernel_Workspace/kernel/src/core-lang/vm.c (line 306)
[declaration] cl_mask(a.b, w, in.bits);
kernel_Workspace/kernel/src/core-lang/vm.c (line 312)
[function] if (!vm->glive[in.a]) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 316)
[declaration] push(vm, &vm->globals[in.a]);
kernel_Workspace/kernel/src/core-lang/vm.c (line 321)
[declaration] cl_mask(a.b, w, in.bits);
kernel_Workspace/kernel/src/core-lang/vm.c (line 335)
[function] switch (in.op) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 336)
[declaration] case OP_ADD: cl_add(r.b, a.b, b.b, w); break;
kernel_Workspace/kernel/src/core-lang/vm.c (line 337)
[declaration] case OP_SUB: cl_sub(r.b, a.b, b.b, w); break;
kernel_Workspace/kernel/src/core-lang/vm.c (line 338)
[declaration] case OP_MUL: cl_mul(r.b, a.b, b.b, w); break;
kernel_Workspace/kernel/src/core-lang/vm.c (line 342)
[declaration] case OP_SHL: cl_shl(r.b, a.b, (uint32_t)b.b[0], w); break;
kernel_Workspace/kernel/src/core-lang/vm.c (line 343)
[declaration] case OP_SHR: cl_shr(r.b, a.b, (uint32_t)b.b[0], w); break;
kernel_Workspace/kernel/src/core-lang/vm.c (line 352)
[function] if (zero) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 357)
[declaration] cl_divmod(a.b, b.b, w, q, m);
kernel_Workspace/kernel/src/core-lang/vm.c (line 366)
[declaration] cl_mask(r.b, w, in.bits);
kernel_Workspace/kernel/src/core-lang/vm.c (line 367)
[declaration] push(vm, &r);
kernel_Workspace/kernel/src/core-lang/vm.c (line 374)
[declaration] cl_sub(r.b, z, a.b, w);
kernel_Workspace/kernel/src/core-lang/vm.c (line 375)
[declaration] cl_mask(r.b, w, in.bits);
kernel_Workspace/kernel/src/core-lang/vm.c (line 377)
[declaration] push(vm, &r);
kernel_Workspace/kernel/src/core-lang/vm.c (line 383)
[declaration] cl_mask(r.b, w, in.bits);
kernel_Workspace/kernel/src/core-lang/vm.c (line 384)
[declaration] push(vm, &r);
kernel_Workspace/kernel/src/core-lang/vm.c (line 395)
[function] switch (in.op) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 407)
[declaration] push(vm, &r);

```

```

kernel_Workspace/kernel/src/core-lang/vm.c (line 425)
[declaration] pop(vm);
kernel_Workspace/kernel/src/core-lang/vm.c (line 429)
[function] if (in.a == 1) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 439)
[function] if (!vm->glive[in.a]) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 445)
[function] if (in.a >= 16 || vm->externs[in.a] == (cl_ext_fn)0) {
kernel_Workspace/kernel/src/core-lang/vm.c (line 464)
[declaration] push(vm, &r);
kernel_Workspace/kernel/src/core-lang/vm.c (line 469)
[function] if (vm->depth >= 8) {
kernel_Workspace/kernel/src/displayport/aux.c (line 50)
[function] if (addr + len > 0x400) {
kernel_Workspace/kernel/src/displayport/aux.c (line 65)
[function] if (addr + len > 0x400) {
kernel_Workspace/kernel/src/displayport/aux.c (line 79)
[function] if (pat == 1) {
kernel_Workspace/kernel/src/displayport/aux.c (line 85)
[function] if (train_step >= 2) {
kernel_Workspace/kernel/src/displayport/aux.c (line 100)
[function] if (len > 128) {
kernel_Workspace/kernel/src/displayport/edid.c (line 8)
[function] if (b[0] != 0x00 || b[1] != 0xFF || b[7] != 0x00) {
kernel_Workspace/kernel/src/displayport/edid.c (line 17)
[function] if (clock == 0) {
kernel_Workspace/kernel/src/displayport/edid.c (line 32)
[function] if (hact == 0 || vact == 0) {
kernel_Workspace/kernel/src/displayport/edid.c (line 36)
[function] if (*count >= max) {
kernel_Workspace/kernel/src/displayport/init.c (line 13)
[function] if (dp_is_ready) {
kernel_Workspace/kernel/src/displayport/init.c (line 17)
[declaration] dp_aux_sink_init();
kernel_Workspace/kernel/src/displayport/init.c (line 19)
[function] if (!dp_link_train()) {
kernel_Workspace/kernel/src/displayport/init.c (line 25)
[function] if (!dp_aux_read_edid(raw, 128)) {
kernel_Workspace/kernel/src/displayport/init.c (line 29)
[function] if (!edid_parse(raw, MODES, 8, &MODE_COUNT)) {
kernel_Workspace/kernel/src/displayport/init.c (line 51)
[function] if (i >= MODE_COUNT) {
kernel_Workspace/kernel/src/displayport/init.c (line 66)
[function] if (MODES[i].id == id) {
kernel_Workspace/kernel/src/displayport/init.c (line 67)
[declaration] dp_op_set_fb(0, MODES[i].w, MODES[i].h, MODES[i].w);
kernel_Workspace/kernel/src/displayport/init.c (line 77)
[declaration] dp_op_state(fb_phys, w, h, stride);

```

```

kernel_Workspace/kernel/src/displayport/init.c (line 82)
[declaration] dp_op_set_fb(phys, w, h, stride);
kernel_Workspace/kernel/src/displayport/init.c (line 91)
[declaration] dp_op_fill(color, x, y, w, h);
kernel_Workspace/kernel/src/displayport/link.c (line 12)
[declaration] dp_aux_read(DPCD_MAX_LINK_RATE, &max_rate, 1);
kernel_Workspace/kernel/src/displayport/link.c (line 13)
[declaration] dp_aux_read(DPCD_MAX_LANE_COUNT, &max_lanes, 1);
kernel_Workspace/kernel/src/displayport/link.c (line 18)
[function] if (lanes > 4) {
kernel_Workspace/kernel/src/displayport/link.c (line 22)
[declaration] dp_aux_write(DPCD_LINK_BW_SET, &bw, 1);
kernel_Workspace/kernel/src/displayport/link.c (line 23)
[declaration] dp_aux_write(DPCD_LANE_COUNT_SET, &lanes, 1);
kernel_Workspace/kernel/src/displayport/link.c (line 26)
[declaration] dp_aux_write(DPCD_TRAINING_SET, &tp, 1);
kernel_Workspace/kernel/src/displayport/link.c (line 31)
[declaration] dp_aux_read(DPCD_LANE_STATUS, st, 2);
kernel_Workspace/kernel/src/displayport/link.c (line 33)
[function] if ((st[0] & 0x11) == 0x11 && (st[1] & 0x11) == 0x11) {
kernel_Workspace/kernel/src/displayport/link.c (line 39)
[declaration] dp_aux_write(DPCD_TRAINING_SET, &tp, 1);
kernel_Workspace/kernel/src/displayport/link.c (line 42)
[declaration] dp_aux_read(DPCD_LANE_STATUS, st, 2);
kernel_Workspace/kernel/src/displayport/link.c (line 44)
[function] if ((st[0] & 0x66) == 0x66 && (st[1] & 0x66) == 0x66) {
kernel_Workspace/kernel/src/displayport/link.c (line 50)
[declaration] dp_aux_read(DPCD_SYMBOL_LOCK, &lock, 1);
kernel_Workspace/kernel/src/displayport/link.c (line 52)
[function] if (lock != 1) {
kernel_Workspace/kernel/src/displayport/link.c (line 57)
[declaration] dp_aux_write(DPCD_TRAINING_SET, &tp, 1);
kernel_Workspace/kernel/src/displayport/link.c (line 59)
[function] switch (bw) {
kernel_Workspace/kernel/src/displayport/operation.c (line 27)
[function] if (fb_phys == 0) {
kernel_Workspace/kernel/src/editor/editor.c (line 11)
[declaration] extern void console_set_enabled(int on);
kernel_Workspace/kernel/src/editor/editor.c (line 12)
[declaration] extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);
kernel_Workspace/kernel/src/editor/editor.c (line 13)
[declaration] extern uint32_t k_input_keycode(void);
kernel_Workspace/kernel/src/editor/editor.c (line 88)
[function] if (bits & (0x80 >> rx)) {
kernel_Workspace/kernel/src/editor/editor.c (line 92)
[declaration] ed_fill(px, py, SCALE, SCALE, color);
kernel_Workspace/kernel/src/editor/editor.c (line 100)
[function] while (*s) {

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 101)
[declaration] ed_char(x, y, *s, color);
kernel_Workspace/kernel/src/editor/editor.c (line 111)
[function] if (v == 0) { buf[0] = '0'; buf[1] = 0; return; }
kernel_Workspace/kernel/src/editor/editor.c (line 116)
[function] while (v > 0 && n < 11) { tmp[n++] = (char)('0' + v % 10); v /= 10; }
kernel_Workspace/kernel/src/editor/editor.c (line 147)
[function] if (v == 0) { tmp[n++] = '0'; }
kernel_Workspace/kernel/src/editor/editor.c (line 149)
[function] while (v > 0 && n < 20) { tmp[n++] = (char)('0' + v % 10); v /= 10; }
kernel_Workspace/kernel/src/editor/editor.c (line 154)
[declaration] log_append(buf);
kernel_Workspace/kernel/src/editor/editor.c (line 165)
[declaration] log_append(buf);
kernel_Workspace/kernel/src/editor/editor.c (line 172)
[function] switch (k) {
kernel_Workspace/kernel/src/editor/editor.c (line 201)
[function] if (tmp[k] == '/' && tmp[k + 1] == '/') { com_at = k; break; }
kernel_Workspace/kernel/src/editor/editor.c (line 208)
[declaration] cl_lexer_init(&l, tmp, ed_strlen(tmp));
kernel_Workspace/kernel/src/editor/editor.c (line 210)
[function] if (com_at >= 0) {
kernel_Workspace/kernel/src/editor/editor.c (line 216)
[declaration] cl_lexer_init(&l, cut, (size_t)com_at);
kernel_Workspace/kernel/src/editor/editor.c (line 221)
[function] if (cl_lex_all(&l, toks, 64, &n) == 0) {
kernel_Workspace/kernel/src/editor/editor.c (line 225)
[function] while (pos < ts) {
kernel_Workspace/kernel/src/editor/editor.c (line 226)
[declaration] ed_char(x + (uint32_t)pos * CHAR_W, y, ' ', COL_FG);
kernel_Workspace/kernel/src/editor/editor.c (line 240)
[function] while (tmp[pos] && (com_at < 0 || pos < com_at)) {
kernel_Workspace/kernel/src/editor/editor.c (line 241)
[declaration] ed_char(x + (uint32_t)pos * CHAR_W, y, tmp[pos], COL_FG);
kernel_Workspace/kernel/src/editor/editor.c (line 245)
[function] if (com_at >= 0) {
kernel_Workspace/kernel/src/editor/editor.c (line 247)
[declaration] ed_char(x + (uint32_t)k * CHAR_W, y, tmp[k], COL_COM);
kernel_Workspace/kernel/src/editor/editor.c (line 285)
[function] if (sx >= 0 && sx < (int)fw && sy >= 0 && sy < (int)fh) {
kernel_Workspace/kernel/src/editor/editor.c (line 296)
[declaration] ed_fill(0, 0, fw, fh, COL_BG);
kernel_Workspace/kernel/src/editor/editor.c (line 298)
[declaration] ed_fill(0, 0, fw, TOP_BAR, COL_BAR);
kernel_Workspace/kernel/src/editor/editor.c (line 299)
[declaration] ed_text(8, 4, "TrangEdit [core-lang]", COL_TYPE);
kernel_Workspace/kernel/src/editor/editor.c (line 300)
[declaration] ed_text(GUTTER + 160, 4, msg, COL_GUTTER);

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 312)
[function] if (row == caret_l) {
kernel_Workspace/kernel/src/editor/editor.c (line 313)
[declaration] ed_fill(GUTTER, y, (uint32_t)cols * CHAR_W, CHAR_H, 0xFF2D2D2D);
kernel_Workspace/kernel/src/editor/editor.c (line 317)
[declaration] ed_num(nb, row + 1);
kernel_Workspace/kernel/src/editor/editor.c (line 318)
[declaration] ed_text(8, y, nb, COL_GUTTER);
kernel_Workspace/kernel/src/editor/editor.c (line 320)
[declaration] draw_hl(row, GUTTER + 4, y);
kernel_Workspace/kernel/src/editor/editor.c (line 326)
[function] if (r >= 0 && r < rows) {
kernel_Workspace/kernel/src/editor/editor.c (line 330)
[declaration] ed_fill(cx, cy, 2, CHAR_H, COL_CARET);
kernel_Workspace/kernel/src/editor/editor.c (line 334)
[function] if (show_log) {
kernel_Workspace/kernel/src/editor/editor.c (line 337)
[declaration] ed_fill(0, ly, fw, 8 * CHAR_H, 0xFF101010);
kernel_Workspace/kernel/src/editor/editor.c (line 340)
[declaration] ed_text(8, ly + (uint32_t)i * CHAR_H, log_lines[i], COL_NUM);
kernel_Workspace/kernel/src/editor/editor.c (line 344)
[declaration] ed_fill(0, fh - BOT_BAR, fw, BOT_BAR, COL_BAR);
kernel_Workspace/kernel/src/editor/editor.c (line 351)
[declaration] ed_num(tmp, caret_l + 1);
kernel_Workspace/kernel/src/editor/editor.c (line 355)
[declaration] ed_num(tmp, caret_c + 1);
kernel_Workspace/kernel/src/editor/editor.c (line 359)
[declaration] ed_text(8, fh - BOT_BAR + 4, st, COL_FG);
kernel_Workspace/kernel/src/editor/editor.c (line 360)
[declaration] ed_text(GUTTER + 200, fh - BOT_BAR + 4, msg, COL_GUTTER);
kernel_Workspace/kernel/src/editor/editor.c (line 362)
[declaration] draw_mouse();
kernel_Workspace/kernel/src/editor/editor.c (line 413)
[function] if (caret_c > 0) {
kernel_Workspace/kernel/src/editor/editor.c (line 425)
[function] if (plen + slen < ED_LINE_LEN - 1) {
kernel_Workspace/kernel/src/editor/editor.c (line 448)
[function] if (caret_c < len) {
kernel_Workspace/kernel/src/editor/editor.c (line 485)
[declaration] render();
kernel_Workspace/kernel/src/editor/editor.c (line 496)
[function] if (P == (void *)0) {
kernel_Workspace/kernel/src/editor/editor.c (line 499)
[function] while (pre[i]) { msg[i] = pre[i]; i++; }
kernel_Workspace/kernel/src/editor/editor.c (line 501)
[declaration] ed_num(tmp, (int)el);
kernel_Workspace/kernel/src/editor/editor.c (line 507)
[declaration] if (em) log_append(em);

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 511)
[declaration] cl_vm_init(&ed_vm, P);
kernel_Workspace/kernel/src/editor/editor.c (line 512)
[declaration] cl_bridge_init(&ed_vm, 0);
kernel_Workspace/kernel/src/editor/editor.c (line 514)
[declaration] cl_vm_register_extern(&ed_vm, "put", ed_put);
kernel_Workspace/kernel/src/editor/editor.c (line 515)
[declaration] cl_vm_register_extern(&ed_vm, "putc", ed_putc);
kernel_Workspace/kernel/src/editor/editor.c (line 517)
[declaration] log_append("== run ==");
kernel_Workspace/kernel/src/editor/editor.c (line 523)
[function] while (ok[i]) { msg[i] = ok[i]; i++; }
kernel_Workspace/kernel/src/editor/editor.c (line 536)
[function] if (gfx_fb_info_raw(&w, &h, &s, &base, &fliprows) != 0 || base == 0) {
kernel_Workspace/kernel/src/editor/editor.c (line 549)
[function] if (path != (void *)0) {
kernel_Workspace/kernel/src/editor/editor.c (line 554)
[function] if (n > 0) {
kernel_Workspace/kernel/src/editor/editor.c (line 563)
[function] if (c == '\n') {
kernel_Workspace/kernel/src/editor/editor.c (line 572)
[function] if (col > 0 || nlines == 0) {
kernel_Workspace/kernel/src/editor/editor.c (line 579)
[declaration] console_set_enabled(0);
kernel_Workspace/kernel/src/editor/editor.c (line 580)
[declaration] mouse_init();
kernel_Workspace/kernel/src/editor/editor.c (line 584)
[function] while (running) {
kernel_Workspace/kernel/src/editor/editor.c (line 590)
[function] if (mouse_poll(&dx, &dy, &dz, &btn)) {
kernel_Workspace/kernel/src/editor/editor.c (line 599)
[function] if (dz != 0) {
kernel_Workspace/kernel/src/editor/editor.c (line 601)
[declaration] clamp_scroll(rows);
kernel_Workspace/kernel/src/editor/editor.c (line 605)
[function] if (btn & 1 && !(mbuttons & 1)) {
kernel_Workspace/kernel/src/editor/editor.c (line 606)
[function] if (my >= (int)TOP_BAR && my < (int)(fh - BOT_BAR)) {
kernel_Workspace/kernel/src/editor/editor.c (line 618)
[declaration] clamp_scroll(rows);
kernel_Workspace/kernel/src/editor/editor.c (line 629)
[function] if (k != 0) {
kernel_Workspace/kernel/src/editor/editor.c (line 630)
[function] switch (k) {
kernel_Workspace/kernel/src/editor/editor.c (line 632)
[declaration] case EDK_ENTER: do_enter(); dirty = 1; break;
kernel_Workspace/kernel/src/editor/editor.c (line 633)
[declaration] case EDK_BACKSPACE: do_backspace(); dirty = 1; break;

```

```

kernel_Workspace/kernel/src/editor/editor.c (line 634)
[declaration] case EDK_DELETE: do_delete(); dirty = 1; break;
kernel_Workspace/kernel/src/editor/editor.c (line 635)
[declaration] case EDK_TAB: ins_char(' '); ins_char(' ');
kernel_Workspace/kernel/src/editor/editor.c (line 636)
[declaration] ins_char(' '); ins_char(' '); dirty = 1; break;
kernel_Workspace/kernel/src/editor/editor.c (line 643)
[declaration] case EDK_F5: ed_compile_run(); dirty = 1; break;
kernel_Workspace/kernel/src/editor/editor.c (line 646)
[function] if (k >= 32 && k < 0x100) {
kernel_Workspace/kernel/src/editor/editor.c (line 647)
[declaration] ins_char((char)k);
kernel_Workspace/kernel/src/editor/editor.c (line 653)
[declaration] clamp_scroll(rows);
kernel_Workspace/kernel/src/editor/editor.c (line 656)
[function] if (dirty) {
kernel_Workspace/kernel/src/editor/editor.c (line 657)
[declaration] render();
kernel_Workspace/kernel/src/editor/editor.c (line 660)
[declaration] __asm__ volatile("pause");
kernel_Workspace/kernel/src/editor/editor.c (line 663)
[declaration] console_set_enabled(1);
kernel_Workspace/kernel/src/editor/mouse.c (line 5)
[declaration] __asm__ volatile("outb %0, %1" :: "a"(val), "Nd"(port));
kernel_Workspace/kernel/src/editor/mouse.c (line 11)
[declaration] __asm__ volatile("inb %1, %0" : "=a"(v) : "Nd"(port));
kernel_Workspace/kernel/src/editor/mouse.c (line 17)
[function] while (inb(0x64) & 0x02) { }
kernel_Workspace/kernel/src/editor/mouse.c (line 22)
[function] while (!(inb(0x64) & 0x01)) { }
kernel_Workspace/kernel/src/editor/mouse.c (line 27)
[declaration] wait_write(); outb(0x64, 0xD4);
kernel_Workspace/kernel/src/editor/mouse.c (line 28)
[declaration] wait_write(); outb(0x60, cmd);
kernel_Workspace/kernel/src/editor/mouse.c (line 29)
[declaration] wait_read(); inb(0x60);
kernel_Workspace/kernel/src/editor/mouse.c (line 34)
[declaration] wait_write(); outb(0x64, 0xD4);
kernel_Workspace/kernel/src/editor/mouse.c (line 35)
[declaration] wait_write(); outb(0x60, 0xF3);
kernel_Workspace/kernel/src/editor/mouse.c (line 36)
[declaration] wait_read(); inb(0x60);
kernel_Workspace/kernel/src/editor/mouse.c (line 38)
[declaration] wait_write(); outb(0x64, 0xD4);
kernel_Workspace/kernel/src/editor/mouse.c (line 39)
[declaration] wait_write(); outb(0x60, r);
kernel_Workspace/kernel/src/editor/mouse.c (line 40)
[declaration] wait_read(); inb(0x60);

```

```

kernel_Workspace/kernel/src/editor/mouse.c (line 45)
[declaration] wait_write(); outb(0x64, 0xA8);
kernel_Workspace/kernel/src/editor/mouse.c (line 47)
[declaration] mouse_cmd(0xF6);
kernel_Workspace/kernel/src/editor/mouse.c (line 49)
[declaration] mouse_rate(200);
kernel_Workspace/kernel/src/editor/mouse.c (line 50)
[declaration] mouse_rate(100);
kernel_Workspace/kernel/src/editor/mouse.c (line 51)
[declaration] mouse_rate(80);
kernel_Workspace/kernel/src/editor/mouse.c (line 53)
[declaration] mouse_cmd(0xF4);
kernel_Workspace/kernel/src/editor/mouse.c (line 62)
[function] while (inb(0x64) & 0x01) {
kernel_Workspace/kernel/src/editor/mouse.c (line 66)
[function] if (!(st & 0x20)) {
kernel_Workspace/kernel/src/editor/mouse.c (line 70)
[function] if (idx == 0 && !(d & 0x08)) {
kernel_Workspace/kernel/src/editor/mouse.c (line 76)
[function] if (idx == 4) {
kernel_Workspace/kernel/src/hdmi/init.c (line 8)
[function] if (hdmi_is_ready) {
kernel_Workspace/kernel/src/hdmi/init.c (line 12)
[function] if (fb_phys == 0 || w == 0 || h == 0 || stride < w) {
kernel_Workspace/kernel/src/hdmi/init.c (line 16)
[declaration] hdmi_op_set_fb(fb_phys, w, h, stride);
kernel_Workspace/kernel/src/hdmi/init.c (line 17)
[function] if (!hdmi_op_ready()) {
kernel_Workspace/kernel/src/hdmi/init.c (line 22)
[function] if (mode != NULL && !hdmi_mode_apply(mode)) {
kernel_Workspace/kernel/src/hdmi/init.c (line 32)
[declaration] return hdmi_is_ready && hdmi_op_ready();
kernel_Workspace/kernel/src/hdmi/init.c (line 37)
[declaration] hdmi_op_state(out);
kernel_Workspace/kernel/src/hdmi/init.c (line 42)
[function] if (phys == 0 || w == 0 || h == 0 || stride < w) {
kernel_Workspace/kernel/src/hdmi/init.c (line 46)
[declaration] hdmi_op_set_fb(phys, w, h, stride);
kernel_Workspace/kernel/src/hdmi/init.c (line 53)
[function] if (!hdmi_ready()) {
kernel_Workspace/kernel/src/hdmi/init.c (line 59)
[declaration] hdmi_op_state(&caps);
kernel_Workspace/kernel/src/hdmi/init.c (line 60)
[function] if (mode == NULL || mode->w != caps.w || mode->h != caps.h) {
kernel_Workspace/kernel/src/hdmi/init.c (line 64)
[declaration] return hdmi_mode_apply(mode);
kernel_Workspace/kernel/src/hdmi/init.c (line 70)
[function] if (mode == NULL || id == NULL || w == NULL || h == NULL || r == NULL) {

```

```

kernel_Workspace/kernel/src/hdmi/interface.c (line 9)
[function] if (owner == 0) {
kernel_Workspace/kernel/src/hdmi/interface.c (line 14)
[function] if (atomic_compare_exchange_strong(&port_owner, &expected, owner)) {
kernel_Workspace/kernel/src/hdmi/interface.c (line 29)
[declaration] return atomic_load(&port_owner);
kernel_Workspace/kernel/src/hdmi/interface.c (line 39)
[function] if (owner != 0 && !hdmi_iface_check(owner)) {
kernel_Workspace/kernel/src/hdmi/interface.c (line 43)
[declaration] return hdmi_init_with(fb_phys, w, h, stride);
kernel_Workspace/kernel/src/hdmi/interface.c (line 48)
[declaration] return hdmi_ready();
kernel_Workspace/kernel/src/hdmi/interface.c (line 53)
[function] if (!hdmi_iface_check(owner)) {
kernel_Workspace/kernel/src/hdmi/interface.c (line 57)
[declaration] return hdmi_mode_set_by_id(id);
kernel_Workspace/kernel/src/hdmi/interface.c (line 62)
[declaration] return hdmi_mode_current_raw(id, w, h, r);
kernel_Workspace/kernel/src/hdmi/interface.c (line 67)
[declaration] return hdmi_mode_at_raw(i, id, w, h, r);
kernel_Workspace/kernel/src/hdmi/interface.c (line 72)
[function] if (!hdmi_iface_check(owner)) {
kernel_Workspace/kernel/src/hdmi/interface.c (line 76)
[declaration] return hdmi_submit_fill(color, x, y, w, h);
kernel_Workspace/kernel/src/hdmi/interface.c (line 81)
[function] if (!hdmi_iface_check(owner)) {
kernel_Workspace/kernel/src/hdmi/interface.c (line 85)
[declaration] return hdmi_poll(out_seq);
kernel_Workspace/kernel/src/hdmi/interface.c (line 90)
[declaration] hdmi_caps_raw(w, h, s, phys);
kernel_Workspace/kernel/src/hdmi/interface.c (line 95)
[function] if (!hdmi_iface_check(owner) || phys == NULL || w == NULL || h == NULL ||
s == NULL) {
kernel_Workspace/kernel/src/hdmi/interface.c (line 101)
[declaration] hdmi_caps_raw(&ww, &hh, &ss, &p);
kernel_Workspace/kernel/src/hdmi/interface.c (line 103)
[function] if (p == 0 || !hdmi_fb_grant()) {
kernel_Workspace/kernel/src/hdmi/interface.c (line 116)
[function] if (!hdmi_iface_check(owner)) {
kernel_Workspace/kernel/src/hdmi/interface.c (line 120)
[declaration] hdmi_fb_revoke();
kernel_Workspace/kernel/src/hdmi/mode.c (line 17)
[declaration] return (uint32_t)(sizeof(MODES) / sizeof(MODES[0]));
kernel_Workspace/kernel/src/hdmi/mode.c (line 22)
[function] if (i >= hdmi_mode_count()) {
kernel_Workspace/kernel/src/hdmi/mode.c (line 32)
[function] if (MODES[i].w == w && MODES[i].h == h && MODES[i].refresh == refresh) {
kernel_Workspace/kernel/src/hdmi/mode.c (line 43)
[function] if (MODES[i].id == id) {

```

```

kernel_Workspace/kernel/src/hdmi/mode.c (line 53)
[function] if (m == NULL || m->w == 0 || m->h == 0 || m->refresh == 0 || m->clock_khz
== 0) {
kernel_Workspace/kernel/src/hdmi/mode.c (line 58)
[declaration] hdmi_op_state(&caps);
kernel_Workspace/kernel/src/hdmi/mode.c (line 64)
[function] if (!hdmi_mode_valid(m)) {
kernel_Workspace/kernel/src/hdmi/mode.c (line 68)
[declaration] hdmi_op_set_mode(m);
kernel_Workspace/kernel/src/hdmi/mode.c (line 81)
[function] if (m == NULL || id == NULL || w == NULL || h == NULL || r == NULL) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 7)
[declaration] return (void *) (DIRECT_BASE + phys);
kernel_Workspace/kernel/src/hdmi/operation.c (line 22)
[function] if (out != NULL) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 44)
[function] if (t == NULL || !op_ready) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 50)
[function] if (stride < cur.w || dst_phys == 0) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 55)
[function] if (dst == NULL) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 59)
[function] if (t->kind == HDMI_TR_FILL) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 60)
[function] if (t->x >= cur.w || t->y >= cur.h) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 73)
[function] if (t->kind == HDMI_TR_BLIT) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 74)
[function] if (t->src_phys == 0 || t->stride == 0 || t->x >= cur.w || t->y >= cur.h)
{
kernel_Workspace/kernel/src/hdmi/operation.c (line 78)
[function] if (src == NULL) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 91)
[function] if (t->kind == HDMI_TR_FLIP && t->dst_phys != 0) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 98)
[function] if (t == NULL || !op_ready || pending >= HDMI_MAX_TRANSFERS) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 102)
[function] if (t->kind != HDMI_TR_FILL && t->kind != HDMI_TR_BLIT && t->kind !=
HDMI_TR_FLIP) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 107)
[function] if (!slot_used[i]) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 110)
[function] if (next_seq == 0) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 116)
[declaration] hdmi_op_exec(&slots[i]);
kernel_Workspace/kernel/src/hdmi/operation.c (line 127)
[function] if (out_seq == NULL) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 132)
[function] if (slot_used[i] && slot_done[i]) {

```

```

kernel_Workspace/kernel/src/hdmi/operation.c (line 159)
[declaration] return hdmi_submit(&t);
kernel_Workspace/kernel/src/hdmi/operation.c (line 164)
[function] if (m == NULL) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 178)
[function] if (cur.fb_phys == 0) {
kernel_Workspace/kernel/src/hdmi/operation.c (line 198)
[function] if (w == NULL || h == NULL || s == NULL || phys == NULL) {
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 8)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 32)
[declaration] return heap_usable_size(ptr);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 38)
[declaration] return dbg_alloc(size);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 40)
[declaration] return heap_alloc(size);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 44)
[function] void kfree(void *ptr) {
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 63)
[declaration] return kzalloc(count * size);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 70)
[declaration] return heap_alloc_aligned(size, align);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 74)
[function] size_t kmalloc_usable_size(void *ptr) { return k_usable_size(ptr); }
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 79)
[declaration] kfree(ptr);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 95)
[declaration] return heap_alloc(pages * ARCH_PAGE_SIZE);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 98)
[function] void kfree_pages(void *ptr, const size_t pages) {
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 103)
[declaration] kprintf("kfree_pages: nieprawidlowy parametr pages=%zu (pojemnosc
bloku=%zu B) dla ptr=%p – odmowa zwolnienia\n", pages, usable, ptr);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 118)
[function] uint64_t kvirt_to_phys(void *ptr) {
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 123)
[declaration] return arch_virt_to_phys(ptr);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.c (line 126)
[function] void kalloc_dump(void) {
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 11)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 56)
[function] void dbg_free(void *ptr) {
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 61)
[declaration] kprintf("dbg_free: bad magic at %p\n", ptr);
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 73)
[function] if (!leak_untrack(ptr, &leaked_size, &alloc_caller)) {
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 74)
[declaration] kprintf("dbg_free: double free or unknown ptr %p\n", ptr);

```

```

kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 90)
[function] size_t dbg_usable_size(void *ptr) {
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 97)
[function] bool dbg_verify(void *ptr) {
kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.c (line 104)
[function] void mm_debug_dump(void) {
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 4)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 22)
[function] bool leak_track_id(void *ptr, size_t size, uint64_t caller, uint64_t
*out_id) {
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 25)
[function] if (!leak_table[i].used) {
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 33)
[declaration] smp_lock_release(&leak_smp_lock);
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 41)
[function] bool leak_track(void *ptr, size_t size, uint64_t caller) { return
leak_track_id(ptr, size, caller, NULL); }
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 42)
[function] bool leak_untrack(void *ptr, size_t *out_size, uint64_t *out_caller) {
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 45)
[function] if (leak_table[i].used && leak_table[i].ptr == ptr) {
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 49)
[declaration] smp_lock_release(&leak_smp_lock);
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 56)
[function] bool leak_contains(void *ptr) {
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 59)
[function] if (leak_table[i].used && leak_table[i].ptr == ptr) {
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 60)
[declaration] smp_lock_release(&leak_smp_lock);
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 67)
[function] size_t leak_count(void) {
kernel_Workspace/kernel/src/mm/alloc/debug/leak.c (line 74)
[function] void leak_dump(void) {
kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 4)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 15)
[function] void alloc_stats_note_alloc(size_t size) {
kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 23)
[function] void alloc_stats_note_free(size_t size) {
kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 31)
[function] size_t alloc_stats_live_count(void) {
kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 37)
[function] size_t alloc_stats_live_bytes(void) {
kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 43)
[function] size_t alloc_stats_peak_bytes(void) {
kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 49)
[function] size_t alloc_stats_total_allocs(void) {

```

```

kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 55)
[function] size_t alloc_stats_total_frees(void) {
kernel_Workspace/kernel/src/mm/alloc/debug/stats.c (line 61)
[function] void alloc_stats_dump(void) {
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 6)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 49)
[function] if ((size_t)*p == idx) {
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 56)
[function] bool buddy_init(uint64_t base, size_t size, buddy_map_cb map_cb,
buddy_unmap_cb unmap_cb) {
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 95)
[function] bool buddy_ready(void) { return buddy_initialized; }
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 109)
[declaration] fl_push(cur, buddy_idx);
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 116)
[declaration] fl_push(o, idx);
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 133)
[declaration] return buddy_alloc(p);
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 136)
[function] void buddy_free(void *ptr) {
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 153)
[declaration] fl_remove(cur, buddy_idx);
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 163)
[function] size_t buddy_block_size(void *ptr) {
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 171)
[function] size_t buddy_stat_used_bytes(void) { return buddy_used_bytes; }
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 172)
[function] size_t buddy_stat_free_bytes(void) { return buddy_free_bytes; }
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 173)
[function] void buddy_dump(void) {
kernel_Workspace/kernel/src/mm/alloc/heap/buddy.c (line 175)
[declaration] kprintf("BUDDY: not initialized\n");
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 32)
[function] if (!pmm_alloc_frame(&phys)) {
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 37)
[declaration] mapping_unmap_range(MAPPING_KERNEL, v, ARCH_PAGE_SIZE);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 39)
[function] if (p != UINT64_MAX) {
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 40)
[declaration] pmm_free_frame(p);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 52)
[declaration] pmm_free_frame(phys);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 58)
[declaration] mapping_unmap_range(MAPPING_KERNEL, v, ARCH_PAGE_SIZE);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 60)
[function] if (p != UINT64_MAX) {
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 61)
[declaration] pmm_free_frame(p);

```

```

kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 80)
[function] if (p != UINT64_MAX) {
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 81)
[declaration] mapping_unmap_range(MAPPING_KERNEL, v, ARCH_PAGE_SIZE);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 82)
[declaration] pmm_free_frame(p);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 147)
[declaration] return slab_alloc(size);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 151)
[declaration] return buddy_alloc(size);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 167)
[declaration] return slab_alloc(size);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 171)
[declaration] return buddy_alloc_aligned(size, align);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 197)
[declaration] slab_free(ptr);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 203)
[declaration] buddy_free(ptr);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 217)
[declaration] return slab_usable_size(ptr);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 222)
[declaration] return buddy_block_size(ptr);
kernel_Workspace/kernel/src/mm/alloc/heap/heap.c (line 238)
[declaration] slab_dump();
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 9)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 92)
[function] if ((uint32_t)*p == page) {
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 116)
[declaration] bitmap_free(&slab_page_bitmap, page);
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 125)
[declaration] pmm_free_frame(phys);
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 126)
[declaration] bitmap_free(&slab_page_bitmap, page);
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 207)
[declaration] slab_bitmap_clear_all(&slab_descs[i]);
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 218)
[function] if (slab_caches[i].objects_per_slab > SLAB_MAX_OBJECTS_PER_SLAB) {
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 255)
[function] if (!slab_grow(cache_id)) {
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 256)
[declaration] slab_unlock();
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 328)
[declaration] slab_unlock();
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 346)
[declaration] partial_remove(c, page);
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 350)
[declaration] mapping_unmap_range(MAPPING_KERNEL, slab_va, ARCH_PAGE_SIZE);

```

```

kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 352)
[function] if (phys != UINT64_MAX) {
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 353)
[declaration] pmm_free_frame(phys);
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 356)
[declaration] bitmap_free(&slab_page_bitmap, page);
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 400)
[function] if (d->state != 1) {
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 424)
[function] if (d->state != 1) {
kernel_Workspace/kernel/src/mm/alloc/heap/slab.c (line 439)
[declaration] kprintf("SLAB: not initialized\n");
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 65)
[declaration] return bitmap_words_for_bits(bit_count) * sizeof(uint64_t);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 132)
[declaration] bitmap_mark_tail_used(bm);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 298)
[function] if (bm->bits[i] != 0) {
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 329)
[function] if (idx < bm->bit_count) {
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 339)
[function] if (~w != 0) {
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 343)
[function] if (idx < bm->bit_count) {
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 377)
[function] if (i == start_word && start_bit != 0) {
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 381)
[function] if (i == end_word) {
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 385)
[function] if (w != 0) {
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 389)
[function] if (idx >= start && idx < limit) {
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 474)
[declaration] return bitmap_alloc(bm);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 489)
[function] if (free_bit == BITMAP_INVALID) {
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 500)
[function] if (bitmap_test_range_free(bm, candidate, count)) {
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 501)
[declaration] bitmap_set_range(bm, candidate, count);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.c (line 509)
[function] if (used == BITMAP_INVALID) {
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 64)
[declaration] return arch_phys_to_virt(frame);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 159)
[function] if (r->type != ARCH_MEM_TYPE_USABLE) {
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 171)
[function] if (frame_count == 0) {

```

```

kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 175)
[declaration] bitmap_clear_range(&frame_bitmap, pfn_start, frame_count);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 177)
[function] if (allocated_frames >= frame_count) {
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 183)
[declaration] dma32_account_free(pfn_start, frame_count);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 230)
[declaration] frame_free(frame);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 345)
[declaration] __asm__ volatile("" ::: "memory");
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 377)
[function] if (aligned > limit - count) {
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 381)
[function] if (bitmap_test_range_free(&frame_bitmap, aligned, count)) {
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 382)
[declaration] bitmap_set_range(&frame_bitmap, aligned, count);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.c (line 385)
[declaration] dma32_account_alloc(aligned, count);
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 6)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 8)
[declaration] static void pmm_panic(const char *msg) __attribute__((noreturn));
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 31)
[declaration] pmm_panic("pmm_unlock without lock");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 80)
[function] if (!frame_zero(phys)) {
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 93)
[declaration] pmm_unlock();
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 98)
[declaration] pmm_unlock();
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 99)
[declaration] kprintf("pmm_init: arch memory not initialized\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 104)
[declaration] pmm_unlock();
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 105)
[declaration] kprintf("pmm_init: frame allocator initialization failed\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 220)
[declaration] return pmm_alloc_frames(frames, out_phys);
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 236)
[declaration] return pmm_alloc_zero_frame(out_phys);
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 251)
[declaration] pmm_free_frames(frame, frames);
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 275)
[declaration] return pmm_alloc_frames_aligned(frames, align_frames, out_phys);
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 328)
[declaration] return pmm_free_frames(phys, frames);
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 337)
[declaration] return frame_total();

```

```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 346)
[declaration] return frame_free_count();
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 355)
[declaration] return frame_allocated();
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 394)
[declaration] return frame_zone_dma32_total();
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 403)
[declaration] return frame_zone_dma32_free();
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 412)
[declaration] return frame_zone_normal_total();
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 421)
[declaration] return frame_zone_normal_free();
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 452)
[declaration] kprintf("PMM: not initialized\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 490)
[function] if (!pmm_alloc_zero_frame(&frames[i])) {
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 491)
[declaration] kprintf("pmm_self_test: single frame alloc failed\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 495)
[function] if (!arch_is_page_aligned(frames[i])) {
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 496)
[declaration] kprintf("pmm_self_test: frame not aligned\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 504)
[function] if (p[j] != 0) {
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 505)
[declaration] kprintf("pmm_self_test: zero frame not zero\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 512)
[function] if (!pmm_free_frame(frames[i])) {
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 513)
[declaration] kprintf("pmm_self_test: single frame free failed\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 522)
[declaration] kprintf("pmm_self_test: contiguous alloc failed\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 527)
[declaration] kprintf("pmm_self_test: contiguous not aligned\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 532)
[declaration] kprintf("pmm_self_test: contiguous free failed\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 540)
[declaration] kprintf("pmm_self_test: 2M aligned alloc failed\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 547)
[declaration] kprintf("pmm_self_test: 2M alignment failed\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 552)
[declaration] kprintf("pmm_self_test: 2M aligned free failed\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 560)
[declaration] kprintf("pmm_self_test: bytes alloc failed\n");
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 572)
[function] if (p[i] != 0) {
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 573)
[declaration] kprintf("pmm_self_test: bytes alloc not zero\n");

```

```

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.c (line 579)
[declaration] kprintf("pmm_self_test: bytes free failed\n");
kernel_Workspace/kernel/src/mm/alloc/special/contiguous.c (line 9)
[declaration] return size_bytes_to_pages_checked(bytes, ARCH_PAGE_SIZE, out_frames);
kernel_Workspace/kernel/src/mm/alloc/special/dma.c (line 10)
[declaration] return size_bytes_to_pages_checked(bytes, ARCH_PAGE_SIZE, out_frames);
kernel_Workspace/kernel/src/mm/alloc/special/dma.c (line 47)
[declaration] pmm_free_frames(phys, frames);
kernel_Workspace/kernel/src/mm/alloc/special/dma.c (line 76)
[declaration] vmm_unmap_device((uint64_t)(uintptr_t)virt, len);
kernel_Workspace/kernel/src/mm/alloc/special/dma.c (line 88)
[declaration] __asm__ volatile("clflush (%0)" :: "r"(a) : "memory");
kernel_Workspace/kernel/src/mm/alloc/special/dma.c (line 91)
[declaration] __asm__ volatile("" ::: "memory");
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.c (line 27)
[declaration] mapping_init();
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.c (line 70)
[function] if (!paging_map_page_in(pml4, v, p, pte_flags)) {
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.c (line 102)
[declaration] paging_unmap_page_in(pml4, virt + (uint64_t)i * ARCH_PAGE_SIZE);
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.c (line 128)
[function] if (!paging_is_mapped_in(pml4, v)) {
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.c (line 136)
[function] if (!paging_set_flags_in(pml4, v, pte_flags)) {
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.c (line 147)
[declaration] return paging_translate_in(resolve_space(space), virt);
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.c (line 152)
[declaration] return paging_is_mapped_in(resolve_space(space), virt);
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.c (line 164)
[declaration] __asm__ volatile("" ::: "memory");
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.c (line 199)
[declaration] copy_page_by_phys(dp, sp);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 5)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 78)
[declaration] kprintf("page_init: pmm not ready\n");
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 85)
[declaration] kprintf("page_init: no arch mem info\n");
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 141)
[declaration] page_unlock();
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 168)
[declaration] page_unlock();
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 205)
[declaration] page_unlock();
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 226)
[declaration] page_unlock();
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 246)
[declaration] page_unlock();

```

```

kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 253)
[declaration] page_unlock();
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 289)
[declaration] return arch_phys_to_virt(phys);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 331)
[declaration] return page_from_phys(arch_virt_to_phys(va));
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 379)
[function] if (page_descs[i].refcount != 0) {
kernel_Workspace/kernel/src/mm/alloc/virtual/page.c (line 390)
[declaration] kprintf("PAGE: not initialized\n");
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 7)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 201)
[declaration] vmm_unlock();
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 213)
[declaration] : pmm_alloc_frame(&phys);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 215)
[function] if (!got) {
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 222)
[function] if (!paging_map_page(virt, phys, pte)) {
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 223)
[declaration] pmm_free_frame(phys);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 235)
[function] if (paging_is_mapped(virt)) {
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 237)
[declaration] paging_unmap_page(virt);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 238)
[declaration] pmm_free_frame(phys);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 242)
[declaration] bitmap_free_range(&vmm_bitmap, start, pages);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 243)
[declaration] vmm_unlock();
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 258)
[declaration] return vmm_alloc_aligned(bytes, ARCH_PAGE_SIZE, flags, out_virt);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 284)
[declaration] vmm_unlock();
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 295)
[function] if (!paging_map_page(virt, frame_phys, pte)) {
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 305)
[declaration] paging_unmap_page(vmm_page_virt(start + i));
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 308)
[declaration] bitmap_free_range(&vmm_bitmap, start, pages);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 309)
[declaration] vmm_unlock();
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 349)
[function] if (paging_is_mapped(v)) {
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 351)
[declaration] paging_unmap_page(v);

```

```

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 352)
[declaration] pmm_free_frame(phys);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 395)
[function] if (paging_is_mapped(v)) {
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 396)
[declaration] paging_unmap_page(v);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 419)
[declaration] return paging_translate(virt);
kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.c (line 466)
[declaration] kprintf("VMM: not initialized\n");
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 115)
[declaration] arch_memory_boot_alloc(4096, 4096, &phys);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 139)
[declaration] tlb_flush_page_addr(virt);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 172)
[declaration] tlb_flush_page_addr(virt);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 184)
[declaration] tlb_flush_page_addr(virt);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 216)
[declaration] return (e1 & 0x0000FFFFFFFFE00000ULL) | (virt & 0x1FFFFFFULL);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 227)
[declaration] return (e2 & 0x0000FFFFFFFFFFFF0000ULL) | (virt & 0xFFFFFULL);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 273)
[declaration] __asm__ volatile("mrs %0, ttbr1_el1" : "=r"(cur));
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 297)
[function] if (idx == 0 && pa >= 0x200000 * 512) {
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 306)
[declaration] __asm__ volatile("msr mair_el1, %0" :: "r"(mair));
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 320)
[declaration] __asm__ volatile("msr tcr_el1, %0" :: "r"(tcr));
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 322)
[declaration] __asm__ volatile("msr ttbr0_el1, %0" :: "r"(kernel_l0));
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 323)
[declaration] __asm__ volatile("msr ttbr1_el1, %0" :: "r"(kernel_l0));
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 325)
[declaration] __asm__ volatile("isb");
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 326)
[declaration] __asm__ volatile("tlbi vmalle1");
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 327)
[declaration] __asm__ volatile("dsb sy");
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 328)
[declaration] __asm__ volatile("isb");
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 340)
[declaration] __asm__ volatile("mrs %0, ttbr1_el1" : "=r"(v));
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 347)
[declaration] __asm__ volatile("msr ttbr0_el1, %0" :: "r"(pml4_phys));
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 348)
[declaration] __asm__ volatile("msr ttbr1_el1, %0" :: "r"(pml4_phys));

```

```

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.c (line 349)
[declaration] __asm__ volatile("isb");
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 18)
[declaration] __asm__ volatile("dsb sy");
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 19)
[declaration] __asm__ volatile("tlbi vmalle1");
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 20)
[declaration] __asm__ volatile("dsb sy");
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 21)
[declaration] __asm__ volatile("isb");
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 31)
[declaration] __asm__ volatile("dsb ishst");
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 32)
[declaration] __asm__ volatile("tlbi vaae1is, %0" :: "r"(addr >> 12));
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 33)
[declaration] __asm__ volatile("dsb ish");
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 34)
[declaration] __asm__ volatile("isb");
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 45)
[declaration] tlb_flush_all();
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.c (line 50)
[declaration] tlb_flush_page_addr(addr + i * 4096);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 37)
[declaration] arch_mem_panic("bad multiboot2 magic");
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 41)
[declaration] arch_mem_panic("bad multiboot2 info address");
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 50)
[declaration] arch_mem_panic("multiboot2 info too small");
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 60)
[function] if (tag->size < 8) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 64)
[function] if (tag->type == MULTIBOOT_TAG_TYPE_END) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 68)
[function] if (tag->type == MULTIBOOT_TAG_TYPE_MMAP) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 75)
[function] if (mmap->entry_size < sizeof(struct multiboot_mmap_entry)) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 79)
[function] while (entry_addr + sizeof(struct multiboot_mmap_entry) <= mmap_end) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 83)
[function] if (n < ARCH_MAX_MEM_REGIONS) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 106)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 108)
[declaration] static void arch_mem_panic(const char *msg) __attribute__((noreturn));
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 221)
[function] while (j > 0 && entries[j - 1].base > key.base) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 244)
[function] if (r.base < last_end) {

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 247)
[function] if (r.len <= delta) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 264)
[declaration] arch_mem_panic("too many memory regions");
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 293)
[function] if (r_end <= base || r.base >= end) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 294)
[declaration] map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, r);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 299)
[function] if (r.base < base) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 305)
[declaration] map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, before);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 312)
[function] if (ov_end > ov_base) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 318)
[declaration] map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, reserved);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 322)
[function] if (r_end > end) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 328)
[declaration] map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, after);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 333)
[declaration] arch_mem_panic("memory reserve produced too many regions");
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 351)
[function] if (r.type != ARCH_MEM_TYPE_USABLE) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 352)
[declaration] map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, r);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 359)
[function] if (aligned_end <= aligned_base) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 365)
[declaration] map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, tiny);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 370)
[function] if (r.base < aligned_base) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 376)
[declaration] map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, before);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 386)
[declaration] map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, usable);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 390)
[function] if (aligned_end < r_end) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 396)
[declaration] map_add(tmp_regions, &tcnt, ARCH_TMP_REGIONS, after);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 401)
[declaration] arch_mem_panic("memory alignment produced too many regions");
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 421)
[function] if (end > max_address) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 425)
[function] if (r->type == ARCH_MEM_TYPE_USABLE) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 426)
[function] if (total_usable > ARCH_U64_MAX - r->len) {

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 432)
[function] if (end > max_usable_address) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 452)
[declaration] arch_mem_panic("arch_memory_init() called twice");
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 456)
[declaration] arch_mem_panic("empty memory map");
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 460)
[declaration] arch_mem_panic("memory map too large");
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 475)
[function] if (base >= end) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 479)
[function] if (mem.count > 0) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 483)
[function] if (base < last_end) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 487)
[function] if (base >= end) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 497)
[declaration] map_add(mem.regions, &mem.count, ARCH_MAX_MEM_REGIONS, r);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 501)
[declaration] arch_mem_panic("no valid memory regions after parsing");
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 571)
[function] if (r->type != ARCH_MEM_TYPE_USABLE) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 577)
[function] if (base >= r->base && end <= r_end) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 619)
[function] if (r->type != ARCH_MEM_TYPE_USABLE) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 626)
[function] if (candidate < r->base) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.c (line 630)
[function] if (candidate <= r_end && (r_end - candidate) >= len) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 5)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 7)
[declaration] static void paging_panic(const char *msg) __attribute__((noreturn));
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 10)
[declaration] kprintf("arch/x86_64/paging.c panic: %s\n", msg);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 24)
[function] if (!boot_phys_offset_valid) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 25)
[declaration] paging_panic("boot physical offset not set");
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 28)
[declaration] return (void *) (uintptr_t) (phys + boot_phys_offset);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 39)
[declaration] __asm__ volatile("" ::: "memory");
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 44)
[declaration] return (size_t) ((virt >> 39) & 0x1FF);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 49)
[declaration] return (size_t) ((virt >> 30) & 0x1FF);

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 54)
[declaration] return (size_t)((virt >> 21) & 0x1FF);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 59)
[declaration] return (size_t)((virt >> 12) & 0x1FF);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 66)
[declaration] __asm__ volatile("mov %%cr3, %0" : "=r"(cr3));
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 73)
[declaration] tlb_flush_all();
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 78)
[declaration] tlb_flush_page_addr(addr);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 88)
[declaration] paging_panic("cannot allocate page table page");
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 91)
[declaration] zero_page_phys(phys);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 102)
[function] if (entry & PTE_PRESENT) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 103)
[function] if (entry & PTE_PAGE_SIZE) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 121)
[function] if (i < pages) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 135)
[function] if (large_flags & PTE_USER) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 154)
[function] if (flags & PTE_USER) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 165)
[function] if (!boot_phys_offset_valid) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 197)
[declaration] paging_flush_page(virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 205)
[function] if (!boot_phys_offset_valid) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 233)
[function] if ((old & PTE_PRESENT) && !(old & PTE_PAGE_SIZE)) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 234)
[declaration] paging_panic("cannot map 2M page over existing 4K table");
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 248)
[function] if (!boot_phys_offset_valid) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 252)
[function] if (len == 0) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 262)
[function] if (len > UINT64_MAX - virt || len > UINT64_MAX - phys) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 269)
[function] while (virt < virt_end) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 277)
[function] if (can_use_2m) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 278)
[function] if (!map_page_2m(virt, phys, flags)) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 285)
[function] if (!paging_map_page(virt, phys, flags)) {

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 294)
[declaration] paging_flush_tlb_all();
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 301)
[function] if ((phys_offset & PAGING_PAGE_MASK) != 0) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 302)
[declaration] paging_panic("boot physical offset must be page aligned");
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 316)
[function] if (!arch_memory_ready()) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 317)
[declaration] paging_panic("arch memory not initialized");
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 320)
[function] if (!boot_phys_offset_valid) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 321)
[declaration] paging_panic("boot physical offset not set");
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 328)
[function] if (boot_phys_offset == ARCH_DIRECT_MAP_BASE) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 334)
[function] if (info == NULL) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 335)
[declaration] paging_panic("no arch memory info");
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 340)
[function] if (phys_limit == 0 || phys_limit == UINT64_MAX) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 341)
[declaration] paging_panic("invalid physical memory limit");
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 348)
[declaration] paging_panic("failed to create direct map");
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 351)
[declaration] paging_flush_tlb_all();
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 356)
[declaration] paging_set_boot_phys_offset(phys_offset);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 357)
[declaration] paging_init_direct_map();
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 367)
[declaration] * tables are allocated (with kernel RW flags); otherwise the walk stops
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 382)
[function] if (!(entry & PTE_PRESENT)) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 383)
[function] if (!create) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 386)
[declaration] ensure_table_entry(pml4, idx, PTE_PRESENT | PTE_WRITABLE);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 393)
[function] if (!(entry & PTE_PRESENT)) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 394)
[function] if (!create) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 397)
[declaration] ensure_table_entry(pdpt, idx, PTE_PRESENT | PTE_WRITABLE);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 400)
[function] if (entry & PTE_PAGE_SIZE) {

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 407)
[function] if (!(entry & PTE_PRESENT)) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 408)
[function] if (!create) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 411)
[declaration] ensure_table_entry(pd, idx, PTE_PRESENT | PTE_WRITABLE);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 414)
[function] if (entry & PTE_PAGE_SIZE) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 430)
[function] if (!boot_phys_offset_valid) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 433)
[function] if ((virt & PAGING_PAGE_MASK) != 0 || (phys & PAGING_PAGE_MASK) != 0) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 439)
[function] if (pte == NULL) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 444)
[declaration] paging_flush_page(virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 452)
[function] if (pte == NULL || !(*pte & PTE_PRESENT)) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 456)
[declaration] paging_flush_page(virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 464)
[function] if (pte == NULL || !(*pte & PTE_PRESENT)) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 467)
[declaration] return (*pte & PAGING_ADDR_MASK) | (virt & PAGING_PAGE_MASK);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 481)
[function] if (pte == NULL || !(*pte & PTE_PRESENT)) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 486)
[declaration] paging_flush_page(virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 494)
[function] if (pte == NULL || !(*pte & PTE_PRESENT)) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 497)
[function] if (out_flags != NULL) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 505)
[declaration] return paging_is_mapped_in(paging_read_cr3(), virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 510)
[declaration] return paging_translate_in(paging_read_cr3(), virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 515)
[declaration] return paging_unmap_page_in(paging_read_cr3(), virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 520)
[declaration] return paging_set_flags_in(paging_read_cr3(), virt, flags);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 525)
[declaration] return paging_get_flags_in(paging_read_cr3(), virt, out_flags);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 535)
[function] if (nx_enabled) {
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 538)
[declaration] __asm__ volatile("rdmsr" : "=A"(efer) : "c"(0xC0000080));
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 540)
[declaration] __asm__ volatile("wrmsr" : : "A"(efer), "c"(0xC0000080));

```

```

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.c (line 551)
[declaration] __asm__ volatile("mov %0, %%cr3" : : "r"(pml4_phys) : "memory");
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 77)
[declaration] tlb_asm_invpcid(INVPCID_ALL, &desc);
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 87)
[declaration] tlb_flush_all();
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 94)
[declaration] tlb_asm_write_cr4(cr4 & ~CR4_PGE);
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 95)
[declaration] tlb_asm_write_cr4(cr4);
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 97)
[declaration] tlb_flush_all();
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 109)
[declaration] tlb_flush_all();
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 123)
[declaration] tlb_flush_all();
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 130)
[declaration] tlb_flush_all();
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 135)
[declaration] tlb_asm_invlpg(a);
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 143)
[declaration] tlb_flush_all();
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 154)
[declaration] tlb_asm_invpcid(INVPCID_SINGLE_PCID, &desc);
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 159)
[declaration] tlb_asm_write_cr3(tlb_asm_read_cr3());
kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.c (line 167)
[declaration] tlb_asm_invpcid(INVPCID_ADDR_IN_PCID, &desc);
kernel_Workspace/kernel/src/mm/cache/cache.c (line 3)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/cache/cache.c (line 10)
[function] if (cache_initialized) {
kernel_Workspace/kernel/src/mm/cache/cache.c (line 14)
[declaration] per_cpu_init();
kernel_Workspace/kernel/src/mm/cache/cache.c (line 29)
[function] if (c == NULL) {
kernel_Workspace/kernel/src/mm/cache/cache.c (line 41)
[function] while (c != NULL) {
kernel_Workspace/kernel/src/mm/cache/cache.c (line 45)
[function] while (*a != '\0' && *a == *b) {
kernel_Workspace/kernel/src/mm/cache/cache.c (line 50)
[function] if (*a == '\0' && *b == '\0') {
kernel_Workspace/kernel/src/mm/cache/cache.c (line 62)
[function] if (!cache_initialized) {
kernel_Workspace/kernel/src/mm/cache/cache.c (line 63)
[declaration] kprintf("CACHE: not initialized\n");
kernel_Workspace/kernel/src/mm/cache/cache.c (line 67)
[declaration] kprintf("CACHE:\n");

```

```

kernel_Workspace/kernel/src/mm/cache/cache.c (line 71)
[function] while (c != NULL) {
kernel_Workspace/kernel/src/mm/cache/object_cache.c (line 82)
[declaration] buddy_free(va);
kernel_Workspace/kernel/src/mm/cache/object_cache.c (line 96)
[function] if (c->ctor != NULL) {
kernel_Workspace/kernel/src/mm/cache/object_cache.c (line 180)
[declaration] buddy_free((void *)(uintptr_t)ch->va);
kernel_Workspace/kernel/src/mm/cache/object_cache.c (line 181)
[declaration] heap_free(ch);
kernel_Workspace/kernel/src/mm/cache/object_cache.c (line 208)
[declaration] cache_unlock();
kernel_Workspace/kernel/src/mm/cache/object_cache.c (line 213)
[function] if (!kcache_grow(c)) {
kernel_Workspace/kernel/src/mm/cache/object_cache.c (line 214)
[declaration] cache_unlock();
kernel_Workspace/kernel/src/mm/cache/object_cache.c (line 253)
[declaration] cache_unlock();
kernel_Workspace/kernel/src/mm/core/mm.c (line 15)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/core/mm.c (line 29)
[declaration] kprintf("mm_init: bad params\n");
kernel_Workspace/kernel/src/mm/core/mm.c (line 46)
[declaration] kprintf("mm_init: pmm_init FAILED\n");
kernel_Workspace/kernel/src/mm/core/mm.c (line 52)
[declaration] kprintf("mm_init: vmm_init FAILED\n");
kernel_Workspace/kernel/src/mm/core/mm.c (line 58)
[declaration] kprintf("mm_init: page_init FAILED\n");
kernel_Workspace/kernel/src/mm/core/mm.c (line 66)
[declaration] kprintf("mm_init: heap_init FAILED\n");
kernel_Workspace/kernel/src/mm/core/mm.c (line 72)
[declaration] kprintf("mm_init: cache_init FAILED\n");
kernel_Workspace/kernel/src/mm/core/mm.c (line 78)
[declaration] kprintf("mm_init: paging_subsystem_init FAILED\n");
kernel_Workspace/kernel/src/mm/core/mm.c (line 86)
[declaration] kprintf("mm_init: aspace_subsystem_init FAILED\n");
kernel_Workspace/kernel/src/mm/core/mm.c (line 104)
[declaration] return pmm_stat_total_bytes();
kernel_Workspace/kernel/src/mm/core/mm.c (line 109)
[declaration] return pmm_stat_free_bytes();
kernel_Workspace/kernel/src/mm/core/range.c (line 53)
[function] if (raw_end > UINT64_MAX - pad) {
kernel_Workspace/kernel/src/mm/core/range.c (line 65)
[function] if (!range_is_canonical(start) || !range_is_canonical(end - 1)) {
kernel_Workspace/kernel/src/mm/core/region.c (line 64)
[function] if (region_valid(before)) {
kernel_Workspace/kernel/src/mm/core/region.c (line 75)
[function] if (region_valid(after)) {

```

```

kernel_Workspace/kernel/src/mm/core/smp_lock.c (line 5)
[declaration] __asm__ volatile("pause" ::: "memory");
kernel_Workspace/kernel/src/mm/core/smp_lock.c (line 64)
[function] __atomic_load_n(&lock->owner_cpu, __ATOMIC_ACQUIRE) == me) {
kernel_Workspace/kernel/src/mm/core/smp_lock.c (line 73)
[declaration] cpu_pause();
kernel_Workspace/kernel/src/mm/kprintf.c (line 18)
[declaration] __asm__ volatile("inb %%dx, %0" : "=a"(v) : "d"(port));
kernel_Workspace/kernel/src/mm/kprintf.c (line 25)
[declaration] __asm__ volatile("outb %0, %%dx" : : "a"(val), "d"(port));
kernel_Workspace/kernel/src/mm/kprintf.c (line 58)
[declaration] serial_putc(*s++);
kernel_Workspace/kernel/src/mm/kprintf.c (line 68)
[declaration] serial_putc('0');
kernel_Workspace/kernel/src/mm/kprintf.c (line 80)
[declaration] serial_putc(buf[i++]);
kernel_Workspace/kernel/src/mm/kprintf.c (line 93)
[function] if (*p != '%') {
kernel_Workspace/kernel/src/mm/kprintf.c (line 94)
[declaration] serial_putc(*p);
kernel_Workspace/kernel/src/mm/kprintf.c (line 99)
[function] if (*p == '\\0') {
kernel_Workspace/kernel/src/mm/kprintf.c (line 107)
[function] while (*p == 'l' || *p == 'z' || *p == 'h') {
kernel_Workspace/kernel/src/mm/kprintf.c (line 108)
[function] if (*p == 'l') {
kernel_Workspace/kernel/src/mm/kprintf.c (line 109)
[function] if (p[1] == 'l') {
kernel_Workspace/kernel/src/mm/kprintf.c (line 121)
[function] switch (*p) {
kernel_Workspace/kernel/src/mm/kprintf.c (line 125)
[declaration] serial_puts(s ? s : "(null)");
kernel_Workspace/kernel/src/mm/kprintf.c (line 129)
[declaration] serial_putc((char)va_arg(ap, int));
kernel_Workspace/kernel/src/mm/kprintf.c (line 137)
[function] if (v < 0) {
kernel_Workspace/kernel/src/mm/kprintf.c (line 138)
[declaration] serial_putc('-');
kernel_Workspace/kernel/src/mm/kprintf.c (line 139)
[declaration] serial_put_uint((uint64_t)(-v), 10, false);
kernel_Workspace/kernel/src/mm/kprintf.c (line 141)
[declaration] serial_put_uint((uint64_t)v, 10, false);
kernel_Workspace/kernel/src/mm/kprintf.c (line 151)
[declaration] serial_put_uint(v, 10, false);
kernel_Workspace/kernel/src/mm/kprintf.c (line 161)
[declaration] serial_put_uint(v, 16, *p == 'X');
kernel_Workspace/kernel/src/mm/kprintf.c (line 167)
[declaration] serial_puts("0x");

```

```

kernel_Workspace/kernel/src/mm/kprintf.c (line 168)
[declaration] serial_put_uint((uint64_t)(uintptr_t)ptr, 16, false);
kernel_Workspace/kernel/src/mm/kprintf.c (line 172)
[declaration] serial_putc('%');
kernel_Workspace/kernel/src/mm/kprintf.c (line 175)
[declaration] serial_putc('%');
kernel_Workspace/kernel/src/mm/kprintf.c (line 176)
[declaration] serial_putc(*p);
kernel_Workspace/kernel/src/mm/paging/paging.c (line 68)
[function] if (pml_entry_present(e) && !pml_entry_large(e)) {
kernel_Workspace/kernel/src/mm/paging/paging.c (line 69)
[declaration] free_tables(pml_entry_addr(e), level - 1);
kernel_Workspace/kernel/src/mm/paging/paging.c (line 99)
[declaration] pmm_free_frame(pml4_phys);
kernel_Workspace/kernel/src/mm/paging/paging.c (line 120)
[function] if (pml_entry_present(e) && !pml_entry_large(e)) {
kernel_Workspace/kernel/src/mm/paging/paging.c (line 121)
[declaration] free_tables(pml_entry_addr(e), PML_LEVEL_PDPT);
kernel_Workspace/kernel/src/mm/paging/paging.c (line 173)
[function] if (!paging_map_page_in(pml4, v, p, pte)) {
kernel_Workspace/kernel/src/mm/paging/paging.c (line 175)
[declaration] paging_unmap_page_in(pml4, virt + (uint64_t)j * ARCH_PAGE_SIZE);
kernel_Workspace/kernel/src/mm/paging/paging.c (line 207)
[function] if (paging_unmap_page_in(pml4, v)) {
kernel_Workspace/kernel/src/mm/paging/paging.c (line 208)
[declaration] tlb_batch_add(&batch, v);
kernel_Workspace/kernel/src/mm/paging/paging.c (line 238)
[function] if (!paging_is_mapped_in(pml4, v)) {
kernel_Workspace/kernel/src/mm/paging/paging.c (line 249)
[function] if (!paging_set_flags_in(pml4, v, pte)) {
kernel_Workspace/kernel/src/mm/paging/paging.c (line 250)
[declaration] tlb_batch_commit(&batch);
kernel_Workspace/kernel/src/mm/paging/paging.c (line 254)
[declaration] tlb_batch_add(&batch, v);
kernel_Workspace/kernel/src/mm/paging/paging.c (line 265)
[declaration] return paging_translate_in(paging_kernel_pml4, virt);
kernel_Workspace/kernel/src/mm/paging/paging.c (line 268)
[declaration] return paging_translate_in(as->pml4_phys, virt);
kernel_Workspace/kernel/src/mm/paging/paging.c (line 277)
[function] return paging_aspace_map((address_space_t *)&(address_space_t){
kernel_Workspace/kernel/src/mm/paging/paging.c (line 290)
[function] return paging_aspace_unmap((address_space_t *)&(address_space_t){
kernel_Workspace/kernel/src/mm/paging/tlb.c (line 33)
[declaration] tlb_flush_all();
kernel_Workspace/kernel/src/mm/paging/tlb.c (line 40)
[declaration] tlb_flush_page_addr(batch->pages[i]);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 63)
[function] if (!pmm_alloc_zero_frame(&phys)) {

```

```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 69)
[declaration] pmm_free_frame(phys);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 82)
[declaration] paging_aspace_unmap(pa->as, v, ARCH_PAGE_SIZE);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 84)
[function] if (phys != UINT64_MAX) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 85)
[declaration] pmm_free_frame(phys);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 97)
[declaration] paging_aspace_unmap(pa->as, v, ARCH_PAGE_SIZE);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 99)
[function] if (phys != UINT64_MAX) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 100)
[declaration] pmm_free_frame(phys);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 122)
[function] if (*p == v) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 136)
[function] if (addr >= v->start && addr < v->end) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 163)
[function] if (v->end <= cand) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 168)
[function] if (cand <= USER_MMAP_MAX - len && cand + len <= v->start) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 174)
[function] if (cand > USER_MMAP_MAX - len) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 199)
[declaration] heap_free(pa);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 212)
[declaration] paging_aspace_destroy(pa->as);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 213)
[declaration] heap_free(pa);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 226)
[declaration] heap_free(sv);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 227)
[declaration] paging_aspace_destroy(pa->as);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 228)
[declaration] heap_free(pa);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 246)
[declaration] unmap_pages_free(pa, v->start, v->end);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 249)
[declaration] heap_free(v);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 281)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 288)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 300)
[declaration] vma_remove(pa, v);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 301)
[declaration] heap_free(v);

```

```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 302)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 330)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 342)
[declaration] vma_remove(pa, v);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 343)
[declaration] heap_free(v);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 344)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 371)
[function] while (v != NULL) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 372)
[function] if (v->start < b && v->end > a) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 380)
[function] if (hit == NULL) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 390)
[function] if (needs_split) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 393)
[function] if (tail == NULL) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 394)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 399)
[declaration] unmap_pages_free(pa, s, e);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 401)
[function] if (needs_split) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 415)
[declaration] vma_remove(pa, hit);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 416)
[declaration] heap_free(hit);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 445)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 450)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 455)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 468)
[declaration] return aspace_protect_checked(pa, addr, len, prot, prot, NULL);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 496)
[function] if (v->start < b && v->end > a) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 497)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 507)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 533)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 538)
[declaration] as_unlock();

```

```

kernel_Workspace/kernel/src/mm/process/address_space.c (line 545)
[function] if (!map_anon_pages(pa, pa->brk, nb, PROT_READ | PROT_WRITE)) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 546)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 552)
[function] if (v == NULL) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 555)
[function] if (v == NULL) {
kernel_Workspace/kernel/src/mm/process/address_space.c (line 556)
[declaration] unmap_pages_free(pa, pa->brk, nb);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 557)
[declaration] as_unlock();
kernel_Workspace/kernel/src/mm/process/address_space.c (line 566)
[declaration] vma_insert(pa, v);
kernel_Workspace/kernel/src/mm/process/address_space.c (line 573)
[declaration] aspace_unmap(pa, nb, pa->brk - nb);
kernel_Workspace/kernel/src/mm/process/mmap.c (line 17)
[declaration] return aspace_map_at(pa, addr, len, prot);
kernel_Workspace/kernel/src/mm/process/mmap.c (line 20)
[declaration] return aspace_map_anon(pa, addr, len, prot);
kernel_Workspace/kernel/src/mm/process/mmap.c (line 25)
[declaration] return aspace_unmap(pa, addr, len);
kernel_Workspace/kernel/src/mm/protection/guard.c (line 6)
[declaration] return aspace_reserve_at(pa, addr, len, VMA_FLAG_GUARD);
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 5)
[declaration] extern void kprintf(const char *fmt, ...);
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 24)
[declaration] __asm__ volatile("mov %%cr4, %0" : "=r"(v));
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 31)
[declaration] __asm__ volatile("mov %0, %%cr4" :: "r"(v) : "memory");
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 81)
[function] if (!pml_entry_present(e)) {
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 85)
[function] if (e & PTE_USER) {
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 89)
[function] if (!pml_entry_large(e) && level > PML_LEVEL_PT) {
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 91)
[declaration] audit_tables(pml_entry_addr(e), level - 1, violations);
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 108)
[function] if (!pml_entry_present(e)) {
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 112)
[function] if (e & PTE_USER) {
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 116)
[function] if (!pml_entry_large(e)) {
kernel_Workspace/kernel/src/mm/protection/isolation.c (line 118)
[declaration] audit_tables(pml_entry_addr(e), PML_LEVEL_PDPT, violations);
kstd/src/kernel/audio.c (line 3)
[declaration] extern int32_t k_audio_play(uint64_t phys, uint32_t len);

```

```

kstd/src/kernel/audio.c (line 4)
[declaration] extern int32_t k_audio_stop(void);
kstd/src/kernel/audio.c (line 5)
[declaration] extern int32_t k_audio_jack(void);
kstd/src/kernel/audio.c (line 6)
[declaration] extern int32_t k_audio_amp(int32_t on);
kstd/src/kernel/audio.c (line 7)
[declaration] extern uint64_t kvirt_to_phys(void *ptr);
kstd/src/kernel/audio.c (line 13)
[function] if (phys == (uint64_t)-1) {
kstd/src/kernel/audio.c (line 22)
[declaration] k_audio_stop();
kstd/src/kernel/audio.c (line 28)
[function] if (!present) {
kstd/src/kernel/audio.c (line 38)
[declaration] k_audio_amp(on ? 1 : 0);
kstd/src/kernel/bt.c (line 3)
[declaration] extern bool bt_ready(void);
kstd/src/kernel/bt.c (line 4)
[declaration] extern void bt_info(uint8_t *ver, uint8_t *addr);
kstd/src/kernel/bt.c (line 5)
[declaration] extern bool bt_hci_cmd(uint16_t op, const uint8_t *p, uint8_t len);
kstd/src/kernel/bt.c (line 6)
[declaration] extern bool bt_event_poll(uint8_t *buf, uint8_t *len);
kstd/src/kernel/bt.c (line 7)
[declaration] extern bool bt_acl_send(const uint8_t *d, uint16_t len);
kstd/src/kernel/bt.c (line 8)
[declaration] extern bool bt_acl_recv(uint8_t *d, uint16_t *len);
kstd/src/kernel/bt.c (line 12)
[function] if (!bt_ready()) {
kstd/src/kernel/bt.c (line 16)
[declaration] bt_info(hci_ver, addr6);
kstd/src/kernel/fs.c (line 3)
[declaration] extern int32_t k_fs_read(const char *path, void *buf, uint32_t cap);
kstd/src/kernel/fs.c (line 4)
[declaration] extern int32_t k_fs_exists(const char *path);
kstd/src/kernel/fs.c (line 10)
[function] if (n < 0) {
kstd/src/kernel/fs.c (line 14)
[function] if (out) {
kstd/src/kernel/input.c (line 3)
[declaration] extern int32_t k_input_key(void);
kstd/src/kernel/input.c (line 7)
[declaration] return k_input_key();
kstd/src/kernel/log.c (line 3)
[declaration] extern void kprintf(const char *fmt, ...);
kstd/src/kernel/log.c (line 7)
[declaration] kprintf("%s\n", s);

```

```

kstd/src/kernel/mem.c (line 4)
[declaration] extern void kfree(void *ptr);
kstd/src/kernel/mem.c (line 5)
[declaration] extern bool vmm_map_device(uint64_t phys, unsigned long len, uint64_t
*virt);
kstd/src/kernel/mem.c (line 6)
[declaration] extern uint64_t kvirt_to_phys(void *ptr);
kstd/src/kernel/mem.c (line 10)
[declaration] return kcalloc(bytes);
kstd/src/kernel/mem.c (line 16)
[declaration] kfree(ptr);
kstd/src/kernel/mem.c (line 23)
[function] if (vmm_map_device(phys, len, &v)) {
kstd/src/kernel/video.c (line 8)
[function] tr_status_t tr_video_init(void) {
kstd/src/kernel/video.c (line 9)
[function] if (dp_ready()) {
kstd/src/kernel/video.c (line 13)
[function] if (hdmi_ready()) {
kstd/src/kernel/video.c (line 20)
[function] bool tr_video_is_ready(void) {
kstd/src/kernel/video.c (line 24)
[function] tr_status_t tr_video_get_caps(tr_video_caps_t *caps) {
kstd/src/kernel/video.c (line 27)
[function] if (use_dp) {
kstd/src/kernel/video.c (line 28)
[declaration] dp_caps(&caps->fb_phys, &caps->width, &caps->height, &caps->stride);
kstd/src/kernel/video.c (line 31)
[declaration] hdmi_caps_raw(&caps->width, &caps->height, &caps->stride,
&caps->fb_phys);
kstd/src/kernel/video.c (line 37)
[function] tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y,
uint32_t w, uint32_t h) {
kstd/src/kernel/video.c (line 41)
[function] if (use_dp) {
kstd/src/user/audio.c (line 7)
[declaration] extern void ds_poll(void);
kstd/src/user/audio.c (line 8)
[declaration] extern int ds_take(uint64_t id, ds_msg_t *out);
kstd/src/user/audio.c (line 15)
[function] if (bounce == TR_NULL) {
kstd/src/user/audio.c (line 18)
[function] if (bounce == TR_NULL) {
kstd/src/user/audio.c (line 23)
[declaration] ds_poll();
kstd/src/user/audio.c (line 27)
[function] if (!ds_take(id, &r) || r.status != 0) {
kstd/src/user/audio.c (line 34)
[function] if (len > 4096 * 4) {

```

```

kstd/src/user/audio.c (line 46)
[declaration] ds_poll();
kstd/src/user/audio.c (line 50)
[function] if (ds_take(id, &r) && r.status == 0) {
kstd/src/user/audio.c (line 59)
[declaration] ds_call(SVC_AUDIO, AUD_STOP, 0, 0, 0);
kstd/src/user/audio.c (line 65)
[function] if (!present) {
kstd/src/user/audio.c (line 70)
[declaration] ds_poll();
kstd/src/user/audio.c (line 74)
[function] if (ds_take(id, &r) && r.status == 0) {
kstd/src/user/audio.c (line 84)
[declaration] ds_call(SVC_AUDIO, AUD_AMP, on ? 1 : 0, 0, 0);
kstd/src/user/bt.c (line 6)
[declaration] extern void ds_poll(void);
kstd/src/user/bt.c (line 7)
[declaration] extern int ds_take(uint64_t id, ds_msg_t *out);
kstd/src/user/bt.c (line 14)
[declaration] ds_poll();
kstd/src/user/bt.c (line 18)
[function] if (!ds_take(id, &r) || r.status != 0) {
kstd/src/user/bt.c (line 40)
[declaration] ds_poll();
kstd/src/user/bt.c (line 44)
[function] if (ds_take(id, &r) && r.status == 0) {
kstd/src/user/bt.c (line 54)
[declaration] ds_poll();
kstd/src/user/bt.c (line 58)
[function] if (!ds_take(id, &r) || r.status != 0) {
kstd/src/user/bt.c (line 83)
[declaration] ds_poll();
kstd/src/user/bt.c (line 87)
[function] if (ds_take(id, &r) && r.status == 0) {
kstd/src/user/bt.c (line 97)
[declaration] ds_poll();
kstd/src/user/bt.c (line 101)
[function] if (!ds_take(id, &r) || r.status != 0) {
kstd/src/user/fs.c (line 6)
[declaration] extern void ds_poll(void);
kstd/src/user/fs.c (line 7)
[declaration] extern int ds_take(uint64_t id, ds_msg_t *out);
kstd/src/user/fs.c (line 15)
[function] while (path[n] && n < 255) {
kstd/src/user/fs.c (line 23)
[declaration] ds_poll();
kstd/src/user/fs.c (line 27)
[function] if (!ds_take(id, &r) || r.status != 0) {

```

```

kstd/src/user/fs.c (line 33)
[function] if (got > cap) {
kstd/src/user/fs.c (line 43)
[function] if (out) {
kstd/src/user/fs.c (line 54)
[function] while (path[n] && n < 255) {
kstd/src/user/fs.c (line 62)
[declaration] ds_poll();
kstd/src/user/fs.c (line 66)
[function] if (ds_take(id, &r) && r.status == 0 && r.arg0 == 1) {
kstd/src/user/input.c (line 6)
[declaration] extern void ds_poll(void);
kstd/src/user/input.c (line 7)
[declaration] extern int ds_take(uint64_t id, ds_msg_t *out);
kstd/src/user/input.c (line 12)
[declaration] ds_poll();
kstd/src/user/input.c (line 16)
[function] if (ds_take(id, &r) && r.status == 0 && r.arg0 != 0) {
kstd/src/user/log.c (line 10)
[function] while (s[n] && n < DS_SCRATCH_SIZE - 1) {
kstd/src/user/log.c (line 15)
[declaration] ds_call(SVC_SYS, OP_LOG, n, 0, 0);
kstd/src/user/mem.c (line 6)
[declaration] extern void ds_poll(void);
kstd/src/user/mem.c (line 7)
[declaration] extern int ds_take(uint64_t id, ds_msg_t *out);
kstd/src/user/mem.c (line 16)
[declaration] ds_poll();
kstd/src/user/mem.c (line 20)
[function] if (ds_take(id, &r) && r.status == 0) {
kstd/src/user/mem.c (line 30)
[declaration] ds_call(SVC_SYS, OP_FREE, (uint64_t)ptr, 0, 0);
kstd/src/user/mem.c (line 39)
[declaration] ds_poll();
kstd/src/user/mem.c (line 43)
[function] if (ds_take(id, &r) && r.status == 0) {
kstd/src/user/video.c (line 6)
[function] tr_status_t tr_video_init(void) {
kstd/src/user/video.c (line 11)
[function] bool tr_video_is_ready(void) {
kstd/src/user/video.c (line 13)
[declaration] ds_poll();
kstd/src/user/video.c (line 15)
[function] if (ds_take(id, &r) && r.status == 0) {
kstd/src/user/video.c (line 21)
[function] tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y,
uint32_t w, uint32_t h) {
kstd/src/user/video.c (line 26)
[declaration] ds_poll();

```

```

kstd/src/user/video.c (line 29)
[function] if (ds_take(id, &r)) {
libs/dsclient.c (line 30)
[function] if (next == d2k->tail) {
libs/dsclient.c (line 51)
[function] while (k2d->tail != k2d->head) {
libs/dsclient.c (line 58)
[function] if (!cache_used[i]) {
libs/dsclient.c (line 70)
[function] if (cache_used[i] && cache[i].id == id) {
triang-lang/out/x86_64/prog.c (line 15)
[declaration] memset(buffer, (int)0ull, 16);

```

C (header)

```

drivers/vgpu/vgpu.h (line 22)
[declaration] bool vgpu_init(uint32_t w, uint32_t h);
drivers/vgpu/vgpu.h (line 23)
[declaration] vgpu_info_t vgpu_get_info(void);
drivers/vgpu/vgpu.h (line 25)
[declaration] void vgpu_clear(uint32_t color);
drivers/vgpu/vgpu.h (line 26)
[declaration] void vgpu_pixel(uint32_t x, uint32_t y, uint32_t color);
drivers/vgpu/vgpu.h (line 28)
[declaration] int32_t vgpu_surface_create(uint32_t w, uint32_t h, uint64_t
*phys_out);
drivers/vgpu/vgpu.h (line 29)
[declaration] bool vgpu_surface_present(int32_t id, uint32_t x, uint32_t y);
drivers/vgpu/vgpu.h (line 31)
[declaration] void vgpu_process_ring(volatile void *ring);
kernel_Workspace/kernel/src/battery/battery.h (line 21)
[declaration] bool battery_init(void);
kernel_Workspace/kernel/src/battery/battery.h (line 22)
[declaration] bool battery_present(void);
kernel_Workspace/kernel/src/battery/battery.h (line 23)
[declaration] bool battery_status(battery_status_t *out);
kernel_Workspace/kernel/src/battery/battery.h (line 24)
[declaration] bool battery_status_packed(uint64_t *a0, uint64_t *a1, uint64_t *a2);
kernel_Workspace/kernel/src/battery/battery.h (line 25)
[declaration] bool battery_set_threshold(uint32_t low_pct);
kernel_Workspace/kernel/src/battery/battery.h (line 26)
[declaration] uint32_t battery_threshold(void);
kernel_Workspace/kernel/src/battery/battery.h (line 27)
[declaration] bool battery_low_pending(void);
kernel_Workspace/kernel/src/battery/battery.h (line 28)
[declaration] void battery_sim_tick(void);
kernel_Workspace/kernel/src/battery/operation.h (line 7)
[declaration] bool (*present)(void);
kernel_Workspace/kernel/src/battery/operation.h (line 8)
[declaration] bool (*status)(battery_status_t *out);

```

kernel_Workspace/kernel/src/battery/operation.h (line 11)
[declaration] void battery_register_backend(const battery_backend_t *b);

kernel_Workspace/kernel/src/bluetooth/bt.h (line 8)
[declaration] bool bt_init(void);

kernel_Workspace/kernel/src/bluetooth/bt.h (line 9)
[declaration] bool bt_ready(void);

kernel_Workspace/kernel/src/bluetooth/bt.h (line 10)
[declaration] void bt_info(uint8_t *hci_ver, uint8_t *bdaddr);

kernel_Workspace/kernel/src/bluetooth/bt.h (line 11)
[declaration] bool bt_hci_cmd(uint16_t opcode, const uint8_t *params, uint8_t len);

kernel_Workspace/kernel/src/bluetooth/bt.h (line 12)
[declaration] bool bt_event_poll(uint8_t *buf, uint8_t cap, uint8_t *len);

kernel_Workspace/kernel/src/bluetooth/bt.h (line 13)
[declaration] bool bt_acl_send(const uint8_t *data, uint16_t len);

kernel_Workspace/kernel/src/bluetooth/bt.h (line 14)
[declaration] bool bt_acl_recv(uint8_t *data, uint16_t cap, uint16_t *len);

kernel_Workspace/kernel/src/bluetooth/operation.h (line 11)
[declaration] void bt_op_model_init(void);

kernel_Workspace/kernel/src/bluetooth/operation.h (line 12)
[declaration] bool bt_op_model_run_cmd(uint16_t opcode, const uint8_t *params, uint8_t len);

kernel_Workspace/kernel/src/camera/camera.h (line 19)
[declaration] bool camera_init(void);

kernel_Workspace/kernel/src/camera/camera.h (line 20)
[declaration] bool camera_present(void);

kernel_Workspace/kernel/src/camera/camera.h (line 21)
[declaration] bool camera_caps_get(camera_caps_t *out);

kernel_Workspace/kernel/src/camera/camera.h (line 22)
[declaration] bool camera_start(void);

kernel_Workspace/kernel/src/camera/camera.h (line 23)
[declaration] bool camera_stop(void);

kernel_Workspace/kernel/src/camera/camera.h (line 24)
[declaration] bool camera_frame(void *buf, uint32_t cap, uint64_t *frame_id);

kernel_Workspace/kernel/src/camera/camera.h (line 25)
[declaration] bool camera_frame_to_phys(uint64_t phys, uint32_t cap, uint64_t *frame_id);

kernel_Workspace/kernel/src/camera/operation.h (line 7)
[declaration] bool (*present)(void);

kernel_Workspace/kernel/src/camera/operation.h (line 8)
[declaration] bool (*start)(void);

kernel_Workspace/kernel/src/camera/operation.h (line 9)
[declaration] bool (*stop)(void);

kernel_Workspace/kernel/src/camera/operation.h (line 10)
[declaration] bool (*frame)(void *buf, uint32_t cap, uint64_t *fid);

kernel_Workspace/kernel/src/camera/operation.h (line 13)
[declaration] void camera_register_backend(const camera_backend_t *b);

kernel_Workspace/kernel/src/caps/caps.h (line 10)
[declaration] unsigned int caps_self_bits(void);

kernel_Workspace/kernel/src/caps/caps.h (line 12)

[declaration] unsigned int caps_world_bits(unsigned int world_id);

kernel_Workspace/kernel/src/caps/caps.h (line 14)

[declaration] int caps_has(unsigned int world_id, unsigned char cap_id);

kernel_Workspace/kernel/src/caps/caps.h (line 16)

[declaration] int caps_name(unsigned char cap_id, unsigned char *buf, unsigned int len);

kernel_Workspace/kernel/src/caps/caps.h (line 18)

[declaration] int caps_request(unsigned int target, unsigned char cap_id);

kernel_Workspace/kernel/src/caps/caps.h (line 20)

[declaration] int caps_release(unsigned int world_id, unsigned char cap_id);

kernel_Workspace/kernel/src/caps/caps.h (line 22)

[declaration] unsigned int caps_world_count(void);

kernel_Workspace/kernel/src/caps/caps.h (line 24)

[declaration] unsigned long long caps_audit_count(void);

kernel_Workspace/kernel/src/core-lang/ast.h (line 50)

[declaration] void cl_arena_init(arena_t *a, uint8_t *buf, size_t cap);

kernel_Workspace/kernel/src/core-lang/bridge.h (line 6)

[declaration] int cl_bridge_init(cl_vm_t *vm, uint8_t ring);

kernel_Workspace/kernel/src/core-lang/bridge.h (line 8)

[declaration] int cl_bridge_add(cl_vm_t *vm, const char *name, cl_ext_fn fn);

kernel_Workspace/kernel/src/core-lang/lexer.h (line 16)

[declaration] void cl_lexer_init(lexer_t *l, const char *src, size_t len);

kernel_Workspace/kernel/src/core-lang/lexer.h (line 18)

[declaration] bool cl_lex_next(lexer_t *l, token_t *out);

kernel_Workspace/kernel/src/core-lang/lexer.h (line 20)

[declaration] int cl_lex_all(lexer_t *l, token_t *buf, size_t cap, size_t *count);

kernel_Workspace/kernel/src/core-lang/loader.h (line 15)

[declaration] size_t cl_save_bc(cl_prog_t *P, uint8_t *buf, size_t cap);

kernel_Workspace/kernel/src/core-lang/native.h (line 9)

[declaration] typedef uint64_t (*cl_native_fn)(uint64_t *, uint64_t *, cl_ext_fn *);

kernel_Workspace/kernel/src/core-lang/native.h (line 14)

[declaration] size_t cl_native_size(void);

kernel_Workspace/kernel/src/core-lang/sema.h (line 16)

[declaration] int cl_sema_run(sema_ctx_t *s, ast_node_t *prog);

kernel_Workspace/kernel/src/core-lang/sema.h (line 18)

[declaration] size_t cl_sema_diag_count(const sema_ctx_t *s);

kernel_Workspace/kernel/src/core-lang/vm.h (line 44)

[declaration] void cl_vm_init(cl_vm_t *vm, cl_prog_t *prog);

kernel_Workspace/kernel/src/core-lang/vm.h (line 45)

[declaration] int cl_vm_register_extern(cl_vm_t *vm, const char *name, cl_ext_fn fn);

kernel_Workspace/kernel/src/core-lang/vm.h (line 46)

[declaration] cl_vm_err_t cl_vm_run(cl_vm_t *vm);

kernel_Workspace/kernel/src/core-lang/vm.h (line 48)

[declaration] int cl_wbytes(int bits);

kernel_Workspace/kernel/src/core-lang/vm.h (line 49)

[declaration] void cl_mask(uint8_t *r, int w, int bits);

```

kernel_Workspace/kernel/src/core-lang/vm.h (line 50)
[declaration] void cl_add(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
kernel_Workspace/kernel/src/core-lang/vm.h (line 51)
[declaration] void cl_sub(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
kernel_Workspace/kernel/src/core-lang/vm.h (line 52)
[declaration] void cl_mul(uint8_t *r, const uint8_t *a, const uint8_t *b, int w);
kernel_Workspace/kernel/src/core-lang/vm.h (line 53)
[declaration] int cl_cmp(const uint8_t *a, const uint8_t *b, int w, bool sign);
kernel_Workspace/kernel/src/core-lang/vm.h (line 54)
[declaration] void cl_shl(uint8_t *r, const uint8_t *a, uint32_t n, int w);
kernel_Workspace/kernel/src/core-lang/vm.h (line 55)
[declaration] void cl_shr(uint8_t *r, const uint8_t *a, uint32_t n, int w);
kernel_Workspace/kernel/src/displayport/aux.h (line 17)
[declaration] void dp_aux_sink_init(void);
kernel_Workspace/kernel/src/displayport/aux.h (line 18)
[declaration] bool dp_aux_read(uint32_t addr, void *buf, uint32_t len);
kernel_Workspace/kernel/src/displayport/aux.h (line 19)
[declaration] bool dp_aux_write(uint32_t addr, const void *buf, uint32_t len);
kernel_Workspace/kernel/src/displayport/aux.h (line 20)
[declaration] bool dp_aux_read_edid(void *buf, uint32_t len);
kernel_Workspace/kernel/src/displayport/dp.h (line 10)
[declaration] bool dp_init(void);
kernel_Workspace/kernel/src/displayport/dp.h (line 11)
[declaration] bool dp_ready(void);
kernel_Workspace/kernel/src/displayport/dp.h (line 13)
[declaration] void dp_link_info(uint32_t *rate_mbps, uint32_t *lanes);
kernel_Workspace/kernel/src/displayport/dp.h (line 15)
[declaration] uint32_t dp_mode_count(void);
kernel_Workspace/kernel/src/displayport/dp.h (line 18)
[declaration] bool dp_mode_set_by_id(uint32_t id);
kernel_Workspace/kernel/src/displayport/dp.h (line 20)
[declaration] void dp_caps(uint64_t *fb_phys, uint32_t *w, uint32_t *h, uint32_t
*stride);
kernel_Workspace/kernel/src/displayport/dp.h (line 21)
[declaration] bool dp_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t
stride);
kernel_Workspace/kernel/src/displayport/link.h (line 7)
[declaration] bool dp_link_train(void);
kernel_Workspace/kernel/src/displayport/link.h (line 8)
[declaration] void dp_link_info(uint32_t *rate_mbps, uint32_t *lanes);
kernel_Workspace/kernel/src/displayport/operation.h (line 6)
[declaration] void dp_op_set_fb(uint64_t phys, uint32_t w, uint32_t h, uint32_t
stride);
kernel_Workspace/kernel/src/displayport/operation.h (line 7)
[declaration] void dp_op_state(uint64_t *phys, uint32_t *w, uint32_t *h, uint32_t
*stride);
kernel_Workspace/kernel/src/displayport/operation.h (line 8)
[declaration] void dp_op_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w,
uint32_t h);

```

kernel_Workspace/kernel/src/editor/editor.h (line 21)

[declaration] int editor_run(const char *path);

kernel_Workspace/kernel/src/editor/mouse.h (line 6)

[declaration] void mouse_init(void);

kernel_Workspace/kernel/src/editor/mouse.h (line 8)

[declaration] int mouse_poll(int *dx, int *dy, int *dz, int *buttons);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 41)

[declaration] bool hdmi_init_with(uint64_t fb_phys, uint32_t w, uint32_t h, uint32_t stride);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 42)

[declaration] bool hdmi_ready(void);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 43)

[declaration] void hdmi_caps(hdmi_caps_t *out);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 44)

[declaration] bool hdmi_fb_set(uint64_t phys, uint32_t w, uint32_t h, uint32_t stride);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 45)

[declaration] bool hdmi_mode_set_by_id(uint32_t id);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 46)

[declaration] bool hdmi_mode_current_raw(uint32_t *id, uint32_t *w, uint32_t *h, uint32_t *r);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 47)

[declaration] uint64_t hdmi_submit(const hdmi_transfer_t *t);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 48)

[declaration] bool hdmi_poll(uint64_t *out_seq);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 49)

[declaration] uint32_t hdmi_pending(void);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 50)

[declaration] uint64_t hdmi_submit_fill(uint32_t color, uint32_t x, uint32_t y, uint32_t w, uint32_t h);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 51)

[declaration] void hdmi_caps_raw(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t *phys);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 53)

[declaration] bool hdmi_fb_grant(void);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 54)

[declaration] void hdmi_fb_revoke(void);

kernel_Workspace/kernel/src/hdmi/hdmi.h (line 55)

[declaration] bool hdmi_fb_granted(void);

kernel_Workspace/kernel/src/hdmi/interface.h (line 9)

[declaration] bool hdmi_iface_acquire(uint32_t owner);

kernel_Workspace/kernel/src/hdmi/interface.h (line 10)

[declaration] bool hdmi_iface_release(uint32_t owner);

kernel_Workspace/kernel/src/hdmi/interface.h (line 11)

[declaration] uint32_t hdmi_iface_owner(void);

kernel_Workspace/kernel/src/hdmi/interface.h (line 12)

[declaration] bool hdmi_iface_check(uint32_t owner);

kernel_Workspace/kernel/src/hdmi/interface.h (line 14)

[declaration] bool hdmi_iface_init(uint32_t owner, uint64_t fb_phys, uint32_t w, uint32_t h, uint32_t stride);

```

kernel_Workspace/kernel/src/hdmi/interface.h (line 15)
[declaration] bool hdmi_iface_ready(void);
kernel_Workspace/kernel/src/hdmi/interface.h (line 17)
[declaration] bool hdmi_iface_mode_set(uint32_t owner, uint32_t id);
kernel_Workspace/kernel/src/hdmi/interface.h (line 18)
[declaration] bool hdmi_iface_mode_current(uint32_t *id, uint32_t *w, uint32_t *h,
uint32_t *r);
kernel_Workspace/kernel/src/hdmi/interface.h (line 19)
[declaration] bool hdmi_iface_mode_at(uint32_t i, uint32_t *id, uint32_t *w,
uint32_t *h, uint32_t *r);
kernel_Workspace/kernel/src/hdmi/interface.h (line 21)
[declaration] uint64_t hdmi_iface_submit_fill(uint32_t owner, uint32_t color,
uint32_t x, uint32_t y, uint32_t w, uint32_t h);
kernel_Workspace/kernel/src/hdmi/interface.h (line 22)
[declaration] bool hdmi_iface_poll(uint32_t owner, uint64_t *out_seq);
kernel_Workspace/kernel/src/hdmi/interface.h (line 24)
[declaration] void hdmi_iface_caps(uint32_t *w, uint32_t *h, uint32_t *s, uint64_t
*phys);
kernel_Workspace/kernel/src/hdmi/interface.h (line 26)
[declaration] bool hdmi_iface_fb_grant(uint32_t owner, uint64_t *phys, uint32_t *w,
uint32_t *h, uint32_t *s);
kernel_Workspace/kernel/src/hdmi/interface.h (line 27)
[declaration] bool hdmi_iface_fb_revoke(uint32_t owner);
kernel_Workspace/kernel/src/hdmi/mode.h (line 23)
[declaration] uint32_t hdmi_mode_count(void);
kernel_Workspace/kernel/src/hdmi/mode.h (line 27)
[declaration] bool hdmi_mode_valid(const hdmi_mode_t *m);
kernel_Workspace/kernel/src/hdmi/mode.h (line 28)
[declaration] bool hdmi_mode_apply(const hdmi_mode_t *m);
kernel_Workspace/kernel/src/hdmi/operation.h (line 7)
[declaration] void hdmi_op_set_mode(const hdmi_mode_t *m);
kernel_Workspace/kernel/src/hdmi/operation.h (line 8)
[declaration] void hdmi_op_exec(hdmi_transfer_t *t);
kernel_Workspace/kernel/src/hdmi/operation.h (line 9)
[declaration] void hdmi_op_state(hdmi_caps_t *out);
kernel_Workspace/kernel/src/hdmi/operation.h (line 10)
[declaration] void hdmi_op_set_fb(uint64_t phys, uint32_t w, uint32_t h, uint32_t
stride);
kernel_Workspace/kernel/src/hdmi/operation.h (line 11)
[declaration] bool hdmi_op_ready(void);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.h (line 13)
[declaration] void kfree(void *ptr);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.h (line 15)
[declaration] size_t kmalloc_usable_size(void *ptr);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.h (line 18)
[declaration] void kfree_pages(void *ptr, size_t pages);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.h (line 22)
[declaration] uint64_t kvirt_to_phys(void *ptr);
kernel_Workspace/kernel/src/mm/alloc/api/alloc.h (line 24)
[declaration] void kalloc_dump(void);

```

kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.h (line 9)
[declaration] void dbg_free(void *ptr);

kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.h (line 11)
[declaration] bool dbg_verify(void *ptr);

kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.h (line 12)
[declaration] size_t dbg_usable_size(void *ptr);

kernel_Workspace/kernel/src/mm/alloc/debug/alloc_debug.h (line 13)
[declaration] void mm_debug_dump(void);

kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 8)
[declaration] bool leak_track(void *ptr, size_t size, uint64_t caller);

kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 10)
[declaration] bool leak_track_id(void *ptr, size_t size, uint64_t caller, uint64_t *out_id);

kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 12)
[declaration] bool leak_untrack(void *ptr, size_t *out_size, uint64_t *out_caller);

kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 13)
[declaration] bool leak_contains(void *ptr);

kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 14)
[declaration] size_t leak_count(void);

kernel_Workspace/kernel/src/mm/alloc/debug/leak.h (line 16)
[declaration] void leak_dump(void);

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 8)
[declaration] void alloc_stats_note_alloc(size_t size);

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 9)
[declaration] void alloc_stats_note_free(size_t size);

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 11)
[declaration] size_t alloc_stats_live_count(void);

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 12)
[declaration] size_t alloc_stats_live_bytes(void);

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 13)
[declaration] size_t alloc_stats_peak_bytes(void);

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 14)
[declaration] size_t alloc_stats_total_allocs(void);

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 15)
[declaration] size_t alloc_stats_total_frees(void);

kernel_Workspace/kernel/src/mm/alloc/debug/stats.h (line 17)
[declaration] void alloc_stats_dump(void);

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 8)
[declaration] typedef bool (*buddy_map_cb)(uint64_t virt, size_t size);

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 9)
[declaration] typedef void (*buddy_unmap_cb)(uint64_t virt, size_t size);

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 16)
[declaration] bool buddy_ready(void);

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 20)
[declaration] void buddy_free(void *ptr);

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 22)
[declaration] size_t buddy_block_size(void *ptr);

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 24)
[declaration] size_t buddy_stat_used_bytes(void);

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 25)
[declaration] size_t buddy_stat_free_bytes(void);

kernel_Workspace/kernel/src/mm/alloc/heap/buddy.h (line 27)
[declaration] void buddy_dump(void);

kernel_Workspace/kernel/src/mm/alloc/heap/heap.h (line 8)
[declaration] bool heap_init(void);

kernel_Workspace/kernel/src/mm/alloc/heap/heap.h (line 9)
[declaration] bool heap_ready(void);

kernel_Workspace/kernel/src/mm/alloc/heap/heap.h (line 15)
[declaration] void heap_free(void *ptr);

kernel_Workspace/kernel/src/mm/alloc/heap/heap.h (line 17)
[declaration] size_t heap_usable_size(void *ptr);

kernel_Workspace/kernel/src/mm/alloc/heap/heap.h (line 19)
[declaration] void heap_dump(void);

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 10)
[declaration] bool slab_init(uint64_t base, size_t size);

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 11)
[declaration] bool slab_ready(void);

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 14)
[declaration] void slab_free(void *ptr);

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 16)
[declaration] size_t slab_usable_size(void *ptr);

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 18)
[declaration] size_t slab_stat_used_bytes(void);

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 19)
[declaration] size_t slab_stat_free_bytes(void);

kernel_Workspace/kernel/src/mm/alloc/heap/slab.h (line 21)
[declaration] void slab_dump(void);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 30)
[declaration] size_t bitmap_words_for_bits(size_t bit_count);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 31)
[declaration] size_t bitmap_bytes_for_bits(size_t bit_count);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 33)
[declaration] void bitmap_init_virt(bitmap_t *bm, void *storage, size_t bit_count);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 35)
[declaration] bool bitmap_init_phys(bitmap_t *bm, uint64_t storage_phys, size_t bit_count);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 37)
[declaration] void bitmap_fill(bitmap_t *bm, bool value);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 39)
[declaration] void bitmap_set(bitmap_t *bm, size_t bit);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 40)
[declaration] void bitmap_clear(bitmap_t *bm, size_t bit);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 41)
[declaration] bool bitmap_test(const bitmap_t *bm, size_t bit);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 43)
[declaration] void bitmap_fill(bitmap_t *bm, bool value);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 45)
[declaration] void bitmap_set(bitmap_t *bm, size_t bit);

kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 46)
[declaration] void bitmap_clear(bitmap_t *bm, size_t bit);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 47)
[declaration] bool bitmap_test(const bitmap_t *bm, size_t bit);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 49)
[declaration] void bitmap_set_range(bitmap_t *bm, size_t start, size_t count);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 50)
[declaration] void bitmap_clear_range(bitmap_t *bm, size_t start, size_t count);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 52)
[declaration] bool bitmap_test_range_free(const bitmap_t *bm, size_t start, size_t count);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 54)
[declaration] size_t bitmap_alloc(bitmap_t *bm);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 57)
[declaration] size_t bitmap_alloc_from(bitmap_t *bm, size_t min_bit);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 59)
[declaration] size_t bitmap_alloc_range(bitmap_t *bm, size_t count, size_t align);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 61)
[declaration] void bitmap_free(bitmap_t *bm, size_t bit);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 62)
[declaration] void bitmap_free_range(bitmap_t *bm, size_t start, size_t count);
kernel_Workspace/kernel/src/mm/alloc/physical/bitmap.h (line 64)
[declaration] size_t bitmap_count_free(const bitmap_t *bm);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 12)
[declaration] bool frame_init(uint64_t bitmap_phys, size_t bit_count);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 14)
[declaration] bool frame_init_from_memory(void);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 16)
[declaration] bool frame_ready(void);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 18)
[declaration] bool frame_alloc(frame_t *out);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 19)
[declaration] bool frame_alloc_zero(frame_t *out);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 29)
[declaration] bool frame_free(frame_t frame);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 30)
[declaration] bool frame_free_contiguous(frame_t start, size_t count);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 32)
[declaration] bool frame_is_valid(frame_t frame);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 33)
[declaration] bool frame_zero(frame_t frame);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 36)
[declaration] uint64_t frame_phys(frame_t frame);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 38)
[declaration] size_t frame_to_pfn(frame_t frame);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 39)
[declaration] frame_t frame_from_pfn(size_t pfn);
kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 41)
[declaration] size_t frame_total(void);

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 42)
[declaration] size_t frame_allocated(void);

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 43)
[declaration] size_t frame_free_count(void);

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 46)
[declaration] size_t frame_zone_dma32_total(void);

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 47)
[declaration] size_t frame_zone_dma32_free(void);

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 48)
[declaration] size_t frame_zone_normal_total(void);

kernel_Workspace/kernel/src/mm/alloc/physical/frame.h (line 49)
[declaration] size_t frame_zone_normal_free(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 10)
[declaration] bool pmm_init(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 12)
[declaration] bool pmm_ready(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 14)
[declaration] bool pmm_alloc_frame(uint64_t *out_phys);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 16)
[declaration] bool pmm_alloc_zero_frame(uint64_t *out_phys);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 18)
[declaration] bool pmm_alloc_frames(size_t count, uint64_t *out_phys);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 23)
[declaration] bool pmm_alloc_bytes(size_t bytes, uint64_t *out_phys);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 24)
[declaration] bool pmm_alloc_zero_bytes(size_t bytes, uint64_t *out_phys);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 28)
[declaration] bool pmm_free_frame(uint64_t phys);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 33)
[declaration] bool pmm_free_frames(uint64_t phys, size_t count);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 34)
[declaration] bool pmm_free_bytes(uint64_t phys, size_t bytes);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 35)
[declaration] size_t pmm_stat_total_frames(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 36)
[declaration] size_t pmm_stat_free_frames(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 37)
[declaration] size_t pmm_stat_allocated_frames(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 39)
[declaration] uint64_t pmm_stat_total_bytes(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 40)
[declaration] uint64_t pmm_stat_free_bytes(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 43)
[declaration] size_t pmm_stat_dma32_total_frames(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 44)
[declaration] size_t pmm_stat_dma32_free_frames(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 45)
[declaration] size_t pmm_stat_normal_total_frames(void);

kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 46)
[declaration] size_t pmm_stat_normal_free_frames(void);
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 48)
[declaration] void pmm_dump(void);
kernel_Workspace/kernel/src/mm/alloc/physical/pmm.h (line 51)
[declaration] bool pmm_self_test(void);
kernel_Workspace/kernel/src/mm/alloc/special/contiguous.h (line 13)
[declaration] void contig_free(uint64_t phys, size_t bytes);
kernel_Workspace/kernel/src/mm/alloc/special/dma.h (line 16)
[declaration] void dma_free_coherent(uint64_t phys, void *virt, size_t bytes);
kernel_Workspace/kernel/src/mm/alloc/special/dma.h (line 18)
[declaration] void dma_sync_for_device(void *virt, size_t len);
kernel_Workspace/kernel/src/mm/alloc/special/dma.h (line 19)
[declaration] void dma_sync_for_cpu(void *virt, size_t len);
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.h (line 12)
[declaration] void mapping_init(void);
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.h (line 13)
[declaration] mapping_space_t mapping_kernel_space(void);
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.h (line 30)
[declaration] uint64_t mapping_translate(mapping_space_t space, uint64_t virt);
kernel_Workspace/kernel/src/mm/alloc/virtual/mapping.h (line 31)
[declaration] bool mapping_is_mapped(mapping_space_t space, uint64_t virt);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 22)
[declaration] bool page_init(void);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 23)
[declaration] bool page_ready(void);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 28)
[declaration] bool page_attach(uint64_t phys, uint32_t type);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 30)
[declaration] bool page_get(page_t *page);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 31)
[declaration] bool page_put(page_t *page);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 33)
[declaration] uint64_t page_phys(const page_t *page);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 35)
[declaration] size_t page_pfn(const page_t *page);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 41)
[declaration] uint32_t page_refcount(const page_t *page);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 42)
[declaration] uint32_t page_type(const page_t *page);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 43)
[declaration] void page_set_type(page_t *page, uint32_t type);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 45)
[declaration] size_t page_stat_total(void);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 46)
[declaration] size_t page_stat_referenced(void);
kernel_Workspace/kernel/src/mm/alloc/virtual/page.h (line 48)
[declaration] void page_dump(void);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 18)
[declaration] bool vmm_init(void);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 19)
[declaration] bool vmm_ready(void);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 21)
[declaration] bool vmm_alloc(size_t bytes, uint32_t flags, uint64_t *out_virt);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 22)
[declaration] bool vmm_alloc_aligned(size_t bytes, size_t align, uint32_t flags, uint64_t *out_virt);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 23)
[declaration] bool vmm_map_device(uint64_t phys, size_t len, uint64_t *out_virt);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 25)
[declaration] bool vmm_free(uint64_t virt, size_t bytes);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 26)
[declaration] bool vmm_unmap_device(uint64_t virt, size_t len);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 28)
[declaration] uint64_t vmm_translate(uint64_t virt);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 30)
[declaration] size_t vmm_stat_total_pages(void);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 31)
[declaration] size_t vmm_stat_free_pages(void);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 32)
[declaration] size_t vmm_stat_allocated_pages(void);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 33)
[declaration] uint64_t vmm_stat_total_bytes(void);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 34)
[declaration] uint64_t vmm_stat_free_bytes(void);

kernel_Workspace/kernel/src/mm/alloc/virtual/vmm.h (line 36)
[declaration] void vmm_dump(void);

kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 25)
[declaration] bool arch_memory_ready(void);

kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 27)
[declaration] bool arch_memory_boot_alloc(uint64_t len, uint64_t align, uint64_t *out);

kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 28)
[declaration] void arch_memory_reserve_range(uint64_t base, uint64_t len);

kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 30)
[declaration] bool arch_is_page_aligned(uint64_t a);

kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 31)
[declaration] uint64_t arch_page_align_up(uint64_t a);

kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 32)
[declaration] uint64_t arch_page_align_down(uint64_t a);

kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 35)
[declaration] uint64_t arch_virt_to_phys(void *virt);

kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 37)
[declaration] uint64_t arch_memory_max_phys(void);

kernel_Workspace/kernel/src/mm/arch/aarch64/memory.h (line 38)
[declaration] void arch_memory_dump(void);

kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 17)
[declaration] void paging_init(uint64_t boot_phys_offset);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 19)
[declaration] bool paging_enable_nx(void);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 21)
[declaration] uint64_t paging_read_cr3(void);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 22)
[declaration] void paging_write_cr3(uint64_t pml4_phys);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 24)
[declaration] bool paging_map_page_in(uint64_t pml4, uint64_t virt, uint64_t phys, uint64_t flags);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 25)
[declaration] bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 26)
[declaration] bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 27)
[declaration] bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);
kernel_Workspace/kernel/src/mm/arch/aarch64/paging.h (line 28)
[declaration] uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 8)
[declaration] bool tlb_init(void);
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 9)
[declaration] bool tlb_ready(void);
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 11)
[declaration] void tlb_flush_all(void);
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 12)
[declaration] void tlb_flush_all_including_global(void);
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 14)
[declaration] void tlb_flush_page(const void *addr);
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 15)
[declaration] void tlb_flush_page_addr(uint64_t addr);
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 17)
[declaration] void tlb_flush_range(const void *addr, size_t pages);
kernel_Workspace/kernel/src/mm/arch/aarch64/tlb.h (line 18)
[declaration] void tlb_flush_range_addr(uint64_t addr, size_t pages);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 71)
[declaration] bool arch_memory_ready(void);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 73)
[declaration] bool arch_memory_boot_alloc(uint64_t len, uint64_t align, uint64_t *out_base);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 76)
[declaration] size_t arch_memory_regions(const arch_mem_region_t **out);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 77)
[declaration] uint64_t arch_memory_total_usable(void);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 78)
[declaration] bool arch_memory_range_is_usable(uint64_t base, uint64_t len);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 79)
[declaration] void arch_memory_reserve_range(uint64_t base, uint64_t len);

kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 83)
[declaration] void arch_memory_dump(void);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 92)
[function] if (v > UINT64_MAX - ARCH_PAGE_MASK) {
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 107)
[declaration] return (void *) (uintptr_t) (phys + ARCH_DIRECT_MAP_BASE);
kernel_Workspace/kernel/src/mm/arch/x86_64/memory.h (line 112)
[declaration] return (uint64_t) ((uintptr_t) virt - ARCH_DIRECT_MAP_BASE);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 35)
[declaration] void paging_set_boot_phys_offset(uint64_t phys_offset);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 37)
[declaration] bool paging_boot_phys_offset_valid(void);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 39)
[declaration] void paging_init_direct_map(void);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 41)
[declaration] void paging_init(uint64_t boot_phys_offset);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 43)
[declaration] uint64_t paging_read_cr3(void);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 44)
[declaration] void paging_flush_tlb_all(void);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 45)
[declaration] void paging_flush_page(uint64_t addr);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 47)
[declaration] bool paging_map_page(uint64_t virt, uint64_t phys, uint64_t flags);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 49)
[declaration] bool paging_map_range(uint64_t virt, uint64_t phys, uint64_t len, uint64_t flags);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 51)
[declaration] uint64_t paging_translate(uint64_t virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 52)
[declaration] bool paging_is_mapped(uint64_t virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 53)
[declaration] bool paging_unmap_page(uint64_t virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 54)
[declaration] bool paging_set_flags(uint64_t virt, uint64_t flags);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 55)
[declaration] bool paging_get_flags(uint64_t virt, uint64_t *out_flags);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 56)
[declaration] void paging_enable_nx(void);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 57)
[declaration] bool paging_nx_enabled(void);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 63)
[declaration] bool paging_unmap_page_in(uint64_t pml4, uint64_t virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 64)
[declaration] uint64_t paging_translate_in(uint64_t pml4, uint64_t virt);
kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 65)
[declaration] bool paging_is_mapped_in(uint64_t pml4, uint64_t virt);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 66)
[declaration] bool paging_set_flags_in(uint64_t pml4, uint64_t virt, uint64_t flags);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 67)
[declaration] bool paging_get_flags_in(uint64_t pml4, uint64_t virt, uint64_t *out_flags);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 69)
[declaration] uint64_t paging_create_pml4(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 70)
[declaration] void paging_destroy_pml4(uint64_t pml4_phys);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 71)
[declaration] void paging_switch_pml4(uint64_t pml4_phys);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 73)
[declaration] bool paging_map_mmio(uint64_t virt, uint64_t phys, uint64_t len);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 75)
[declaration] void paging_enable_write_protect(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 76)
[declaration] void paging_disable_write_protect(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 77)
[declaration] bool paging_write_protect_enabled(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 79)
[declaration] void paging_enable_smep(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 80)
[declaration] void paging_enable_smap(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 81)
[declaration] bool paging_smep_enabled(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 82)
[declaration] bool paging_smap_enabled(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 84)
[declaration] bool paging_pcid_supported(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 85)
[declaration] void paging_enable_pcid(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 86)
[declaration] bool paging_pcid_enabled(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 87)
[declaration] void paging_invpcid(uint64_t type, uint64_t pcid, uint64_t addr);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 90)
[declaration] bool paging_la57_supported(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 91)
[declaration] void paging_assert_4level_paging(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/paging.h (line 92)
[declaration] void paging_write_cr3(uint64_t pml4_phys);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 14)
[declaration] bool tlb_init(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 15)
[declaration] bool tlb_ready(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 16)
[declaration] bool tlb_has_pcid(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 17)
[declaration] bool tlb_has_invpcid(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 19)
[declaration] void tlb_flush_all(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 20)
[declaration] void tlb_flush_all_including_global(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 22)
[declaration] void tlb_flush_page(const void *addr);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 23)
[declaration] void tlb_flush_page_addr(uint64_t addr);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 25)
[declaration] void tlb_flush_range(const void *addr, size_t pages);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 26)
[declaration] void tlb_flush_range_addr(uint64_t addr, size_t pages);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 28)
[declaration] void tlb_flush_pcid(uint16_t pcid);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 29)
[declaration] void tlb_flush_pcid_addr(uint16_t pcid, uint64_t addr);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 31)
[declaration] void tlb_wbinvd(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb.h (line 32)
[declaration] void tlb_clflush(uint64_t addr);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 6)
[declaration] void tlb_asm_invlpg(uint64_t addr);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 8)
[declaration] uint64_t tlb_asm_read_cr3(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 9)
[declaration] void tlb_asm_write_cr3(uint64_t value);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 11)
[declaration] uint64_t tlb_asm_read_cr4(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 12)
[declaration] void tlb_asm_write_cr4(uint64_t value);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 14)
[declaration] void tlb_asm_invpcid(uint64_t type, const void *desc);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 16)
[declaration] void tlb_asm_wbinvd(void);

kernel_Workspace/kernel/src/mm/arch/x86_64/tlb_asm.h (line 17)
[declaration] void tlb_asm_clflush(uint64_t addr);

kernel_Workspace/kernel/src/mm/cache/cache.h (line 7)
[declaration] bool cache_init(void);

kernel_Workspace/kernel/src/mm/cache/cache.h (line 8)
[declaration] bool cache_ready(void);

kernel_Workspace/kernel/src/mm/cache/cache.h (line 10)
[declaration] void cache_register(kcache_t *c);

kernel_Workspace/kernel/src/mm/cache/cache.h (line 13)
[declaration] void cache_dump(void);

kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 12)
[declaration] typedef void (*kcache_ctor_t)(void *obj);

kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 47)
[declaration] void kcache_destroy(kcache_t *c);
kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 51)
[declaration] void kcache_free(kcache_t *c, void *ptr);
kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 53)
[declaration] size_t kcache_object_size(kcache_t *c);
kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 54)
[declaration] size_t kcache_free_count(kcache_t *c);
kernel_Workspace/kernel/src/mm/cache/object_cashe.h (line 55)
[declaration] size_t kcache_total_count(kcache_t *c);
kernel_Workspace/kernel/src/mm/cache/per_cpu.h (line 12)
[declaration] bool per_cpu_init(void);
kernel_Workspace/kernel/src/mm/cache/per_cpu.h (line 13)
[declaration] size_t per_cpu_count(void);
kernel_Workspace/kernel/src/mm/cache/per_cpu.h (line 14)
[declaration] size_t per_cpu_id(void);
kernel_Workspace/kernel/src/mm/core/address.h (line 10)
[declaration] bool addr_is_canonical(uint64_t a);
kernel_Workspace/kernel/src/mm/core/address.h (line 11)
[declaration] bool addr_is_user(uint64_t a);
kernel_Workspace/kernel/src/mm/core/address.h (line 12)
[declaration] bool addr_is_kernel(uint64_t a);
kernel_Workspace/kernel/src/mm/core/address.h (line 13)
[declaration] bool addr_is_direct_map(uint64_t a);
kernel_Workspace/kernel/src/mm/core/address.h (line 15)
[declaration] uint64_t addr_align_up(uint64_t a, uint64_t align);
kernel_Workspace/kernel/src/mm/core/address.h (line 16)
[declaration] uint64_t addr_align_down(uint64_t a, uint64_t align);
kernel_Workspace/kernel/src/mm/core/address.h (line 18)
[declaration] uint64_t addr_phys_to_direct(uint64_t phys);
kernel_Workspace/kernel/src/mm/core/address.h (line 19)
[declaration] uint64_t addr_direct_to_phys(uint64_t va);
kernel_Workspace/kernel/src/mm/core/mm.h (line 26)
[declaration] bool mm_init(const mm_boot_params_t *params);
kernel_Workspace/kernel/src/mm/core/mm.h (line 27)
[declaration] bool mm_ready(void);
kernel_Workspace/kernel/src/mm/core/mm.h (line 29)
[declaration] uint64_t mm_total_ram(void);
kernel_Workspace/kernel/src/mm/core/mm.h (line 30)
[declaration] uint64_t mm_free_ram(void);
kernel_Workspace/kernel/src/mm/core/mm.h (line 32)
[declaration] void mm_dump(void);
kernel_Workspace/kernel/src/mm/core/range.h (line 18)
[declaration] bool range_is_canonical(uint64_t addr);
kernel_Workspace/kernel/src/mm/core/region.h (line 13)
[declaration] region_t region_make(uint64_t start, uint64_t len);
kernel_Workspace/kernel/src/mm/core/region.h (line 14)
[declaration] bool region_valid(region_t r);

kernel_Workspace/kernel/src/mm/core/region.h (line 15)
[declaration] uint64_t region_len(region_t r);

kernel_Workspace/kernel/src/mm/core/region.h (line 17)
[declaration] bool region_contains(region_t r, uint64_t addr);

kernel_Workspace/kernel/src/mm/core/region.h (line 18)
[declaration] bool region_overlaps(region_t a, region_t b);

kernel_Workspace/kernel/src/mm/core/region.h (line 19)
[declaration] region_t region_intersect(region_t a, region_t b);

kernel_Workspace/kernel/src/mm/core/sizeutil.h (line 8)
[declaration] bool size_is_pow2(size_t v);

kernel_Workspace/kernel/src/mm/core/sizeutil.h (line 11)
[declaration] bool size_round_up_pow2(size_t v, size_t *out);

kernel_Workspace/kernel/src/mm/core/smp_lock.h (line 21)
[declaration] void smp_lock_init(smp_ticket_lock_t *lock);

kernel_Workspace/kernel/src/mm/core/smp_lock.h (line 24)
[declaration] void smp_lock_acquire(smp_ticket_lock_t *lock);

kernel_Workspace/kernel/src/mm/core/smp_lock.h (line 27)
[declaration] bool smp_lock_release(smp_ticket_lock_t *lock);

kernel_Workspace/kernel/src/mm/core/smp_lock.h (line 30)
[declaration] int64_t smp_current_cpu_id(void);

kernel_Workspace/kernel/src/mm/paging/paging.h (line 19)
[declaration] bool paging_subsystem_init(void);

kernel_Workspace/kernel/src/mm/paging/paging.h (line 22)
[declaration] void paging_aspace_destroy(address_space_t *as);

kernel_Workspace/kernel/src/mm/paging/paging.h (line 23)
[declaration] void paging_aspace_switch(address_space_t *as);

kernel_Workspace/kernel/src/mm/paging/paging.h (line 24)
[declaration] uint64_t paging_aspace_cr3(const address_space_t *as);

kernel_Workspace/kernel/src/mm/paging/paging.h (line 32)
[declaration] bool paging_aspace_unmap(address_space_t *as, uint64_t virt, size_t len);

kernel_Workspace/kernel/src/mm/paging/paging.h (line 39)
[declaration] uint64_t paging_aspace_translate(address_space_t *as, uint64_t virt);

kernel_Workspace/kernel/src/mm/paging/paging.h (line 41)
[declaration] bool paging_kernel_map(uint64_t virt, uint64_t phys, size_t len, uint32_t prot);

kernel_Workspace/kernel/src/mm/paging/paging.h (line 42)
[declaration] bool paging_kernel_unmap(uint64_t virt, size_t len);

kernel_Workspace/kernel/src/mm/paging/pml.h (line 14)
[declaration] uint64_t pml_level_shift(int level);

kernel_Workspace/kernel/src/mm/paging/pml.h (line 15)
[declaration] uint64_t pml_index(uint64_t virt, int level);

kernel_Workspace/kernel/src/mm/paging/pml.h (line 17)
[declaration] bool pml_entry_present(uint64_t entry);

kernel_Workspace/kernel/src/mm/paging/pml.h (line 18)
[declaration] bool pml_entry_large(uint64_t entry);

kernel_Workspace/kernel/src/mm/paging/pml.h (line 19)
[declaration] uint64_t pml_entry_addr(uint64_t entry);

kernel_Workspace/kernel/src/mm/paging/tlb.h (line 16)

[declaration] void tlb_batch_begin(tlb_batch_t *batch);

kernel_Workspace/kernel/src/mm/paging/tlb.h (line 17)

[declaration] void tlb_batch_add(tlb_batch_t *batch, uint64_t virt);

kernel_Workspace/kernel/src/mm/paging/tlb.h (line 18)

[declaration] void tlb_batch_full(tlb_batch_t *batch);

kernel_Workspace/kernel/src/mm/paging/tlb.h (line 19)

[declaration] void tlb_batch_commit(tlb_batch_t *batch);

kernel_Workspace/kernel/src/mm/process/address_space.h (line 30)

[declaration] bool aspace_subsystem_init(void);

kernel_Workspace/kernel/src/mm/process/address_space.h (line 33)

[declaration] void aspace_destroy(proc_aspace_t *pa);

kernel_Workspace/kernel/src/mm/process/address_space.h (line 37)

[declaration] uint64_t aspace_map_anon(proc_aspace_t *pa, uint64_t hint, size_t len, uint32_t prot);

kernel_Workspace/kernel/src/mm/process/address_space.h (line 38)

[declaration] uint64_t aspace_map_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

kernel_Workspace/kernel/src/mm/process/address_space.h (line 39)

[declaration] uint64_t aspace_reserve_at(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t flags);

kernel_Workspace/kernel/src/mm/process/address_space.h (line 40)

[declaration] bool aspace_unmap(proc_aspace_t *pa, uint64_t addr, size_t len);

kernel_Workspace/kernel/src/mm/process/address_space.h (line 41)

[declaration] bool aspace_protect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

kernel_Workspace/kernel/src/mm/process/address_space.h (line 51)

[declaration] uint64_t aspace_stack_base(void);

kernel_Workspace/kernel/src/mm/process/address_space.h (line 53)

[declaration] uint64_t aspace_brk(proc_aspace_t *pa, uint64_t new_brk);

kernel_Workspace/kernel/src/mm/process/mmap.h (line 20)

[declaration] bool munmap(proc_aspace_t *pa, uint64_t addr, size_t len);

kernel_Workspace/kernel/src/mm/process/mmap.h (line 21)

[declaration] bool mprotect(proc_aspace_t *pa, uint64_t addr, size_t len, uint32_t prot);

kernel_Workspace/kernel/src/mm/protection/guard.h (line 9)

[declaration] uint64_t guard_install(proc_aspace_t *pa, uint64_t addr, size_t len);

kernel_Workspace/kernel/src/mm/protection/guard.h (line 10)

[declaration] bool guard_user_stack(proc_aspace_t *pa);

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 8)

[declaration] bool isolation_init(void);

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 10)

[declaration] bool isolation_has_smep(void);

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 11)

[declaration] bool isolation_has_smmap(void);

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 12)

[declaration] bool isolation_enable_smep(void);

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 13)

[declaration] bool isolation_enable_smmap(void);

```

kernel_Workspace/kernel/src/mm/protection/isolation.h (line 15)
[declaration] size_t isolation_audit_kernel(void);
kernel_Workspace/kernel/src/mm/protection/permissions.h (line 8)
[declaration] uint32_t perm_sanitize(uint32_t prot);
kernel_Workspace/kernel/src/mm/protection/permissions.h (line 9)
[declaration] bool perm_is_wx(uint32_t prot);
kernel_Workspace/kernel/src/mm/protection/permissions.h (line 10)
[declaration] bool perm_mprotect_allowed(uint32_t old_prot, uint32_t new_prot);
kernel_Workspace/kernel/src/mm/protection/permissions.h (line 11)
[declaration] void perm_set_strict_wx(bool on);
kernel_Workspace/kernel/src/mm/protection/permissions.h (line 13)
[declaration] uint32_t perm_kernel_default(void);
kernel_Workspace/kernel/src/mm/protection/permissions.h (line 14)
[declaration] uint32_t perm_user_default(void);
kstd/include/kstd.h (line 12)
[declaration] void tr_log(const char *s);
kstd/include/kstd_audio.h (line 7)
[declaration] void tr_free(void *ptr, uint32_t bytes);
kstd/include/kstd_audio.h (line 8)
[declaration] tr_status_t tr_map_mmio(uint64_t phys, uint32_t len, void **out_va);
kstd/include/kstd_bt.h (line 6)
[declaration] tr_status_t tr_bt_info(uint8_t *hci_ver, uint8_t *addr6);
kstd/include/kstd_bt.h (line 7)
[declaration] tr_status_t tr_bt_cmd(uint16_t opcode, const void *params, uint8_t len);
kstd/include/kstd_bt.h (line 8)
[declaration] tr_status_t tr_bt_evt(void *buf, uint8_t *len);
kstd/include/kstd_bt.h (line 9)
[declaration] tr_status_t tr_bt_acl_send(const void *data, uint16_t len);
kstd/include/kstd_bt.h (line 10)
[declaration] tr_status_t tr_bt_acl_recv(void *buf, uint16_t *len);
kstd/include/kstd_fs.h (line 6)
[declaration] tr_status_t tr_fs_read(const char *path, void *buf, uint32_t cap, uint32_t *out);
kstd/include/kstd_fs.h (line 7)
[declaration] tr_status_t tr_fs_exists(const char *path);
kstd/include/kstd_input.h (line 6)
[declaration] int32_t tr_input_key(void);
kstd/include/kstd_mem.h (line 7)
[declaration] void tr_free(void *ptr, uint32_t bytes);
kstd/include/kstd_mem.h (line 8)
[declaration] tr_status_t tr_map_mmio(uint64_t phys, uint32_t len, void **out_va);
kstd/include/kstd_video.h (line 22)
[declaration] tr_status_t tr_video_init(void);
kstd/include/kstd_video.h (line 23)
[declaration] bool tr_video_is_ready(void);
kstd/include/kstd_video.h (line 24)
[declaration] tr_status_t tr_video_get_caps(tr_video_caps_t *caps);

```

```

kstd/include/kstd_video.h (line 27)
[declaration] uint32_t tr_video_mode_count(void);
kstd/include/kstd_video.h (line 28)
[declaration] tr_status_t tr_video_mode_get(uint32_t index, tr_video_mode_t *mode);
kstd/include/kstd_video.h (line 29)
[declaration] tr_status_t tr_video_mode_set(uint32_t mode_id);
kstd/include/kstd_video.h (line 32)
[declaration] tr_status_t tr_video_fill(uint32_t color, uint32_t x, uint32_t y,
uint32_t w, uint32_t h);
libs/dsabi.h (line 93)
[declaration] __asm__ volatile("int $0x80" :: "a"(1));
libs/dsabi.h (line 124)
[declaration] void ds_init(uint64_t params_va);
libs/dsabi.h (line 127)
[declaration] int ds_take(uint64_t id, ds_msg_t *out);
libs/dsabi.h (line 128)
[declaration] void ds_poll(void);
libs/dsasync.h (line 21)
[function] if (r->done) {
libs/dsasync.h (line 25)
[declaration] ds_poll();
libs/dsasync.h (line 27)
[function] if (r->id != 0 && ds_take(r->id, &r->msg)) {
libs/dsaudio.h (line 7)
[declaration] return ds_call(SVC_AUDIO, AUD_PLAY, phys, len, 0);
libs/dsaudio.h (line 12)
[declaration] ds_call(SVC_AUDIO, AUD_STOP, 0, 0, 0);
libs/dsaudio.h (line 17)
[declaration] return ds_call(SVC_AUDIO, AUD_JACK, 0, 0, 0);
libs/dsaudio.h (line 22)
[declaration] ds_call(SVC_AUDIO, AUD_AMP, on ? 1 : 0, 0, 0);
libs/dsinput.h (line 7)
[declaration] return ds_call(SVC_INPUT, IN_KEY_POLL, 0, 0, 0);
libs/dsinput.h (line 14)
[function] if (!ds_take(id, &m) || m.arg0 == 0) {
libs/dslock.h (line 7)
[declaration] return ds_call(SVC_BLOCK, BLK_COUNT, disk, 0, 0);
libs/dslock.h (line 12)
[declaration] return ds_call(SVC_BLOCK, BLK_READ, disk, block, 0);
libs/dslock.h (line 17)
[declaration] return ds_call(SVC_BLOCK, BLK_WRITE, disk, block, 0);
libs/dslog.h (line 10)
[function] while (s[n] != '\0' && n < DS_SCRATCH_SIZE - 1) {
libs/dslog.h (line 15)
[declaration] ds_call(SVC_SYS, OP_LOG, n, 0, 0);
libs/dsmem.h (line 7)
[declaration] return ds_call(SVC_SYS, OP_ALLOC, pages, 0, 0);
libs/dsmem.h (line 12)
[declaration] return ds_call(SVC_SYS, OP_FREE, va, 0, 0);

```

```

libs/dsmem.h (line 17)
[declaration] return ds_call(SVC_SYS, OP_MAPMMIO, phys, len, va);
libs/dsmem.h (line 22)
[declaration] return ds_call(SVC_SYS, OP_PAGEPHYS, va, 0, 0);
libs/dsvgpu.h (line 6)
[declaration] extern uint64_t ds_call(uint32_t, uint32_t, uint64_t, uint64_t,
uint64_t);
libs/dsvgpu.h (line 7)
[declaration] extern void ds_poll(void);
libs/dsvgpu.h (line 8)
[declaration] extern int ds_take(uint64_t, ds_msg_t *);
libs/dsvgpu.h (line 14)
[declaration] ds_poll();
libs/dsvgpu.h (line 18)
[function] if (!ds_take(id, &r) || r.status != 0) {
libs/dsvgpu.h (line 31)
[declaration] ds_poll();
libs/dsvideo.h (line 14)
[declaration] return ds_call(SVC_VIDEO, VID_FB_INFO, 0, 0, 0);
libs/dsvideo.h (line 21)
[function] if (!ds_take(id, &m) || m.status != 0) {
libs/dsvideo.h (line 35)
[declaration] ds_call(SVC_VIDEO, VID_FB_TAKEOVER, 0, 0, 0);
libs/dsvideo.h (line 40)
[declaration] ds_call(SVC_VIDEO, VID_FB_RELEASE, 0, 0, 0);

```

Odin

```

drivers/audiodriver/src/odin/driver.odin (line 8)
[struct] Bdl_Entry :: struct {
drivers/audiodriver/src/odin/driver.odin (line 16)
[proc] nam16 :: proc(off: u32) -> ^u16 {
drivers/audiodriver/src/odin/driver.odin (line 21)
[proc] bm8 :: proc(off: u32) -> ^u8 {
drivers/audiodriver/src/odin/driver.odin (line 25)
[proc] bm16 :: proc(off: u32) -> ^u16 {
drivers/audiodriver/src/odin/driver.odin (line 29)
[proc] bm32 :: proc(off: u32) -> ^u32 {
drivers/audiodriver/src/odin/driver.odin (line 50)
[proc] ad_init :: proc "C" (nam_va: u64, bm_va: u64) -> i32 {
drivers/audiodriver/src/odin/driver.odin (line 73)
[proc] ad_stop :: proc "C" () -> i32 {
drivers/audiodriver/src/odin/driver.odin (line 80)
[proc] ad_position :: proc "C" () -> u32 {
drivers/audiodriver/src/odin/jack.odin (line 8)
[proc] ad_jack_present :: proc "C" () -> i32 {
drivers/audiodriver/src/odin/jack.odin (line 21)
[proc] ad_set_amp :: proc "C" (on: i32) -> i32 {

```

drivers/audiodriver/src/odin/microphone.odin (line 6)

```
[proc] ad_capture :: proc "C" (data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64)
-> i32 {
```

drivers/audiodriver/src/odin/speaker.odin (line 6)

```
[proc] ad_play :: proc "C" (data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) ->
i32 {
```

Python

doc.py (line 131)

```
[class] class Definition:
```

doc.py (line 144)

```
[class] class MarkdownFile:
```

doc.py (line 153)

```
[function] def iter_source_files(root: str, exts: set[str] | None, exclude_dirs:
set[str]):
```

doc.py (line 170)

```
[function] def read_text(path: str) -> str:
```

doc.py (line 185)

```
[function] def extract_preceding_comment(lines: list[str], lineno_0based: int, ext:
str) -> str:
```

doc.py (line 222)

```
[function] def parse_python(rel_path: str, full_path: str, source: str) ->
list[Definition]:
```

doc.py (line 457)

```
[function] def parse_generic(rel_path: str, ext: str, source: str, max_body_lines:
int) -> list[Definition]:
```

doc.py (line 526)

```
[function] def collect(root: str, exts: set[str] | None, exclude_dirs: set[str],
max_body_lines: int):
```

doc.py (line 550)

```
[function] def build_styles():
```

doc.py (line 587)

```
[function] def group_by_language(defs: list[Definition]):
```

doc.py (line 594)

```
[function] def path_line_label(d: Definition) -> str:
```

doc.py (line 598)

```
[function] def render_signature_section(story, styles, title: str, defs:
list[Definition], max_items: int):
```

doc.py (line 622)

```
[function] def render_fragments_section(story, styles, defs: list[Definition],
max_items: int):
```

doc.py (line 647)

```
[function] def render_markdown_section(story, styles, md_files: list[MarkdownFile]):
```

doc.py (line 680)

```
[function] def build_pdf(output_path: str, root: str, public_defs, internal_defs,
all_defs_for_fragments, ...
```

doc.py (line 734)

```
[function] def main():
```

Rust

```

drivers/audiodriver/src/jacklib.rs (line 4)
[struct] pub struct JackMgr {
drivers/audiodriver/src/jacklib.rs (line 14)
[function] pub fn tick(&mut self) {
drivers/audiodriver/src/jacklib.rs (line 30)
[function] pub fn set_amp(&mut self, on: bool) {
drivers/audiodriver/src/jacklib.rs (line 35)
[function] pub fn present(&self) -> bool {
drivers/audiodriver/src/jacklib.rs (line 40)
[function] pub fn amp_enabled(&self) -> bool {
drivers/audiodriver/src/lib.rs (line 16)
[function] pub fn init(nam_va: u64, bm_va: u64) -> bool {
drivers/audiodriver/src/lib.rs (line 20)
[function] pub fn play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool
{
drivers/audiodriver/src/lib.rs (line 24)
[function] pub fn capture(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) ->
bool {
drivers/audiodriver/src/lib.rs (line 28)
[function] pub fn stop() {
drivers/audiodriver/src/lib.rs (line 32)
[function] pub fn jack_present() -> bool {
drivers/audiodriver/src/lib.rs (line 36)
[function] pub fn set_amp(on: bool) {
drivers/audiodriver/src/lib.rs (line 40)
[function] pub fn position() -> u32 {
drivers/audiodriver/src/tone.rs (line 1)
[function] pub fn fill_square(buf: &mut [u8], periods: u32) {
driverspace/src/drivers/storage.rs (line 9)
[struct] pub struct StorageDrv {
driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 7)
[function] pub fn ds_spark_safe_add_32(a: u32, b: u32) -> u32;
driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 10)
[function] pub fn ds_spark_safe_array_index(index: u32, max_size: u32) -> u32;
driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 21)
[function] pub fn safe_add(a: u32, b: u32) -> u32 {
driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 28)
[function] pub fn ds_spark_buddy_get_buddy(addr: u64, size_exp: u64) -> u64;
driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 29)
[function] pub fn ds_spark_safe_dma_end(base: u64, len: u64) -> u64;
driverspace_workspace/lib/ds-formal-ffi/src/lib.rs (line 35)
[function] pub fn get_buddy(addr: u64, exp: u64) -> u64 {
driverspace_workspace/lib/ds-fw-audio/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
driverspace_workspace/lib/ds-fw-block/src/request.rs (line 7)
[enum] pub enum RequestState {
driverspace_workspace/lib/ds-fw-block/src/request.rs (line 15)
[struct] pub struct IoRequest {

```

```

driverspace_workspace/lib/ds-fw-block/src/request.rs (line 28)
[function] pub fn new(id: u64, lba: u64, block_count: u32, buffer: *mut u8, len:
usize, is_write: bool) -> Self {
driverspace_workspace/lib/ds-fw-block/src/request.rs (line 42)
[function] pub fn state(&self) -> RequestState {
driverspace_workspace/lib/ds-fw-block/src/request.rs (line 52)
[function] pub fn mark_in_progress(&self) {
driverspace_workspace/lib/ds-fw-block/src/request.rs (line 56)
[function] pub fn mark_completed(&self) {
driverspace_workspace/lib/ds-fw-block/src/request.rs (line 63)
[function] pub fn mark_failed(&self) {
driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 6)
[struct] pub struct BlockRequest {
driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 16)
[trait] pub trait BlockDevice {
driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 40)
[struct] pub struct BlockGeometry {
driverspace_workspace/lib/ds-fw-block/src/traits.rs (line 47)
[function] pub fn total_bytes(&self) -> u64 {
driverspace_workspace/lib/ds-fw-gpu/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
driverspace_workspace/lib/ds-fw-input/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
driverspace_workspace/lib/ds-ipc/src/client.rs (line 7)
[struct] pub struct ManagerClient {
driverspace_workspace/lib/ds-ipc/src/client.rs (line 16)
[function] pub fn request_mmio(&self, phys_base: u64, size: u64) -> Result<u64,
DsError> {
driverspace_workspace/lib/ds-ipc/src/client.rs (line 41)
[function] pub fn bind_irq(&self, irq_num: u32) -> Result<(), DsError> {
driverspace_workspace/lib/ds-ipc/src/dispatcher.rs (line 10)
[struct] pub struct MessageDispatcher {
driverspace_workspace/lib/ds-ipc/src/dispatcher.rs (line 21)
[function] pub fn register(&mut self, opcode: Opcode, handler: HandlerFn) ->
Result<(), DsError> {
driverspace_workspace/lib/ds-ipc/src/dispatcher.rs (line 30)
[function] pub fn dispatch(&self, sender: Handle, opcode_raw: u32, payload_ptr:
*const u8, payload_len: u32) -> Result<(), DsError> {
driverspace_workspace/lib/ds-log/src/formatter.rs (line 7)
[struct] pub struct LogBuffer {
driverspace_workspace/lib/ds-log/src/formatter.rs (line 20)
[function] pub fn as_bytes(&self) -> &[u8] {
driverspace_workspace/lib/ds-log/src/formatter.rs (line 24)
[function] pub fn reset(&mut self) {
driverspace_workspace/lib/ds-log/src/levels.rs (line 5)
[enum] pub enum LogLevel {
driverspace_workspace/lib/ds-log/src/lib.rs (line 14)
[struct] pub struct EndpointRef;
driverspace_workspace/lib/ds-log/src/lib.rs (line 19)
[function] pub fn is_valid(&self) -> bool {

```

```

driverspace_workspace/lib/ds-log/src/lib.rs (line 24)
[function] pub fn handle(&self) -> u32 {
driverspace_workspace/lib/ds-log/src/lib.rs (line 28)
[function] pub fn set(&self, handle: Handle) {
driverspace_workspace/lib/ds-log/src/transport.rs (line 19)
[function] pub fn emit_log(level: LogLevel, module: &str, buf: &LogBuffer) {
driverspace_workspace/lib/ds-mem/src/alloc/buddy.rs (line 5)
[struct] pub struct BuddyAllocator {
driverspace_workspace/lib/ds-mem/src/alloc/buddy.rs (line 20)
[function] pub fn alloc_pages(&mut self, order: usize) -> Option<usize> {
driverspace_workspace/lib/ds-mem/src/alloc/buddy.rs (line 28)
[function] pub fn free_pages(&mut self, addr: usize, order: usize) {
driverspace_workspace/lib/ds-mem/src/alloc/heap.rs (line 6)
[struct] pub struct DriverHeap {
driverspace_workspace/lib/ds-mem/src/alloc/heap.rs (line 18)
[function] pub fn init(&mut self, base_addr: usize, total_pages: usize) {
driverspace_workspace/lib/ds-mem/src/alloc/slab.rs (line 1)
[struct] pub struct SlabAllocator {
driverspace_workspace/lib/ds-mem/src/alloc/slab.rs (line 14)
[function] pub fn alloc(&mut self) -> Option<usize> {
driverspace_workspace/lib/ds-mem/src/alloc/slab.rs (line 19)
[function] pub fn free(&mut self, ptr: usize) {
driverspace_workspace/lib/ds-mem/src/dma/buffer.rs (line 10)
[struct] pub struct DmaFlags: u32 {
driverspace_workspace/lib/ds-mem/src/dma/buffer.rs (line 17)
[struct] pub struct DmaBuffer {
driverspace_workspace/lib/ds-mem/src/dma/buffer.rs (line 25)
[function] pub fn allocate(manager_ep: Handle, size: u64, flags: DmaFlags) ->
Result<Self, DsError> {
driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 8)
[struct] pub struct MmioRegion {
driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 16)
[function] pub fn map(manager_ep: Handle, phys_base: u64, size: u64) -> Result<Self,
DsError> {
driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 67)
[struct] pub struct Volatile<T>(pub T);
driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 71)
[function] pub fn read(&self) -> T {
driverspace_workspace/lib/ds-mem/src/dma/mapping.rs (line 76)
[function] pub fn write(&mut self, value: T) {
driverspace_workspace/lib/ds-mem/src/vmm.rs (line 6)
[function] pub fn map_memory(
driverspace_workspace/lib/ds-mem/src/vmm.rs (line 33)
[function] pub fn unmap_memory(manager_ep: Handle, virt_addr: u64, size: u64) ->
Result<(), DsError> {
driverspace_workspace/lib/kapi-abi/src/errors.rs (line 7)
[enum] pub enum DsError {
driverspace_workspace/lib/kapi-abi/src/errors.rs (line 44)
[function] pub fn from_u32(val: u32) -> Self {

```

```

driverspace_workspace/lib/kapi-abi/src/opcodes.rs (line 6)
[enum] pub enum Opcode {
driverspace_workspace/lib/kapi-abi/src/opcodes.rs (line 47)
[function] pub fn from_u32(val: u32) -> Option<Self> {
driverspace_workspace/lib/kapi-abi/src/payloads/dev.rs (line 10)
[enum] pub enum BusType {
driverspace_workspace/lib/kapi-abi/src/payloads/dev.rs (line 22)
[struct] pub struct DeviceFlags: u32 {
driverspace_workspace/lib/kapi-abi/src/payloads/dev.rs (line 38)
[struct] pub struct DevAttachPayload {
driverspace_workspace/lib/kapi-abi/src/payloads/irq.rs (line 6)
[struct] pub struct IrqBindPayload {
driverspace_workspace/lib/kapi-abi/src/payloads/mem.rs (line 8)
[struct] pub struct MmioMapPayload {
driverspace_workspace/lib/kapi-abi/src/payloads/mem.rs (line 25)
[struct] pub struct DmaAllocPayload {
driverspace_workspace/lib/kapi-abi/src/payloads/sys.rs (line 9)
[struct] pub struct LogPayload {
driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 8)
[struct] pub struct Handle(pub u32);
driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 14)
[function] pub fn is_valid(&self) -> bool {
driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 23)
[struct] pub struct CapId(pub u64);
driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 28)
[struct] pub struct PhysAddr(pub u64);
driverspace_workspace/lib/kapi-abi/src/primitives.rs (line 33)
[struct] pub struct VirtAddr(pub u64);
driverspace_workspace/lib/kapi-syscall/src/lib.rs (line 40)
[function] pub fn sys_ipc_send(
driverspace_workspace/lib/kapi-syscall/src/lib.rs (line 50)
[function] pub fn sys_ipc_call(
driverspacelib/src/abi.rs (line 53)
[enum] pub enum DsCmd {
driverspacelib/src/abi.rs (line 83)
[struct] pub struct DsRing {
driverspacelib/src/abi.rs (line 94)
[struct] pub struct DsMsg {
driverspacelib/src/abi.rs (line 110)
[struct] pub struct DsInitParams {
driverspacelib/src/abi.rs (line 123)
[enum] pub enum DsError {
driverspacelib/src/audio.rs (line 4)
[struct] pub struct AudioBars {
driverspacelib/src/audio.rs (line 9)
[function] pub fn info_req() -> u64 {
driverspacelib/src/audio.rs (line 13)
[function] pub fn info_take(id: u64) -> Option<AudioBars> {

```

```

driverspacelib/src/audio.rs (line 23)
[function] pub fn page_phys_req(va: u64) -> u64 {
driverspacelib/src/audio.rs (line 27)
[function] pub fn page_phys_take(id: u64) -> Option<u64> {
driverspacelib/src/driver.rs (line 6)
[struct] pub struct DeviceInfo {
driverspacelib/src/driver.rs (line 14)
[trait] pub trait Driver {
driverspacelib/src/input.rs (line 4)
[function] pub fn key_req() -> u64 {
driverspacelib/src/input.rs (line 8)
[function] pub fn key_take(id: u64) -> Option<u8> {
driverspacelib/src/jack.rs (line 4)
[struct] pub struct JackInfo {
driverspacelib/src/jack.rs (line 9)
[function] pub fn query_req() -> u64 {
driverspacelib/src/jack.rs (line 13)
[function] pub fn query_take(id: u64) -> Option<JackInfo> {
driverspacelib/src/jack.rs (line 22)
[function] pub fn set_amp(on: bool) {
driverspacelib/src/jack.rs (line 26)
[function] pub fn play(va: u64, len: u32) -> u64 {
driverspacelib/src/jack.rs (line 30)
[function] pub fn stop() {
driverspacelib/src/log.rs (line 10)
[function] pub fn ds_log(msg: &str) {
driverspacelib/src/log.rs (line 15)
[function] pub fn ds_log_raw(ptr: *const u8, len: usize) {
driverspacelib/src/runtime.rs (line 38)
[function] pub fn init_once(params_va: u64) {
driverspacelib/src/runtime.rs (line 51)
[function] pub fn register<D: Driver>(drv: &mut D) {
driverspacelib/src/runtime.rs (line 94)
[function] pub fn request(cmd: DsCmd, a0: u64, a1: u64, a2: u64) -> u64 {
driverspacelib/src/runtime.rs (line 99)
[function] pub fn request_raw(cmd: u32, a0: u64, a1: u64, a2: u64) -> u64 {
driverspacelib/src/runtime.rs (line 104)
[function] pub fn tick() {
driverspacelib/src/runtime.rs (line 128)
[function] pub fn take_resp(id: u64) -> Option<DsMsg> {
driverspacelib/src/svc.rs (line 4)
[function] pub fn call(class: u32, op: u32, a0: u64, a1: u64, a2: u64) -> u64 {
driverspacelib/src/svc.rs (line 8)
[function] pub fn take(id: u64) -> Option<DsMsg> {
driverspacelib/src/video.rs (line 4)
[struct] pub struct FbInfo {
driverspacelib/src/video.rs (line 11)
[function] pub fn fb_info_req() -> u64 {

```

```

driverspacelib/src/video.rs (line 15)
[function] pub fn fb_info_take(id: u64) -> Option<FbInfo> {
driverspacelib/src/video.rs (line 26)
[function] pub fn takeover() -> u64 {
driverspacelib/src/video.rs (line 30)
[function] pub fn release() -> u64 {
kernel_Workspace/C_base/kc-abi/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/Odin_base/odin-abi-bridge/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/Odin_base/odin-abi/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/base/kw-C-abi/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/base/kw-libs/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/base/kw-odin-abi/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/core/gluecore/src/lib.rs (line 16)
[function] pub fn alloc_frame() -> Result<PhysAddr, CoreError> {
kernel_Workspace/core/gluecore/src/lib.rs (line 23)
[function] pub fn free_frame(addr: PhysAddr) -> Result<(), CoreError> {
kernel_Workspace/core/kstd_alloc/src/heap.rs (line 13)
[struct] pub struct KernelHeap;
kernel_Workspace/core/kstd_alloc/src/heap.rs (line 45)
[function] pub fn alloc_zeroed(size: usize) -> Result<*mut u8, Status> {
kernel_Workspace/core/kstd_alloc/src/heap.rs (line 50)
[function] pub fn alloc_array(n: usize, elem_size: usize) -> Result<*mut u8, Status>
{
kernel_Workspace/core/kstd_alloc/src/lib.rs (line 14)
[struct] pub struct KernelAllocator;
kernel_Workspace/core/kstd_alloc/src/omm.rs (line 5)
[trait] pub trait OomHandler: Fn() -> ! + Send + Sync {}
kernel_Workspace/core/kstd_alloc/src/omm.rs (line 10)
[function] pub fn set_handler(handler: &'static dyn OomHandler) {
kernel_Workspace/core/kstd_alloc/src/omm.rs (line 14)
[function] pub fn invoke() -> ! {
kernel_Workspace/core/kstd_alloc/src/page.rs (line 9)
[function] pub fn alloc_pages(order: usize) -> Result<VirtAddr, Status> {
kernel_Workspace/core/kstd_alloc/src/page.rs (line 18)
[function] pub fn free_pages(addr: VirtAddr, order: usize) {
kernel_Workspace/core/kstd_alloc/src/page.rs (line 23)
[function] pub fn alloc_single_page() -> Result<VirtAddr, Status> {
kernel_Workspace/core/kstd_alloc/src/page.rs (line 27)
[function] pub fn free_single_page(addr: VirtAddr) {
kernel_Workspace/core/kstd_alloc/src/page.rs (line 31)
[function] pub fn pages_for_size(size: usize) -> usize {
kernel_Workspace/core/kstd_alloc/src/slab.rs (line 11)
[struct] pub struct SlabCache {

```

```

kernel_Workspace/core/kstd_alloc/src/slab.rs (line 19)
[function] pub fn new(name: &str, obj_size: usize, align: usize) -> Result<Self,
Status> {
kernel_Workspace/core/kstd_alloc/src/traits.rs (line 3)
[trait] pub trait Allocator {
kernel_Workspace/core/kstd_alloc/src/traits.rs (line 19)
[struct] pub struct Layout {
kernel_Workspace/core/kstd_alloc/src/traits.rs (line 41)
[function] pub fn padded(&self) -> usize {
kernel_Workspace/core/kstd_base/src/errors.rs (line 3)
[enum] pub enum Status {
kernel_Workspace/core/kstd_base/src/intrinsics.rs (line 12)
[function] pub fn memory_barrier() {
kernel_Workspace/core/kstd_base/src/markers.rs (line 3)
[struct] pub struct SendSync<T>(PhantomData<T>);
kernel_Workspace/core/kstd_base/src/markers.rs (line 7)
[struct] pub struct Immutable<T>(PhantomData<T>);
kernel_Workspace/core/kstd_base/src/markers.rs (line 14)
[struct] pub struct Unpin<T>(PhantomData<T>);
kernel_Workspace/core/kstd_base/src/primitives.rs (line 7)
[enum] pub enum Ring {
kernel_Workspace/core/kstd_base/src/primitives.rs (line 15)
[struct] pub struct ProtFlags(pub u32);
kernel_Workspace/core/kstd_base/src/primitives.rs (line 46)
[struct] pub struct MapFlags(pub u32);
kernel_Workspace/core/kstd_base/src/primitives.rs (line 58)
[struct] pub struct IrqFlags(pub u32);
kernel_Workspace/core/kstd_base/src/trails.rs (line 3)
[trait] pub trait PageAlign {
kernel_Workspace/core/kstd_base/src/trails.rs (line 43)
[trait] pub trait IoPort {
kernel_Workspace/core/kstd_base/src/trails.rs (line 52)
[trait] pub trait MmioRegion {
kernel_Workspace/core/kstd_base/src/types.rs (line 3)
[struct] pub struct PhysAddr(pub u64);
kernel_Workspace/core/kstd_base/src/types.rs (line 7)
[struct] pub struct VirtAddr(pub u64);
kernel_Workspace/core/kstd_base/src/types.rs (line 11)
[struct] pub struct Handle(pub u32);
kernel_Workspace/core/kstd_base/src/types.rs (line 30)
[struct] pub struct CapId(pub u64);
kernel_Workspace/core/kstd_base/src/types.rs (line 34)
[struct] pub struct WorldId(pub u32);
kernel_Workspace/core/kstd_base/src/types.rs (line 38)
[struct] pub struct TaskId(pub u64);
kernel_Workspace/core/kstd_base/src/types.rs (line 42)
[struct] pub struct Size(pub usize);
kernel_Workspace/core/kstd_core/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {

```

```

kernel_Workspace/core/kstd_data/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/core/kstd_io/src/buffer.rs (line 1)
[struct] pub struct RingBuffer<const N: usize> {
kernel_Workspace/core/kstd_io/src/buffer.rs (line 12)
[function] pub fn push(&mut self, b: u8) -> bool {
kernel_Workspace/core/kstd_io/src/buffer.rs (line 20)
[function] pub fn pop(&mut self) -> Option<u8> {
kernel_Workspace/core/kstd_io/src/buffer.rs (line 27)
[function] pub fn is_empty(&self) -> bool { self.head == self.tail }
kernel_Workspace/core/kstd_io/src/buffer.rs (line 28)
[function] pub fn is_full(&self) -> bool { (self.head + 1) % N == self.tail }
kernel_Workspace/core/kstd_io/src/buffer.rs (line 29)
[function] pub fn clear(&mut self) { self.head = 0; self.tail = 0; }
kernel_Workspace/core/kstd_io/src/console.rs (line 8)
[struct] pub struct Serial;
kernel_Workspace/core/kstd_io/src/console.rs (line 24)
[struct] pub struct VgaConsole;
kernel_Workspace/core/kstd_io/src/input.rs (line 7)
[function] pub fn raw_scancode() -> Option<u32> {
kernel_Workspace/core/kstd_io/src/input.rs (line 12)
[function] pub fn cooked_char() -> Option<u8> {
kernel_Workspace/core/kstd_io/src/input.rs (line 17)
[function] pub fn buffered_byte() -> Option<u8> {
kernel_Workspace/core/kstd_io/src/input.rs (line 22)
[struct] pub struct RawInput;
kernel_Workspace/core/kstd_io/src/input.rs (line 29)
[struct] pub struct CookedInput;
kernel_Workspace/core/kstd_io/src/traits.rs (line 1)
[trait] pub trait Write {
kernel_Workspace/core/kstd_io/src/traits.rs (line 22)
[trait] pub trait Read {
kernel_Workspace/core/linux_abi/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/core/linux_abi_com/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/core/linux_abi_driverspace/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/core/windows-com/src/lib.rs (line 2)
[function] pub fn add(left: u64, right: u64) -> u64 {
kernel_Workspace/kernel/src/arch/mod.rs (line 13)
[function] pub fn init() {
kernel_Workspace/kernel/src/arch/mod.rs (line 18)
[function] pub fn hlt_loop() -> ! {
kernel_Workspace/kernel/src/arch/mod.rs (line 23)
[function] pub fn now() -> u64 {
kernel_Workspace/kernel/src/arch/mod.rs (line 28)
[function] pub fn current_cpu() -> usize {

```

```

kernel_Workspace/kernel/src/arch/riscv64.rs (line 20)
[function] pub fn heap_init() {
kernel_Workspace/kernel/src/arch/riscv64.rs (line 27)
[struct] pub struct KernelHeap;
kernel_Workspace/kernel/src/arch/riscv64.rs (line 51)
[function] pub fn now() -> u64 {
kernel_Workspace/kernel/src/arch/riscv64.rs (line 66)
[function] pub fn init() {
kernel_Workspace/kernel/src/arch/riscv64.rs (line 74)
[function] pub fn current_cpu() -> usize {
kernel_Workspace/kernel/src/arch/riscv64.rs (line 78)
[function] pub fn hlt_loop() -> ! {
kernel_Workspace/kernel/src/arch/riscv64.rs (line 86)
[function] pub fn early_uart(s: &str) {
kernel_Workspace/kernel/src/arch/x86_64.rs (line 10)
[function] pub fn init() {
kernel_Workspace/kernel/src/arch/x86_64.rs (line 18)
[function] pub fn hlt_loop() -> ! {
kernel_Workspace/kernel/src/arch/x86_64.rs (line 24)
[function] pub fn current_cpu() -> usize {
kernel_Workspace/kernel/src/arch/x86_64.rs (line 36)
[function] pub fn now() -> u64 {
kernel_Workspace/kernel/src/audio/jack.rs (line 6)
[function] pub fn init(base: u32) -> bool {
kernel_Workspace/kernel/src/audio/jack.rs (line 10)
[function] pub fn poll_jack() -> bool {
kernel_Workspace/kernel/src/audio/jack.rs (line 23)
[function] pub fn query() -> u32 {
kernel_Workspace/kernel/src/audio/jack.rs (line 29)
[function] pub fn set_amp(on: bool) {
kernel_Workspace/kernel/src/audio/jack.rs (line 34)
[function] pub fn play_phys(phys: u64, len: u32) -> bool {
kernel_Workspace/kernel/src/audio/jack.rs (line 38)
[function] pub fn stop() {
kernel_Workspace/kernel/src/battery/aut.rs (line 4)
[function] pub fn authorize(ring: u8, op: u8) -> bool {
kernel_Workspace/kernel/src/battery/bridge.rs (line 9)
[function] pub fn batt_call(op: u32, m: &DsMsg, r: &mut DsMsg, ring: u8) -> i32 {
kernel_Workspace/kernel/src/battery/init.rs (line 5)
[function] pub fn init() -> bool {
kernel_Workspace/kernel/src/bluetooth/aut.rs (line 8)
[function] pub fn authorize(ring: u8, op: u32) -> bool {
kernel_Workspace/kernel/src/bluetooth/bridge.rs (line 20)
[function] pub fn bt_call(op: u32, m: &DsMsg, r: &mut DsMsg, ring: u8) -> i32 {
kernel_Workspace/kernel/src/bluetooth/init.rs (line 6)
[function] pub fn init() -> bool {
kernel_Workspace/kernel/src/bluetooth/init.rs (line 10)
[function] pub fn ready() -> bool {

```

```

kernel_Workspace/kernel/src/camera/aut.rs (line 6)
[function] pub fn authorize(ring: u8, op: u8) -> bool {
kernel_Workspace/kernel/src/camera/bridge.rs (line 15)
[function] pub fn cam_call(op: u32, m: &DsMsg, r: &mut DsMsg, ring: u8) -> i32 {
kernel_Workspace/kernel/src/camera/init.rs (line 5)
[function] pub fn init() -> bool {
kernel_Workspace/kernel/src/caps/audit.rs (line 6)
[enum] pub enum EventKind {
kernel_Workspace/kernel/src/caps/audit.rs (line 16)
[struct] pub struct AuditEvent {
kernel_Workspace/kernel/src/caps/audit.rs (line 41)
[function] pub fn init_audit_log() -> Result<(), &'static str> {
kernel_Workspace/kernel/src/caps/audit.rs (line 73)
[function] pub fn log_check(world: u32, cap: Capability, ok: bool) {
kernel_Workspace/kernel/src/caps/audit.rs (line 78)
[function] pub fn log_grant(granter: u32, target: u32, cap: Capability, ok: bool) {
kernel_Workspace/kernel/src/caps/audit.rs (line 84)
[function] pub fn log_revoke(world: u32, cap: Capability, ok: bool) {
kernel_Workspace/kernel/src/caps/audit.rs (line 90)
[function] pub fn log_register(world: u32) {
kernel_Workspace/kernel/src/caps/audit.rs (line 94)
[function] pub fn log_unregister(world: u32) {
kernel_Workspace/kernel/src/caps/audit.rs (line 98)
[function] pub fn count() -> usize {
kernel_Workspace/kernel/src/caps/audit.rs (line 102)
[function] pub fn recent(n: usize) -> Vec<AuditEvent> {
kernel_Workspace/kernel/src/caps/audit.rs (line 119)
[function] pub fn by_world(world: u32, limit: usize) -> Vec<AuditEvent> {
kernel_Workspace/kernel/src/caps/audit.rs (line 126)
[function] pub fn by_kind(kind: EventKind, limit: usize) -> Vec<AuditEvent> {
kernel_Workspace/kernel/src/caps/audit.rs (line 133)
[function] pub fn deny_count() -> usize {
kernel_Workspace/kernel/src/caps/check.rs (line 10)
[function] pub fn kernel_world_id() -> u32 {
kernel_Workspace/kernel/src/caps/check.rs (line 14)
[function] pub fn set_kernel_world_id(id: u32) {
kernel_Workspace/kernel/src/caps/check.rs (line 18)
[function] pub fn current_world_id_pub() -> u32 {
kernel_Workspace/kernel/src/caps/check.rs (line 22)
[function] pub fn set_current_world(world_id: u32) {
kernel_Workspace/kernel/src/caps/check.rs (line 30)
[function] pub fn has_cap(cap: Capability) -> bool {
kernel_Workspace/kernel/src/caps/check.rs (line 35)
[function] pub fn world_has_cap(world_id: u32, cap: Capability) -> bool {
kernel_Workspace/kernel/src/caps/check.rs (line 39)
[function] pub fn require_cap(cap: Capability) -> CapResult<()> {
kernel_Workspace/kernel/src/caps/check.rs (line 54)
[function] pub fn require_world_cap(world_id: u32, cap: Capability) -> CapResult<()>
{

```

```

kernel_Workspace/kernel/src/caps/check.rs (line 67)
[function] pub fn require_caps(caps: &[Capability]) -> CapResult<()> {
kernel_Workspace/kernel/src/caps/check.rs (line 74)
[function] pub fn with_cap<T, F>(cap: Capability, f: F) -> CapResult<T>
kernel_Workspace/kernel/src/caps/check.rs (line 82)
[function] pub fn with_caps<T, F>(caps: &[Capability], f: F) -> CapResult<T>
kernel_Workspace/kernel/src/caps/check.rs (line 104)
[function] pub fn assert_cap(cap: Capability) {
kernel_Workspace/kernel/src/caps/check.rs (line 111)
[function] pub fn fast_check(cap: Capability) -> bool {
kernel_Workspace/kernel/src/caps/check.rs (line 115)
[function] pub fn if_cap<T, F>(cap: Capability, f: F, default: T) -> T
kernel_Workspace/kernel/src/caps/check.rs (line 126)
[struct] pub struct TemporaryCaps {
kernel_Workspace/kernel/src/caps/check.rs (line 132)
[function] pub fn enter(extra: CapabilitySet) -> CapResult<Self> {
kernel_Workspace/kernel/src/caps/defaults.rs (line 6)
[function] pub fn default_for_ring(ring: u8) -> CapabilitySet {
kernel_Workspace/kernel/src/caps/defaults.rs (line 15)
[function] pub fn install_defaults() -> Result<(), &'static str> {
kernel_Workspace/kernel/src/caps/defaults.rs (line 22)
[function] pub fn register_for_ring(ring: u8, parent: Option<u32>) -> Result<u32,
&'static str> {
kernel_Workspace/kernel/src/caps/grant.rs (line 18)
[function] pub fn grant_cap(granter: u32, target: u32, cap: Capability) ->
CapResult<()> {
kernel_Workspace/kernel/src/caps/grant.rs (line 31)
[function] pub fn delegate_caps(granter: u32, target: u32, caps: CapabilitySet) ->
CapResult<()> {
kernel_Workspace/kernel/src/caps/grant.rs (line 38)
[function] pub fn inherit_set(parent_world: u32) -> CapabilitySet {
kernel_Workspace/kernel/src/caps/grant.rs (line 51)
[function] pub fn spawn_child(parent_world: u32) -> Result<u32, &'static str> {
kernel_Workspace/kernel/src/caps/grant.rs (line 78)
[function] pub fn grant_temporary(granter: u32, target: u32, cap: Capability,
kernel_Workspace/kernel/src/caps/grant.rs (line 97)
[function] pub fn prune_expired() {
kernel_Workspace/kernel/src/caps/grant.rs (line 110)
[function] pub fn is_temporary(world_id: u32, cap: Capability) -> bool {
kernel_Workspace/kernel/src/caps/hierarchy.rs (line 5)
[function] pub fn parent(cap: Capability) -> Option<Capability> {
kernel_Workspace/kernel/src/caps/hierarchy.rs (line 47)
[function] pub fn implies(held: Capability, required: Capability) -> bool {
kernel_Workspace/kernel/src/caps/hierarchy.rs (line 62)
[function] pub fn set_implies(held: CapabilitySet, required: Capability) -> bool {
kernel_Workspace/kernel/src/caps/hierarchy.rs (line 71)
[function] pub fn expand_hierarchy(set: CapabilitySet) -> CapabilitySet {
kernel_Workspace/kernel/src/caps/hierarchy.rs (line 85)
[function] pub fn path_to_root(cap: Capability) -> Vec<Capability> {

```

```

kernel_Workspace/kernel/src/caps/hierarchy.rs (line 96)
[function] pub fn depth(cap: Capability) -> usize {
kernel_Workspace/kernel/src/caps/hierarchy.rs (line 106)
[function] pub fn subtree(root: Capability) -> Vec<Capability> {
kernel_Workspace/kernel/src/caps/hierarchy.rs (line 123)
[function] pub fn can_delegate(holder: CapabilitySet, candidate: Capability) -> bool
{
kernel_Workspace/kernel/src/caps/mod.rs (line 31)
[function] pub fn init() -> Result<(), &'static str> {
kernel_Workspace/kernel/src/caps/mod.rs (line 38)
[function] pub fn snapshot() -> SystemSnapshot {
kernel_Workspace/kernel/src/caps/mod.rs (line 48)
[struct] pub struct SystemSnapshot {
kernel_Workspace/kernel/src/caps/mod.rs (line 55)
[function] pub fn self_test() -> crate::testing::TestResult {
kernel_Workspace/kernel/src/caps/policy.rs (line 9)
[function] pub fn set_hook(h: PolicyHook) {
kernel_Workspace/kernel/src/caps/policy.rs (line 23)
[function] pub fn enforce(world: u32, cap: Capability) -> CapResult<()> {
kernel_Workspace/kernel/src/caps/policy.rs (line 35)
[function] pub fn enforce_self(cap: Capability) -> CapResult<()> {
kernel_Workspace/kernel/src/caps/policy.rs (line 40)
[function] pub fn allowed(world: u32, cap: Capability) -> bool {
kernel_Workspace/kernel/src/caps/revoke.rs (line 7)
[function] pub fn revoke_from_world(world_id: u32, cap: Capability) -> CapResult<()>
{
kernel_Workspace/kernel/src/caps/revoke.rs (line 15)
[function] pub fn revoke_subtree(world_id: u32, cap: Capability) -> CapResult<()> {
kernel_Workspace/kernel/src/caps/revoke.rs (line 23)
[function] pub fn revoke_global(cap: Capability) {
kernel_Workspace/kernel/src/caps/revoke.rs (line 28)
[function] pub fn restore_global(cap: Capability) {
kernel_Workspace/kernel/src/caps/revoke.rs (line 32)
[function] pub fn revoked_list() -> Vec<Capability> {
kernel_Workspace/kernel/src/caps/revoke.rs (line 43)
[function] pub fn is_globally_revoked(cap: Capability) -> bool {
kernel_Workspace/kernel/src/caps/revoke.rs (line 47)
[function] pub fn lockdown(world_id: u32) {
kernel_Workspace/kernel/src/caps/sets.rs (line 4)
[struct] pub struct CapSetBuilder {
kernel_Workspace/kernel/src/caps/sets.rs (line 9)
[function] pub fn new() -> Self {
kernel_Workspace/kernel/src/caps/sets.rs (line 13)
[function] pub fn empty() -> Self {
kernel_Workspace/kernel/src/caps/sets.rs (line 17)
[function] pub fn add(mut self, cap: Capability) -> Self {
kernel_Workspace/kernel/src/caps/sets.rs (line 22)
[function] pub fn add_many(mut self, caps: &[Capability]) -> Self {

```

```

kernel_Workspace/kernel/src/caps/sets.rs (line 29)
[function] pub fn add_category(mut self, cat: super::types::CapCategory) -> Self {
kernel_Workspace/kernel/src/caps/sets.rs (line 38)
[function] pub fn remove(mut self, cap: Capability) -> Self {
kernel_Workspace/kernel/src/caps/sets.rs (line 43)
[function] pub fn with_hierarchy(self) -> Self {
kernel_Workspace/kernel/src/caps/sets.rs (line 47)
[function] pub fn build(self) -> CapabilitySet {
kernel_Workspace/kernel/src/caps/sets.rs (line 55)
[function] pub fn minimal_user() -> CapabilitySet {
kernel_Workspace/kernel/src/caps/sets.rs (line 67)
[function] pub fn standard_user() -> CapabilitySet {
kernel_Workspace/kernel/src/caps/sets.rs (line 84)
[function] pub fn privileged_user() -> CapabilitySet {
kernel_Workspace/kernel/src/caps/sets.rs (line 98)
[function] pub fn driver() -> CapabilitySet {
kernel_Workspace/kernel/src/caps/sets.rs (line 111)
[function] pub fn kernel() -> CapabilitySet {
kernel_Workspace/kernel/src/caps/sets.rs (line 115)
[function] pub fn sandbox() -> CapabilitySet {
kernel_Workspace/kernel/src/caps/sets.rs (line 125)
[function] pub fn validate_hierarchy(set: CapabilitySet) -> Result<(), &'static str>
{
kernel_Workspace/kernel/src/caps/sets.rs (line 141)
[function] pub fn effective(set: CapabilitySet) -> CapabilitySet {
kernel_Workspace/kernel/src/caps/sets.rs (line 145)
[function] pub fn effective_diff(a: CapabilitySet, b: CapabilitySet) ->
CapabilitySet {
kernel_Workspace/kernel/src/caps/sets.rs (line 151)
[function] pub fn is_subset_with_hierarchy(subset: CapabilitySet, superset:
CapabilitySet) -> bool {
kernel_Workspace/kernel/src/caps/store.rs (line 9)
[struct] pub struct WorldCaps {
kernel_Workspace/kernel/src/caps/store.rs (line 28)
[function] pub fn init_store() -> Result<(), &'static str> {
kernel_Workspace/kernel/src/caps/store.rs (line 36)
[function] pub fn global_caps() -> CapabilitySet {
kernel_Workspace/kernel/src/caps/store.rs (line 41)
[function] pub fn register_world(parent_world: Option<u32>, initial: CapabilitySet)
-> Result<u32, &'static str> {
kernel_Workspace/kernel/src/caps/store.rs (line 82)
[function] pub fn get_world_caps(world_id: u32) -> Result<CapabilitySet, &'static
str> {
kernel_Workspace/kernel/src/caps/store.rs (line 88)
[function] pub fn set_world_caps(world_id: u32, new_caps: CapabilitySet) ->
Result<(), &'static str> {
kernel_Workspace/kernel/src/caps/store.rs (line 115)
[function] pub fn add_world_cap(world_id: u32, cap: Capability) -> Result<(),
&'static str> {

```

```

kernel_Workspace/kernel/src/caps/store.rs (line 120)
[function] pub fn remove_world_cap(world_id: u32, cap: Capability) -> Result<(),
&'static str> {
kernel_Workspace/kernel/src/caps/store.rs (line 138)
[function] pub fn unregister_world(world_id: u32) -> Result<(), &'static str> {
kernel_Workspace/kernel/src/caps/store.rs (line 153)
[function] pub fn world_has_cap(world_id: u32, cap: Capability) -> bool {
kernel_Workspace/kernel/src/caps/store.rs (line 163)
[function] pub fn world_count() -> usize {
kernel_Workspace/kernel/src/caps/store.rs (line 168)
[function] pub fn iter_worlds<F: FnMut(u32, CapabilitySet)>(mut f: F) {
kernel_Workspace/kernel/src/caps/store.rs (line 177)
[function] pub fn add_global_revoked(cap: Capability) {
kernel_Workspace/kernel/src/caps/store.rs (line 182)
[function] pub fn remove_global_revoked(cap: Capability) {
kernel_Workspace/kernel/src/caps/syscalls.rs (line 17)
[function] pub fn cap_syscall(num: u64, a0: u64, a1: u64, a2: u64) -> u64 {
kernel_Workspace/kernel/src/caps/types.rs (line 10)
[enum] pub enum Capability {
kernel_Workspace/kernel/src/caps/types.rs (line 99)
[function] pub fn iter_all() -> impl Iterator<Item = Capability> {
kernel_Workspace/kernel/src/caps/types.rs (line 112)
[function] pub fn category(self) -> CapCategory {
kernel_Workspace/kernel/src/caps/types.rs (line 127)
[enum] pub enum CapCategory {
kernel_Workspace/kernel/src/caps/types.rs (line 154)
[struct] pub struct CapabilitySet {
kernel_Workspace/kernel/src/caps/types.rs (line 215)
[function] pub fn iter(self) -> impl Iterator<Item = Capability> {
kernel_Workspace/kernel/src/caps/types.rs (line 234)
[struct] pub struct CapabilityError {
kernel_Workspace/kernel/src/cpu/acpi.rs (line 4)
[struct] pub struct CpuEntry {
kernel_Workspace/kernel/src/cpu/acpi.rs (line 9)
[struct] pub struct IoApic {
kernel_Workspace/kernel/src/cpu/acpi.rs (line 15)
[struct] pub struct MadtInfo {
kernel_Workspace/kernel/src/cpu/acpi.rs (line 21)
[struct] pub struct FadtInfo {
kernel_Workspace/kernel/src/cpu/acpi.rs (line 25)
[struct] pub struct RsdP {
kernel_Workspace/kernel/src/cpu/acpi.rs (line 57)
[function] pub fn find_rsdP(phys_offset: u64) -> Option<u64> {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 24)
[function] pub fn init(base_phys: u64) -> bool {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 46)
[function] pub fn is_x2apic() -> bool {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 50)
[function] pub fn read(reg: u32) -> u32 {

```

```

kernel_Workspace/kernel/src/cpu/lapic.rs (line 60)
[function] pub fn write(reg: u32, val: u32) {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 70)
[function] pub fn id() -> u32 {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 78)
[function] pub fn version() -> u32 {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 82)
[function] pub fn enable_bsp() {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 87)
[function] pub fn enable_ap() {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 93)
[function] pub fn eoi() {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 97)
[function] pub fn send_ipi(icr: u32, dest_apic_id: u32) {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 109)
[function] pub fn send_init_ipi() {
kernel_Workspace/kernel/src/cpu/lapic.rs (line 116)
[function] pub fn send_startup_ipi(apic_id: u32, vector: u8) {
kernel_Workspace/kernel/src/cpu/mod.rs (line 18)
[function] pub fn init_riscv() {
kernel_Workspace/kernel/src/cpu/mod.rs (line 24)
[function] pub fn total_cpus() -> u32 {
kernel_Workspace/kernel/src/cpu/mod.rs (line 29)
[function] pub fn total_cpus() -> u32 {
kernel_Workspace/kernel/src/cpu/riscv.rs (line 7)
[function] pub fn init() {
kernel_Workspace/kernel/src/cpu/riscv.rs (line 12)
[function] pub fn current_hart() -> u64 {
kernel_Workspace/kernel/src/cpu/riscv.rs (line 17)
[function] pub fn poweroff() -> ! {
kernel_Workspace/kernel/src/cpu/riscv.rs (line 22)
[function] pub fn reboot() -> ! {
kernel_Workspace/kernel/src/cpu/scheduler/arch/risc_v/context_switch.rs (line 639)
[function] pub fn get_switch_count() -> u64 {
kernel_Workspace/kernel/src/cpu/scheduler/arch/risc_v/context_switch.rs (line 643)
[function] pub fn get_fpu_switch_count() -> u64 {
kernel_Workspace/kernel/src/cpu/scheduler/arch/risc_v/context_switch.rs (line 647)
[function] pub fn enable_lazy_fpu() {
kernel_Workspace/kernel/src/cpu/scheduler/arch/risc_v/context_switch.rs (line 651)
[function] pub fn disable_lazy_fpu() {
kernel_Workspace/kernel/src/cpu/scheduler/arch/risc_v/context_switch.rs (line 655)
[function] pub fn is_lazy_fpu_enabled() -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/arch/risc_v/context_switch.rs (line 692)
[function] pub fn satp_mode_name(satp: u64) -> &'static str {
kernel_Workspace/kernel/src/cpu/scheduler/arch/risc_v/context_switch.rs (line 703)
[function] pub fn satp_asid(satp: u64) -> u64 {
kernel_Workspace/kernel/src/cpu/scheduler/arch/risc_v/context_switch.rs (line 707)
[function] pub fn satp_ppn(satp: u64) -> u64 {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/arch/x86_64/context_swich.rs (line 731)
[function] pub fn get_switch_count() -> u64 {
kernel_Workspace/kernel/src/cpu/scheduler/arch/x86_64/context_swich.rs (line 735)
[function] pub fn get_fpu_switch_count() -> u64 {
kernel_Workspace/kernel/src/cpu/scheduler/arch/x86_64/context_swich.rs (line 739)
[function] pub fn enable_lazy_fpu() {
kernel_Workspace/kernel/src/cpu/scheduler/arch/x86_64/context_swich.rs (line 743)
[function] pub fn disable_lazy_fpu() {
kernel_Workspace/kernel/src/cpu/scheduler/arch/x86_64/context_swich.rs (line 747)
[function] pub fn is_lazy_fpu_enabled() -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 49)
[function] pub fn register_runqueue(cpu: u32, rq: *mut RunQueue) {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 55)
[function] pub fn register_apic_id(cpu: u32, apic_id: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 61)
[function] pub fn snapshot_registry() -> [*mut RunQueue; MAX_CPUS] {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 81)
[function] pub fn set_tsc_frequency(hz: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 85)
[function] pub fn tsc_frequency() -> u64 {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 89)
[function] pub fn booted_cpu_count() -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 183)
[function] pub fn now_ns() -> u64 {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 270)
[function] pub fn current_cpu_id() -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 338)
[struct] pub struct IdtEntry {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 353)
[function] pub fn set(&mut self, handler: usize, selector: u16, ist: u8, type_attr:
u8) {
kernel_Workspace/kernel/src/cpu/scheduler/arch_hooks.rs (line 363)
[function] pub fn is_present(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/class.rs (line 12)
[struct] pub struct SchedClassOps {
kernel_Workspace/kernel/src/cpu/scheduler/class.rs (line 336)
[function] pub fn class_of(class: SchedClass) -> &'static SchedClassOps {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 27)
[struct] pub struct Bitmap<const WORDS: usize> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 43)
[function] pub fn set(&mut self, bit: usize) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 50)
[function] pub fn clear(&mut self, bit: usize) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 57)
[function] pub fn toggle(&mut self, bit: usize) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 64)
[function] pub fn test(&self, bit: usize) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 68)
[function] pub fn set_range(&mut self, start: usize, end: usize) {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 77)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 81)
[function] pub fn is_full(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 85)
[function] pub fn weight(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 89)
[function] pub fn find_first_set(&self) -> Option<usize> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 98)
[function] pub fn find_first_zero(&self) -> Option<usize> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 111)
[function] pub fn find_next_set(&self, after: usize) -> Option<usize> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 126)
[function] pub fn and(&self, other: &Self) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 134)
[function] pub fn or(&self, other: &Self) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 142)
[function] pub fn xor(&self, other: &Self) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 150)
[function] pub fn andnot(&self, other: &Self) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 158)
[function] pub fn intersects(&self, other: &Self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 162)
[function] pub fn iter(&self) -> BitmapIter<'_> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 173)
[struct] pub struct BitmapIter<'a> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 200)
[struct] pub struct BitmapSlice<'a> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 206)
[function] pub fn new(words: &'a mut [u64], nbits: usize) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 211)
[function] pub fn capacity(&self) -> usize {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 215)
[function] pub fn set(&mut self, bit: usize) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 221)
[function] pub fn clear(&mut self, bit: usize) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 227)
[function] pub fn test(&self, bit: usize) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 231)
[function] pub fn weight(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 235)
[function] pub fn find_first_zero(&self) -> Option<usize> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 248)
[function] pub fn clear_all(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 263)
[struct] pub struct AtomicBitmap<const WORDS: usize> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 275)
[function] pub fn test(&self, bit: usize) -> bool {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 280)
[function] pub fn test_and_set(&self, bit: usize) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 289)
[function] pub fn test_and_clear(&self, bit: usize) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 297)
[function] pub fn weight(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 301)
[function] pub fn is_full(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/bitmap.rs (line 311)
[function] pub fn find_first_zero_and_set(&self) -> Option<usize> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 17)
[struct] pub struct AtomicCpuMask {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 29)
[function] pub fn set(&self, cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 37)
[function] pub fn clear(&self, cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 45)
[function] pub fn test(&self, cpu: u32) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 51)
[function] pub fn snapshot(&self) -> CpuMask {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 74)
[function] pub fn mark_possible(cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 79)
[function] pub fn mark_present(cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 84)
[function] pub fn mark_online(cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 90)
[function] pub fn mark_offline(cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 96)
[function] pub fn is_online(cpu: u32) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 101)
[function] pub fn online_mask() -> CpuMask {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 106)
[function] pub fn active_mask() -> CpuMask {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 111)
[function] pub fn set_topology(cpu: u32, pkg: u32, core: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 119)
[function] pub fn sibling_mask(cpu: u32) -> CpuMask {
kernel_Workspace/kernel/src/cpu/scheduler/collections/cpumask.rs (line 137)
[function] pub fn package_mask(cpu: u32) -> CpuMask {
kernel_Workspace/kernel/src/cpu/scheduler/collections/plist.rs (line 44)
[struct] pub struct PList {
kernel_Workspace/kernel/src/cpu/scheduler/collections/plist.rs (line 53)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/plist.rs (line 152)
[function] pub fn first(&self) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/collections/plist.rs (line 167)
[function] pub fn last(&self) -> *mut TaskStruct {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/collections/rbtree.rs (line 62)
[struct] pub struct RbTree {
kernel_Workspace/kernel/src/cpu/scheduler/collections/rt_array.rs (line 4)
[struct] pub struct RtArray {
kernel_Workspace/kernel/src/cpu/scheduler/collections/rt_array.rs (line 20)
[function] pub fn set_bit(&mut self, prio: usize) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/rt_array.rs (line 27)
[function] pub fn clear_bit(&mut self, prio: usize) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/rt_array.rs (line 41)
[function] pub fn highest_prio(&self) -> Option<usize> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/rt_array.rs (line 149)
[function] pub fn active_levels(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 15)
[struct] pub struct PidAllocator {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 31)
[function] pub fn alloc(&self) -> Option<TaskId> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 54)
[function] pub fn free(&self, pid: TaskId) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 64)
[function] pub fn reserve(&self, pid: TaskId) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 77)
[function] pub fn is_used(&self, pid: TaskId) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 85)
[function] pub fn count_allocated(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 94)
[struct] pub struct TaskTable {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 100)
[struct] pub struct RtFields {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 119)
[function] pub fn insert(&self, task: *mut TaskStruct) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 154)
[function] pub fn remove(&self, pid: TaskId) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 185)
[function] pub fn lookup_pid(&self, pid: TaskId) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 196)
[function] pub fn lookup_tgid(&self, tgid: TaskId) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 207)
[function] pub fn count(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 221)
[struct] pub struct TaskIterator<'a> {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 279)
[function] pub fn new(table: &'a TaskTable) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 299)
[struct] pub struct ThreadGroupIterator {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 334)
[struct] pub struct ChildIterator {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 358)
[struct] pub struct DescendantIterator {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 386)
[struct] pub struct ProcessGroup {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 397)
[function] pub fn new(pgid: TaskId, leader: *mut TaskStruct) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 427)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 443)
[struct] pub struct Session {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 453)
[function] pub fn new(sid: TaskId, leader: *mut TaskStruct) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 473)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 477)
[function] pub fn set_foreground(&self, pgid: TaskId) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 481)
[function] pub fn get_foreground(&self) -> TaskId {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 488)
[struct] pub struct UidTaskCount {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 498)
[function] pub fn inc(&self, uid: u32) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 503)
[function] pub fn dec(&self, uid: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 510)
[function] pub fn get(&self, uid: u32) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 516)
[enum] pub enum TaskRegistryError {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 524)
[struct] pub struct TaskRegistry {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 585)
[function] pub fn find_by_pid(&self, pid: TaskId) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 589)
[function] pub fn find_by_tgid(&self, tgid: TaskId) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 685)
[function] pub fn iter(&self) -> TaskIterator {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 700)
[function] pub fn set_init_task(&self, task: *mut TaskStruct) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 704)
[function] pub fn stats(&self) -> (u32, u64, u64) {
kernel_Workspace/kernel/src/cpu/scheduler/collections/task.rs (line 713)
[struct] pub struct PidNamespace {
kernel_Workspace/kernel/src/cpu/scheduler/core.rs (line 22)
[function] pub fn alloc_pid() -> Option<TaskId> {
kernel_Workspace/kernel/src/cpu/scheduler/core.rs (line 26)
[function] pub fn free_pid(pid: TaskId) {
kernel_Workspace/kernel/src/cpu/scheduler/core.rs (line 34)
[function] pub fn reserve_pid(pid: TaskId) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/core.rs (line 41)
[function] pub fn pid_in_use(pid: TaskId) -> bool {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/core.rs (line 219)
[function] pub fn sys_sched_get_priority_max(policy: SchedPolicy) -> i32 {
kernel_Workspace/kernel/src/cpu/scheduler/core.rs (line 227)
[function] pub fn sys_sched_get_priority_min(policy: SchedPolicy) -> i32 {
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 1)
[struct] pub struct Tracepoint {}
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 3)
[struct] pub struct TracePointId {}
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 5)
[struct] pub struct Tracepointkind{}
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 7)
[struct] pub struct tracepoints {
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 11)
[struct] pub struct TraceEvent{
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 15)
[enum] pub enum TraceData{
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 19)
[struct] pub struct ContextSwichTraceData{
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 23)
[struct] pub struct TraceContext{
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 27)
[struct] pub struct TracepointRegistry{
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 31)
[struct] pub struct RegisteredTracepoint{}
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 33)
[struct] pub struct TraceBuffer{}
kernel_Workspace/kernel/src/cpu/scheduler/debug/tracepoints.rs (line 35)
[struct] pub struct PerCpuTracebuffer{}
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 19)
[struct] pub struct EnqueueFlags: u32 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 30)
[struct] pub struct DequeueFlags: u32 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 240)
[function] pub fn insert(root: &mut *mut TaskStruct, leftmost: &mut *mut TaskStruct,
node: *mut TaskStruct, key_of: KeyOf) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 357)
[function] pub fn delete(root: &mut *mut TaskStruct, leftmost: &mut *mut TaskStruct,
z: *mut TaskStruct) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 457)
[struct] pub struct RqFair {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 478)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 482)
[function] pub fn leftmost(&self) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 499)
[function] pub fn enqueue(&mut self, task: *mut TaskStruct, flags: EnqueueFlags) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 510)
[function] pub fn dequeue(&mut self, task: *mut TaskStruct) {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 520)
[function] pub fn pick_first(&self) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 524)
[function] pub fn charge_exec(&mut self, task: *mut TaskStruct, delta_exec: u64,
now: u64) -> u64 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 540)
[function] pub fn should_preempt(&self, current: *const TaskStruct, candidate:
*const TaskStruct) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 552)
[function] pub fn rb_invariants_ok(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 557)
[function] pub fn rb_count(&self) -> usize {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 568)
[struct] pub struct RqDl {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 591)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 595)
[function] pub fn leftmost(&self) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 599)
[function] pub fn admission_control(&self, dl_runtime: u64, dl_period: u64) -> bool
{
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 618)
[function] pub fn enqueue(&mut self, task: *mut TaskStruct, now: u64, flags:
EnqueueFlags) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 639)
[function] pub fn dequeue(&mut self, task: *mut TaskStruct, removing_permanently:
bool) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 649)
[function] pub fn pick_first(&self) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 653)
[function] pub fn update_curr(&mut self, task: *mut TaskStruct, delta_exec: u64,
now: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 670)
[function] pub fn should_preempt(&self, current: *const TaskStruct, candidate:
*const TaskStruct) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 675)
[function] pub fn rb_invariants_ok(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 684)
[struct] pub struct RqRt {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 694)
[function] pub fn new() -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 710)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 735)
[function] pub fn enqueue(&mut self, task: *mut TaskStruct, flags: EnqueueFlags) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 769)
[function] pub fn dequeue(&mut self, task: *mut TaskStruct) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 793)
[function] pub fn pick_first(&self) -> *mut TaskStruct {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 809)
[function] pub fn requeue(&mut self, task: *mut TaskStruct) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 820)
[function] pub fn highest_priority(&self) -> i32 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 828)
[struct] pub struct RqStop {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 833)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 837)
[function] pub fn enqueue(&mut self, task: *mut TaskStruct) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 843)
[function] pub fn dequeue(&mut self, task: *mut TaskStruct) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 850)
[function] pub fn pick_first(&self) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 857)
[struct] pub struct RunQueue {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 882)
[function] pub fn new(cpu: u32, idle_task: *mut TaskStruct) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 914)
[function] pub fn bind_idle_task(&self) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 930)
[function] pub fn current(&self) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 938)
[function] pub fn nr_running(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 942)
[function] pub fn is_idle(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/runqueue.rs (line 1214)
[function] pub fn fair_root_for_test(&self) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 21)
[function] pub fn account_switch() {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 25)
[function] pub fn account_voluntary_switch(task: &mut TaskStruct) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 30)
[function] pub fn account_involuntary_switch(task: &mut TaskStruct) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 35)
[function] pub fn account_migration(task: &mut TaskStruct) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 41)
[function] pub fn account_user_time(task: &mut TaskStruct, delta_ns: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 48)
[function] pub fn account_system_time(task: &mut TaskStruct, delta_ns: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 55)
[function] pub fn account_idle_time(delta_ns: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 59)
[function] pub fn account_iowait_time(delta_ns: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 64)
[function] pub fn total_switches() -> u64 { TOTAL_SWITCHES.load(Ordering::Relaxed) }
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 65)
[function] pub fn total_voluntary_switches() -> u64 {
TOTAL_VOLUNTARY_SWITCHES.load(Ordering::Relaxed) }

```

```

kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 66)
[function] pub fn total_involuntary_switches() -> u64 {
TOTAL_INVOLUNTARY_SWITCHES.load(Ordering::Relaxed) }
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 67)
[function] pub fn total_migrations() -> u64 {
TOTAL_MIGRATIONS.load(Ordering::Relaxed) }
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 68)
[function] pub fn total_user_time_ns() -> u64 {
TOTAL_USER_TIME_NS.load(Ordering::Relaxed) }
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 69)
[function] pub fn total_system_time_ns() -> u64 {
TOTAL_SYSTEM_TIME_NS.load(Ordering::Relaxed) }
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 70)
[function] pub fn total_idle_time_ns() -> u64 {
TOTAL_IDLE_TIME_NS.load(Ordering::Relaxed) }
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 71)
[function] pub fn total_iowait_time_ns() -> u64 {
TOTAL_IOWAIT_TIME_NS.load(Ordering::Relaxed) }
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 72)
[function] pub fn total_running_time_ns() -> u64 {
TOTAL_RUNNING_TIME_NS.load(Ordering::Relaxed) }
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 77)
[struct] pub struct RqStats {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 126)
[function] pub fn record_switch(&mut self, voluntary: bool) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 143)
[function] pub fn record_migration(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 149)
[function] pub fn add_user_time(&mut self, delta_ns: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 157)
[function] pub fn add_system_time(&mut self, delta_ns: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 165)
[function] pub fn add_idle_time(&mut self, delta_ns: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 171)
[function] pub fn add_iowait_time(&mut self, delta_ns: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 177)
[function] pub fn set_nr_running(&mut self, nr: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 182)
[function] pub fn set_nr_uninterruptible(&mut self, nr: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 187)
[function] pub fn update_load_stats(&mut self, sum_vruntime: u64, sum_weight: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 193)
[function] pub fn load_percent(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 204)
[function] pub fn load_avg_1min(&self) -> f32 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 213)
[struct] pub struct ExecTimeHistogram {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 229)
[function] pub fn add_sample(&mut self, exec_ns: u64) {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 251)
[function] pub fn average_ns(&self) -> f32 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 259)
[function] pub fn total_samples(&self) -> u64 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 273)
[struct] pub struct SystemStats {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 287)
[struct] pub struct CpuStatsSnapshot {
kernel_Workspace/kernel/src/cpu/scheduler/entities/stats.rs (line 298)
[function] pub fn collect_system_stats() -> SystemStats {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 44)
[struct] pub struct TaskFlags: u32 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 73)
[struct] pub struct AtomicTaskFlags(AtomicU32);
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 85)
[function] pub fn load(&self, order: Ordering) -> TaskFlags {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 90)
[function] pub fn store(&self, flags: TaskFlags, order: Ordering) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 95)
[function] pub fn fetch_insert(&self, flags: TaskFlags) -> TaskFlags {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 102)
[function] pub fn fetch_remove(&self, flags: TaskFlags) -> TaskFlags {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 108)
[function] pub fn contains(&self, flags: TaskFlags) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 113)
[function] pub fn test_and_set(&self, flags: TaskFlags) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 127)
[enum] pub enum TaskState {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 208)
[enum] pub enum SchedPolicy {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 230)
[enum] pub enum SchedClass {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 253)
[struct] pub struct CpuMask {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 266)
[function] pub fn single(cpu: u32) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 272)
[function] pub fn first_n(n: u32) -> Self {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 281)
[function] pub fn set(&mut self, cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 288)
[function] pub fn clear(&mut self, cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 295)
[function] pub fn is_set(&self, cpu: u32) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 300)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 304)
[function] pub fn count(&self) -> u32 {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 308)
[function] pub fn intersects(&self, other: &CpuMask) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 312)
[function] pub fn and(&self, other: &CpuMask) -> CpuMask {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 320)
[function] pub fn or(&self, other: &CpuMask) -> CpuMask {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 328)
[function] pub fn first(&self) -> Option<u32> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 337)
[function] pub fn next_after(&self, after: u32) -> Option<u32> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 352)
[function] pub fn iter(&self) -> CpuMaskIter {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 363)
[struct] pub struct CpuMaskIter {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 406)
[function] pub fn nice_to_weight(nice: i8) -> u64 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 411)
[function] pub fn nice_to_wmult(nice: i8) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 416)
[function] pub fn weight_to_nice(weight: u64) -> i8 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 431)
[function] pub fn calc_delta_fair(delta_exec: u64, weight: u64, inv_weight: u32) ->
u64 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 442)
[struct] pub struct InterruptFrame {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 452)
[struct] pub struct FxSaveArea {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 476)
[struct] pub struct CpuContext {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 548)
[function] pub fn xsave_area_size() -> usize {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 557)
[function] pub fn rb_parent(pc: usize) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 562)
[function] pub fn rb_color(pc: usize) -> usize {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 567)
[function] pub fn rb_is_red(pc: usize) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 572)
[function] pub fn rb_is_black(pc: usize) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 577)
[function] pub fn rb_make_parent_color(parent: *mut TaskStruct, color: usize) ->
usize {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 583)
[struct] pub struct LoadAvg {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 595)
[function] pub fn accumulate(&mut self, now: u64, delta_ns: u64, weight: u64,
running: bool) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 624)
[struct] pub struct SchedEntity {

```

```
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 671)
[struct] pub struct RtSchedEntity {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 703)
[struct] pub struct PListNode {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 711)
[function] pub fn init(&mut self, owner: *mut TaskStruct) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 731)
[struct] pub struct DLSchedEntity {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 762)
[struct] pub struct TaskStats {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 781)
[struct] pub struct Credentials {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 831)
[struct] pub struct RLimit {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 848)
[enum] pub enum RlimitResource {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 872)
[struct] pub struct SignalState {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 878)
[function] pub fn has_pending(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 882)
[function] pub fn raise(&mut self, signum: u8) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 888)
[function] pub fn clear(&mut self, signum: u8) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 898)
[struct] pub struct ListHead {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 908)
[function] pub fn init(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 914)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 949)
[function] pub fn is_linked(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 997)
[struct] pub struct SpinLock {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1012)
[function] pub fn lock(&self) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1024)
[function] pub fn unlock(&self) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1028)
[function] pub fn try_lock(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1034)
[function] pub fn is_locked(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1038)
[function] pub fn lock_irqsave(&self) -> usize {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1043)
[function] pub fn unlock_irqrestore(&self, flags: usize) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1048)
[function] pub fn lock_guard(&self) -> SpinLockGuard<'_> {
```

```

kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1053)
[function] pub fn lock_irqsave_guard(&self) -> SpinLockGuard<'a> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1060)
[struct] pub struct SpinLockGuard<'a> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1076)
[enum] pub enum TaskError {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1112)
[struct] pub struct TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1403)
[function] pub fn state(&self) -> TaskState {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1410)
[function] pub fn set_state(&self, new_state: TaskState) -> Result<(), TaskError> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1433)
[function] pub fn wake_up(&self) -> Result<(), TaskError> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1441)
[function] pub fn sleep(&self, interruptible: bool) -> Result<(), TaskError> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1446)
[function] pub fn is_kernel_thread(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1450)
[function] pub fn is_idle_task(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1454)
[function] pub fn is_zombie(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1458)
[function] pub fn is_runnable(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1462)
[function] pub fn needs_resched(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1466)
[function] pub fn set_need_resched(&self) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1470)
[function] pub fn clear_need_resched(&self) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1474)
[function] pub fn set_nice(&mut self, nice: i8) -> Result<(), TaskError> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1487)
[function] pub fn set_rt_priority(&mut self, rt_priority: u8) -> Result<(), TaskError> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1501)
[function] pub fn effective_prio(&self) -> i32 {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1505)
[function] pub fn set_affinity(&mut self, mask: CpuMask) -> Result<(), TaskError> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1517)
[function] pub fn can_run_on(&self, cpu: u32) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1522)
[function] pub fn rq_ptr(&self) -> *mut core::ffi::c_void {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1526)
[function] pub fn set_rq_ptr(&self, rq: *mut core::ffi::c_void) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1530)
[function] pub fn charge_cputime(&mut self, delta_ns: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1549)
[function] pub fn record_voluntary_switch(&mut self) {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1553)
[function] pub fn record_involuntary_switch(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1557)
[function] pub fn record_migration(&mut self, new_cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1564)
[function] pub fn set_comm(&mut self, name: &str) {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1571)
[function] pub fn comm_str(&self) -> &str {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1608)
[struct] pub struct ThreadInfoView<'a> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1618)
[function] pub fn thread_info(&mut self) -> ThreadInfoView<'_> {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1641)
[function] pub fn fair_has_priority(a: &TaskStruct, b: &TaskStruct) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1645)
[function] pub fn deadline_has_priority(a: &TaskStruct, b: &TaskStruct) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1652)
[function] pub fn blank() -> TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/entities/task.rs (line 1696)
[function] pub fn init_test_stub(&mut self, pid: TaskId, policy: SchedPolicy, nice:
i8) {
kernel_Workspace/kernel/src/cpu/scheduler/mod.rs (line 57)
[function] pub fn self_test() -> Result<&'static str, &'static str> {
kernel_Workspace/kernel/src/cpu/scheduler/mod.rs (line 68)
[function] pub fn current_cpu_id() -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 16)
[struct] pub struct CapacityState {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 54)
[function] pub fn compute_cost(&mut self, max_cap: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 64)
[function] pub fn is_valid(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 70)
[struct] pub struct PerformanceDomain {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 120)
[function] pub fn init(&mut self, id: u32, cpus: CpuMask) {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 132)
[function] pub fn add_state(&mut self, state: CapacityState) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 151)
[function] pub fn finalize(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 173)
[function] pub fn find_state_for_capacity(&self, target_cap: u32) ->
Option<&CapacityState> {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 186)
[function] pub fn find_state_for_frequency(&self, target_freq: u32) ->
Option<&CapacityState> {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 203)
[function] pub fn get_effective_capacity(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 207)
[function] pub fn update_thermal_pressure(&self, pressure: u32) {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 218)
[function] pub fn compute_power_at_state(&self, state_idx: usize, temp_millideg:
u32) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 237)
[function] pub fn update_temperature(&self, power_mw: u32, delta_time_ms: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 252)
[function] pub fn contains_cpu(&self, cpu: u32) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 256)
[function] pub fn get_active_state_idx(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 260)
[function] pub fn set_active_state_idx(&self, idx: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 268)
[struct] pub struct EnergyModel {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 298)
[function] pub fn init(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 307)
[function] pub fn register_domain(&mut self, pd: PerformanceDomain) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 329)
[function] pub fn get_pd_for_cpu(&self, cpu: u32) -> Option<&PerformanceDomain> {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 340)
[function] pub fn get_pd_for_cpu_mut(&mut self, cpu: u32) -> Option<&mut
PerformanceDomain> {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 351)
[function] pub fn compute_system_energy(&self, delta_time_ms: u32) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 365)
[function] pub fn compute_task_energy_on_cpu(&self, task_util: u32, cpu: u32) -> u32
{
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 381)
[function] pub fn find_best_state_for_domain(&self, pd_idx: u32, target_util: u32)
-> Option<usize> {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 404)
[function] pub fn invert_capacity_for_thermal(&mut self, cpu: u32, thermal_pressure:
u32) {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 410)
[function] pub fn clear_thermal_pressure(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 416)
[function] pub fn get_max_capacity(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 420)
[function] pub fn get_min_capacity(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 424)
[function] pub fn is_cpu_big(&self, cpu: u32) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 432)
[function] pub fn is_cpu_little(&self, cpu: u32) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 440)
[function] pub fn dump_state(&self) {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 484)
[function] pub fn em_build_synthetic_little_pd(cpus: CpuMask) -> PerformanceDomain {
kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 502)
[function] pub fn em_build_synthetic_big_pd(cpus: CpuMask) -> PerformanceDomain {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/power/em.rs (line 521)
[function] pub fn em_build_synthetic_huge_pd(cpus: CpuMask) -> PerformanceDomain {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 28)
[struct] pub struct PidState {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 45)
[function] pub fn update(&mut self, measurement: i64) -> i64 {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 54)
[function] pub fn reset(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 62)
[struct] pub struct ThermalZone {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 89)
[function] pub fn init(&mut self, pd_idx: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 99)
[function] pub fn update_temp(&self, temp_millideg: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 103)
[function] pub fn get_temp(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 107)
[function] pub fn get_level(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 111)
[function] pub fn evaluate_level(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 148)
[function] pub fn compute_pid_output(&mut self) -> i64 {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 155)
[struct] pub struct PowerScheduler {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 186)
[function] pub fn init(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 199)
[function] pub fn register_zone(&mut self, pd_idx: u32) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 209)
[function] pub fn get_zone_for_pd(&self, pd_idx: u32) -> Option<&ThermalZone> {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 218)
[function] pub fn get_zone_for_pd_mut(&mut self, pd_idx: u32) -> Option<&mut
ThermalZone> {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 227)
[function] pub fn update_tick(&mut self, current_time_ns: u64) {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 258)
[function] pub fn apply_mitigation(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 308)
[function] pub fn get_max_allowed_capacity(&self, cpu: u32) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 315)
[function] pub fn is_eas_enabled(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 319)
[function] pub fn enable_eas(&self) {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 323)
[function] pub fn disable_eas(&self) {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 327)
[function] pub fn is_emergency_shutdown(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/power/mod.rs (line 331)
[function] pub fn dump_thermal_state(&self) {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/power/placement.rs (line 14)
[struct] pub struct CpuSnapshot {
kernel_Workspace/kernel/src/cpu/scheduler/power/placement.rs (line 35)
[struct] pub struct DomainSnapshot {
kernel_Workspace/kernel/src/cpu/scheduler/power/placement.rs (line 54)
[struct] pub struct EnergyCtx {
kernel_Workspace/kernel/src/cpu/scheduler/power/placement.rs (line 77)
[struct] pub struct PlacementResult {
kernel_Workspace/kernel/src/cpu/scheduler/smp/balancing/calculate.rs (line 17)
[enum] pub enum GroupType {
kernel_Workspace/kernel/src/cpu/scheduler/smp/balancing/calculate.rs (line 28)
[struct] pub struct LoadCalculation {
kernel_Workspace/kernel/src/cpu/scheduler/smp/balancing/env.rs (line 18)
[struct] pub struct LoadBalanceEnv {
kernel_Workspace/kernel/src/cpu/scheduler/smp/balancing/env.rs (line 41)
[function] pub fn set_flag(&mut self, flag: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/smp/balancing/env.rs (line 45)
[function] pub fn clear_flag(&mut self, flag: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/smp/balancing/env.rs (line 49)
[function] pub fn has_flag(&self, flag: u32) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/smp/balancing/env.rs (line 55)
[function] pub fn should_stop(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/smp/balancing/env.rs (line 59)
[function] pub fn record_attempt(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/smp/balancing/env.rs (line 63)
[function] pub fn record_failure(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/smp/balancing/env.rs (line 68)
[function] pub fn reset_for_next_iteration(&mut self) {
kernel_Workspace/kernel/src/cpu/scheduler/smp/ipi.rs (line 8)
[enum] pub enum IpiType {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 10)
[enum] pub enum StopperState {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 19)
[struct] pub struct StopperWork {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 41)
[function] pub fn get_state(&self) -> StopperState {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 52)
[function] pub fn set_state(&self, new_state: StopperState) {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 60)
[struct] pub struct CpuStopper {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 83)
[function] pub fn init(&mut self, cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 88)
[function] pub fn queue_work(&self, func: StopperFn, arg: *mut core::ffi::c_void) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 120)
[function] pub fn execute_work(&self) {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 146)
[function] pub fn wait_for_completion(&self) -> u32 {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 166)
[struct] pub struct StopperRegistry {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 178)
[function] pub fn init_cpu(&mut self, cpu: u32) {
kernel_Workspace/kernel/src/cpu/scheduler/smp/migration/stopper.rs (line 184)
[function] pub fn get(&self, cpu: u32) -> Option<&CpuStopper> {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 24)
[enum] pub enum CacheLevel {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 33)
[struct] pub struct CacheTopology {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 41)
[struct] pub struct NumaNode {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 60)
[function] pub fn distance_to(&self, other_node: u32) -> u8 {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 67)
[struct] pub struct SchedGroup {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 96)
[struct] pub struct SchedDomain {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 137)
[function] pub fn has_flag(&self, flag: u32) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 141)
[function] pub fn is_numa(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 145)
[function] pub fn shares_cache(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 149)
[function] pub fn is_smt(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 155)
[struct] pub struct CpuTopology {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 192)
[struct] pub struct TopologyState {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 226)
[function] pub fn alloc_domain(&mut self) -> Option<&mut SchedDomain> {
kernel_Workspace/kernel/src/cpu/scheduler/smp/topology.rs (line 234)
[function] pub fn alloc_group(&mut self) -> Option<&mut SchedGroup> {
kernel_Workspace/kernel/src/cpu/scheduler/time/plist.rs (line 42)
[struct] pub struct PList {
kernel_Workspace/kernel/src/cpu/scheduler/time/plist.rs (line 51)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/time/plist.rs (line 150)
[function] pub fn first(&self) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/time/plist.rs (line 165)
[function] pub fn last(&self) -> *mut TaskStruct {
kernel_Workspace/kernel/src/cpu/scheduler/time/rbtree.rs (line 62)
[struct] pub struct RbTree {
kernel_Workspace/kernel/src/cpu/scheduler/time/rt_array.rs (line 4)
[struct] pub struct RtArray {
kernel_Workspace/kernel/src/cpu/scheduler/time/rt_array.rs (line 20)
[function] pub fn set_bit(&mut self, prio: usize) {

```

```

kernel_Workspace/kernel/src/cpu/scheduler/time/rt_array.rs (line 27)
[function] pub fn clear_bit(&mut self, prio: usize) {
kernel_Workspace/kernel/src/cpu/scheduler/time/rt_array.rs (line 41)
[function] pub fn highest_prio(&self) -> Option<usize> {
kernel_Workspace/kernel/src/cpu/scheduler/time/rt_array.rs (line 149)
[function] pub fn active_levels(&self) -> u32 {
kernel_Workspace/kernel/src/cpu/scheduler/time/task.rs (line 7)
[struct] pub struct ListHead {
kernel_Workspace/kernel/src/cpu/scheduler/time/task.rs (line 18)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/cpu/scheduler/time/task.rs (line 52)
[struct] pub struct RtFields {
kernel_Workspace/kernel/src/cpu/scheduler/time/task.rs (line 58)
[struct] pub struct FairFields {
kernel_Workspace/kernel/src/cpu/scheduler/time/task.rs (line 63)
[struct] pub struct TaskStruct {
kernel_Workspace/kernel/src/cpu/smp.rs (line 91)
[function] pub fn flush_serial_no_panic() {
kernel_Workspace/kernel/src/cpu/smp.rs (line 141)
[function] pub fn init(boot_info: &'static bootloader::BootInfo) {
kernel_Workspace/kernel/src/cpu/smp.rs (line 248)
[function] pub fn total_cpus() -> u32 {
kernel_Workspace/kernel/src/cpu/smp.rs (line 252)
[function] pub fn poweroff() -> bool {
kernel_Workspace/kernel/src/cpu/smp.rs (line 276)
[function] pub fn reboot() -> ! {
kernel_Workspace/kernel/src/cpu/smp.rs (line 293)
[function] pub fn self_test() -> TestResult {
kernel_Workspace/kernel/src/cpu/trampoline.rs (line 23)
[function] pub fn install(cr3: u64, entry: u64) {
kernel_Workspace/kernel/src/cpu/trampoline.rs (line 48)
[function] pub fn set_stack_and_arg(stack_top: u64, arg: u64) {
kernel_Workspace/kernel/src/ctrlinstall/common.rs (line 2)
[struct] pub struct PackageId {
kernel_Workspace/kernel/src/ctrlinstall/common.rs (line 8)
[struct] pub struct Version {
kernel_Workspace/kernel/src/ctrlinstall/common.rs (line 15)
[enum] pub enum ComponentKind {
kernel_Workspace/kernel/src/ctrlinstall/common.rs (line 23)
[enum] pub enum PackageStatus {
kernel_Workspace/kernel/src/ctrlinstall/common.rs (line 31)
[enum] pub enum CtrlInstallError {
kernel_Workspace/kernel/src/ctrlinstall/init/bootstrap.rs (line 1)
[struct] pub struct BootstrapResult {
kernel_Workspace/kernel/src/ctrlinstall/init/bootstrap.rs (line 6)
[function] pub fn bootstrap() -> Result<BootstrapResult, CtrlInstallError> {
kernel_Workspace/kernel/src/ctrlinstall/init/state.rs (line 1)
[struct] pub struct SystemState {

```

```

kernel_Workspace/kernel/src/ctrlinstall/init/state.rs (line 6)
[struct] pub struct InstalledPackage {
kernel_Workspace/kernel/src/ctrlinstall/init/state.rs (line 16)
[function] pub fn is_installed(&self, name: &str) -> bool { todo!() }
kernel_Workspace/kernel/src/ctrlinstall/init/state.rs (line 17)
[function] pub fn get_package(&self, name: &str) -> Option<&InstalledPackage> {
todo!() }
kernel_Workspace/kernel/src/ctrlinstall/init/state.rs (line 18)
[function] pub fn register(&mut self, pkg: InstalledPackage) { todo!() }
kernel_Workspace/kernel/src/ctrlinstall/init/state.rs (line 19)
[function] pub fn unregister(&mut self, name: &str) -> Result<(), CtrlInstallError>
{ todo!() }
kernel_Workspace/kernel/src/ctrlinstall/install/executor.rs (line 1)
[struct] pub struct InstallExecutor<'a> {
kernel_Workspace/kernel/src/ctrlinstall/install/executor.rs (line 6)
[function] pub fn execute(&mut self, tx: &Transaction) -> Result<(),
CtrlInstallError> {
kernel_Workspace/kernel/src/ctrlinstall/install/resolver.rs (line 1)
[struct] pub struct ResolvedPlan {
kernel_Workspace/kernel/src/ctrlinstall/install/resolver.rs (line 5)
[function] pub fn resolve(
kernel_Workspace/kernel/src/ctrlinstall/install/transaction.rs (line 1)
[enum] pub enum TransactionStep {
kernel_Workspace/kernel/src/ctrlinstall/install/transaction.rs (line 8)
[struct] pub struct Transaction {
kernel_Workspace/kernel/src/ctrlinstall/mod.rs (line 9)
[struct] pub struct CtrlInstall {
kernel_Workspace/kernel/src/ctrlinstall/mod.rs (line 15)
[function] pub fn new() -> Result<Self, CtrlInstallError> {
kernel_Workspace/kernel/src/ctrlinstall/mod.rs (line 21)
[function] pub fn install(&mut self, name: &str) -> Result<(), CtrlInstallError> {
kernel_Workspace/kernel/src/ctrlinstall/mod.rs (line 33)
[function] pub fn update(&mut self) -> Result<(), CtrlInstallError> {
kernel_Workspace/kernel/src/ctrlinstall/mod.rs (line 39)
[function] pub fn list_installed(&self) -> &[init::state::InstalledPackage] {
kernel_Workspace/kernel/src/ctrlinstall/repo/fetch.rs (line 1)
[trait] pub trait PackageFetcher {
kernel_Workspace/kernel/src/ctrlinstall/repo/fetch.rs (line 5)
[struct] pub struct LocalFetcher {
kernel_Workspace/kernel/src/ctrlinstall/repo/fetch.rs (line 9)
[struct] pub struct RemoteFetcher {
kernel_Workspace/kernel/src/ctrlinstall/repo/index.rs (line 1)
[struct] pub struct RepositoryIndex {
kernel_Workspace/kernel/src/ctrlinstall/repo/index.rs (line 6)
[function] pub fn find(&self, name: &str) -> Option<&PackageManifest> { todo!() }
kernel_Workspace/kernel/src/ctrlinstall/repo/index.rs (line 7)
[function] pub fn find_by_kind(&self, kind: ComponentKind) ->
alloc::vec::Vec<&PackageManifest> { todo!() }

```

```

kernel_Workspace/kernel/src/ctrlinstall/repo/index.rs (line 8)
[function] pub fn search(&self, query: &str) -> alloc::vec::Vec<&PackageManifest> {
todo!() }
kernel_Workspace/kernel/src/ctrlinstall/repo/manifest.rs (line 1)
[struct] pub struct PackageManifest {
kernel_Workspace/kernel/src/ctrlinstall/repo/manifest.rs (line 10)
[struct] pub struct Dependency {
kernel_Workspace/kernel/src/ctrlinstall/repo/manifest.rs (line 15)
[struct] pub struct FileEntry {
kernel_Workspace/kernel/src/ctrlinstall/update/diff.rs (line 1)
[enum] pub enum UpdateAction {
kernel_Workspace/kernel/src/ctrlinstall/update/diff.rs (line 7)
[function] pub fn diff(
kernel_Workspace/kernel/src/ctrlinstall/update/upgrade.rs (line 1)
[struct] pub struct UpgradePlan {
kernel_Workspace/kernel/src/ctrlinstall/update/upgrade.rs (line 6)
[function] pub fn plan_upgrade(
kernel_Workspace/kernel/src/ctrlinstall/update/upgrade.rs (line 13)
[function] pub fn execute_upgrade(plan: &UpgradePlan, executor: &mut
InstallExecutor) -> Result<(), CtrlInstallError> {
kernel_Workspace/kernel/src/displayport/aut.rs (line 7)
[function] pub fn authorize(ring: u8, op: u8) -> bool {
kernel_Workspace/kernel/src/displayport/bridge.rs (line 13)
[function] pub fn dp_call(op: u32, m: &DsMsg, r: &mut DsMsg, ring: u8) -> i32 {
kernel_Workspace/kernel/src/displayport/init.rs (line 6)
[function] pub fn init() -> bool {
kernel_Workspace/kernel/src/displayport/init.rs (line 10)
[function] pub fn ready() -> bool {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 2)
[struct] pub struct PciDev {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 22)
[function] pub fn read32(d: PciDev, off: u32) -> u32 {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 29)
[function] pub fn write32(d: PciDev, off: u32, v: u32) {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 36)
[function] pub fn port_out(port: u16, val: u32, size: u32) {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 46)
[function] pub fn port_in(port: u16, size: u32) -> u32 {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 68)
[function] pub fn read16(d: PciDev, off: u32) -> u16 {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 72)
[function] pub fn read8(d: PciDev, off: u32) -> u8 {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 77)
[function] pub fn vendor(self) -> u16 {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 81)
[function] pub fn device_id(self) -> u16 {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 85)
[function] pub fn class(self) -> u8 {

```

```

kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 89)
[function] pub fn subclass(self) -> u8 {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 93)
[function] pub fn prog_if(self) -> u8 {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 97)
[function] pub fn bar(self, idx: u32) -> u64 {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 108)
[function] pub fn enable_mmio(self) {
kernel_Workspace/kernel/src/drivers/pci/mod.rs (line 114)
[function] pub fn find_class(class: u8, subclass: u8, prog_if: u8) -> Option<PciDev>
{
kernel_Workspace/kernel/src/drivers/usb/class/hid/keyboard.rs (line 18)
[function] pub fn key_to_ascii(key: u8, shift: bool) -> Option<u8> {
kernel_Workspace/kernel/src/drivers/usb/class/hid/keyboard.rs (line 44)
[function] pub fn push_char(c: u8) {
kernel_Workspace/kernel/src/drivers/usb/class/hid/keyboard.rs (line 55)
[function] pub fn take_char() -> Option<u8> {
kernel_Workspace/kernel/src/drivers/usb/class/hid/mod.rs (line 16)
[struct] pub struct HidKeyboard {
kernel_Workspace/kernel/src/drivers/usb/class/hid/mod.rs (line 31)
[function] pub fn attach(x: &mut Xhci, dev: &mut UsbDevice) -> Result<bool,
UsbError> {
kernel_Workspace/kernel/src/drivers/usb/class/hid/mod.rs (line 105)
[function] pub fn poll(x: &mut Xhci) {
kernel_Workspace/kernel/src/drivers/usb/class/hid/report.rs (line 1)
[struct] pub struct BootReport {
kernel_Workspace/kernel/src/drivers/usb/class/hid/report.rs (line 6)
[function] pub fn parse_boot_keyboard(report: &[u8]) -> Option<BootReport> {
kernel_Workspace/kernel/src/drivers/usb/class/mass/mod.rs (line 17)
[struct] pub struct UsbMass {
kernel_Workspace/kernel/src/drivers/usb/class/mass/mod.rs (line 31)
[function] pub fn bulk_out(&mut self, x: &mut Xhci, phys: u64, len: u32) ->
Result<(), UsbError> {
kernel_Workspace/kernel/src/drivers/usb/class/mass/mod.rs (line 44)
[function] pub fn bulk_in(&mut self, x: &mut Xhci, phys: u64, len: u32) -> Result<(),
UsbError> {
kernel_Workspace/kernel/src/drivers/usb/class/mass/mod.rs (line 57)
[function] pub fn scsi_cmd(&mut self, x: &mut Xhci, cdb: &[u8],
kernel_Workspace/kernel/src/drivers/usb/class/mass/mod.rs (line 101)
[function] pub fn attach(x: &mut Xhci, dev: &mut UsbDevice) -> Result<bool,
UsbError> {
kernel_Workspace/kernel/src/drivers/usb/class/mass/mod.rs (line 186)
[function] pub fn with_controller<F: FnOnce(&mut Xhci) -> R, R>(f: F) -> R {
kernel_Workspace/kernel/src/drivers/usb/class/mass/scsi.rs (line 5)
[function] pub fn read_capacity(x: &mut Xhci, m: &mut UsbMass) -> Result<(u32, u32),
UsbError> {
kernel_Workspace/kernel/src/drivers/usb/class/mass/scsi.rs (line 20)
[function] pub fn read10(x: &mut Xhci, m: &mut UsbMass,
kernel_Workspace/kernel/src/drivers/usb/class/mass/scsi.rs (line 33)
[function] pub fn write10(x: &mut Xhci, m: &mut UsbMass,

```

```

kernel_Workspace/kernel/src/drivers/usb/class/mod.rs (line 5)
[trait] pub trait ClassDriver {
kernel_Workspace/kernel/src/drivers/usb/core/descriptor.rs (line 2)
[struct] pub struct DeviceDesc {
kernel_Workspace/kernel/src/drivers/usb/core/descriptor.rs (line 13)
[struct] pub struct InterfaceDesc {
kernel_Workspace/kernel/src/drivers/usb/core/descriptor.rs (line 22)
[struct] pub struct EndpointDesc {
kernel_Workspace/kernel/src/drivers/usb/core/descriptor.rs (line 29)
[function] pub fn parse_device(buf: &[u8]) -> Option<DeviceDesc> {
kernel_Workspace/kernel/src/drivers/usb/core/descriptor.rs (line 45)
[function] pub fn parse_config(buf: &[u8],
kernel_Workspace/kernel/src/drivers/usb/core/device.rs (line 7)
[struct] pub struct UsbDevice {
kernel_Workspace/kernel/src/drivers/usb/core/device.rs (line 24)
[function] pub fn new(slot: u8, speed: u32, ctx_size: usize) -> Result<Self,
UsbError> {
kernel_Workspace/kernel/src/drivers/usb/core/enumerate.rs (line 14)
[function] pub fn enumerate(x: &mut Xhci, port: u32) -> Result<UsbDevice, UsbError>
{
kernel_Workspace/kernel/src/drivers/usb/core/speed.rs (line 6)
[function] pub fn default_ep0_mps(speed: u32) -> u16 {
kernel_Workspace/kernel/src/drivers/usb/dma.rs (line 5)
[struct] pub struct DmaBuf {
kernel_Workspace/kernel/src/drivers/usb/dma.rs (line 12)
[function] pub fn new(len: usize) -> Result<Self, UsbError> {
kernel_Workspace/kernel/src/drivers/usb/dma.rs (line 31)
[function] pub fn zero(&mut self) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/context.rs (line 4)
[struct] pub struct Contexts {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/context.rs (line 11)
[function] pub fn setup_configure_ep(&mut self, speed: u32, port: u32,
kernel_Workspace/kernel/src/drivers/usb/host/xhci/context.rs (line 37)
[function] pub fn setup_configure_bulk_pair(&mut self, speed: u32, port: u32,
kernel_Workspace/kernel/src/drivers/usb/host/xhci/context.rs (line 71)
[function] pub fn new(ctx_size: usize) -> Result<Self, UsbError> {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/context.rs (line 97)
[function] pub fn setup_address_device(&mut self, speed: u32, port: u32,
kernel_Workspace/kernel/src/drivers/usb/host/xhci/context.rs (line 120)
[function] pub fn setup_evaluate_mps(&mut self, mps: u16) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/control.rs (line 6)
[function] pub fn wait_transfer(x: &mut Xhci, slot: u8) -> Result<u8, UsbError> {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/control.rs (line 26)
[function] pub fn wait_transfer_ep(x: &mut Xhci, slot: u8, ep: u8) -> Result<u8,
UsbError> {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/control.rs (line 46)
[function] pub fn control(x: &mut Xhci,
kernel_Workspace/kernel/src/drivers/usb/host/xhci/control.rs (line 94)
[function] pub fn control_in(x: &mut Xhci, dev: &mut UsbDevice,

```

```

kernel_Workspace/kernel/src/drivers/usb/host/xhci/control.rs (line 102)
[function] pub fn control_out(x: &mut Xhci, dev: &mut UsbDevice,
kernel_Workspace/kernel/src/drivers/usb/host/xhci/event.rs (line 9)
[function] pub fn drain_events(&mut self) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/init.rs (line 17)
[struct] pub struct Xhci {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/init.rs (line 44)
[function] pub(super) fn rt_write64(regs: &XhciRegs, off: usize, v: u64) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/init.rs (line 49)
[function] pub fn init(regs: XhciRegs) -> Result<Xhci, UsbError> {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/init.rs (line 103)
[function] pub fn command(&mut self, trb: Trb) -> Result<Trb, UsbError> {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/init.rs (line 130)
[function] pub fn scan_ports(&mut self) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/regs.rs (line 36)
[struct] pub struct XhciRegs {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/regs.rs (line 49)
[function] pub fn new(phys: u64) -> Result<Self, UsbError> {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/regs.rs (line 97)
[function] pub fn op_read(&self, off: usize) -> u32 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/regs.rs (line 101)
[function] pub fn op_write(&self, off: usize, v: u32) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/regs.rs (line 105)
[function] pub fn rt_read(&self, off: usize) -> u32 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/regs.rs (line 109)
[function] pub fn rt_write(&self, off: usize, v: u32) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/regs.rs (line 113)
[function] pub fn doorbell(&self, slot: u32, target: u32, task: u32) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/regs.rs (line 120)
[function] pub fn port_sc(&self, port: u32) -> u32 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/regs.rs (line 124)
[function] pub fn port_sc_write(&self, port: u32, v: u32) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/regs.rs (line 128)
[function] pub fn port_speed(&self, port: u32) -> u32 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 5)
[struct] pub struct CmdRing {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 13)
[function] pub fn new(count: usize) -> Result<Self, UsbError> {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 24)
[function] pub fn phys(&self) -> u64 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 28)
[function] pub fn enqueue(&mut self, mut trb: Trb) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 50)
[struct] pub struct EventRing {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 58)
[struct] pub struct TransferRing {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 66)
[function] pub fn new(count: usize) -> Result<Self, UsbError> {

```

```

kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 77)
[function] pub fn phys(&self) -> u64 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 81)
[function] pub fn enqueue(&mut self, mut trb: Trb) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 102)
[function] pub fn new(count: usize) -> Result<Self, UsbError> {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 115)
[function] pub fn erst_phys(&self) -> u64 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 119)
[function] pub fn pending(&self) -> Option<Trb> {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 131)
[function] pub fn pop(&mut self) {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/ring.rs (line 140)
[function] pub fn erdp(&self) -> u64 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 29)
[struct] pub struct Trb {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 36)
[function] pub fn typ(&self) -> u32 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 40)
[function] pub fn cycle(&self) -> bool {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 44)
[function] pub fn completion_code(&self) -> u8 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 48)
[function] pub fn slot_id(&self) -> u8 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 52)
[function] pub fn ep_id(&self) -> u8 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 56)
[function] pub fn transfer_len(&self) -> u32 {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 60)
[function] pub fn link(addr: u64) -> Self {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 68)
[function] pub fn enable_slot() -> Self {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 72)
[function] pub fn address_device(slot: u8, ctx_phys: u64) -> Self {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 80)
[function] pub fn configure_ep(slot: u8, ctx_phys: u64) -> Self {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 88)
[function] pub fn noop_cmd() -> Self {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 92)
[function] pub fn setup_stage(raw_setup: u64, trt: u32) -> Self {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 100)
[function] pub fn data_stage(addr: u64, len: u32, dir_in: bool) -> Self {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 108)
[function] pub fn status_stage(dir_in: bool) -> Self {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 116)
[function] pub fn normal(addr: u64, len: u32) -> Self {
kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 124)
[function] pub fn evaluate_ctx(slot: u8, ctx_phys: u64) -> Self {

```

```

kernel_Workspace/kernel/src/drivers/usb/host/xhci/trb.rs (line 133)
[function] pub fn pack_setup(bm_request: u8, b_request: u8, value: u16,
kernel_Workspace/kernel/src/drivers/usb/mod.rs (line 10)
[enum] pub enum UsbError {
kernel_Workspace/kernel/src/drivers/usb/mod.rs (line 24)
[function] pub fn init() -> Result<(), UsbError> {
kernel_Workspace/kernel/src/drivers/usb/mod.rs (line 61)
[function] pub fn poll() {
kernel_Workspace/kernel/src/drivers/usb/mod.rs (line 67)
[function] pub fn self_test() -> TestResult {
kernel_Workspace/kernel/src/drivers/usb/pci_glue.rs (line 8)
[struct] pub struct XhciPci {
kernel_Workspace/kernel/src/drivers/usb/pci_glue.rs (line 13)
[function] pub fn find_xhci() -> Result<XhciPci, UsbError> {
kernel_Workspace/kernel/src/driverspaceinit/abi/abi.rs (line 9)
[enum] pub enum DsCmd {
kernel_Workspace/kernel/src/driverspaceinit/abi/abi.rs (line 38)
[struct] pub struct DsMsg {
kernel_Workspace/kernel/src/driverspaceinit/abi/abi.rs (line 53)
[struct] pub struct DsRing {
kernel_Workspace/kernel/src/driverspaceinit/abi/src.rs (line 3)
[function] pub fn ring_bytes(cap: u64) -> usize {
kernel_Workspace/kernel/src/driverspaceinit/abi/src.rs (line 7)
[struct] pub struct RingView {
kernel_Workspace/kernel/src/driverspaceinit/abi/src.rs (line 20)
[function] pub fn init(&self, cap: u64) {
kernel_Workspace/kernel/src/driverspaceinit/abi/src.rs (line 28)
[function] pub fn push(&self, msg: &DsMsg) -> bool {
kernel_Workspace/kernel/src/driverspaceinit/abi/src.rs (line 49)
[function] pub fn pop(&self) -> Option<DsMsg> {
kernel_Workspace/kernel/src/driverspaceinit/init/enter.rs (line 61)
[function] pub fn enter() -> Result<(), DsError> {
kernel_Workspace/kernel/src/driverspaceinit/init/init.rs (line 9)
[struct] pub struct Driverspace {
kernel_Workspace/kernel/src/driverspaceinit/init/init.rs (line 24)
[function] pub fn prepare() -> Result<(), DsError> {
kernel_Workspace/kernel/src/driverspaceinit/init/init.rs (line 91)
[function] pub fn self_test() -> Result<(), DsError> {
kernel_Workspace/kernel/src/driverspaceinit/init/init.rs (line 134)
[function] pub fn ready() -> bool {
kernel_Workspace/kernel/src/driverspaceinit/init/init.rs (line 138)
[function] pub fn k2d_view() -> Option<RingView> {
kernel_Workspace/kernel/src/driverspaceinit/init/init.rs (line 142)
[function] pub fn d2k_view() -> Option<RingView> {
kernel_Workspace/kernel/src/driverspaceinit/init/init.rs (line 146)
[function] pub fn scratch_view() -> Option<*mut u8> {
kernel_Workspace/kernel/src/driverspaceinit/init/init.rs (line 150)
[function] pub fn map_into_ds(va: u64, phys: u64, len: usize, prot:
space::ProtFlags) -> bool {

```

```

kernel_Workspace/kernel/src/driverspaceinit/init/init.rs (line 159)
[function] pub fn unmap_from_ds(va: u64, len: usize) -> bool {
kernel_Workspace/kernel/src/driverspaceinit/init/initabi.rs (line 7)
[struct] pub struct DsSwitch {
kernel_Workspace/kernel/src/driverspaceinit/init/initabi.rs (line 47)
[struct] pub struct DsInitParams {
kernel_Workspace/kernel/src/driverspaceinit/init/initcommand.rs (line 5)
[enum] pub enum DsError {
kernel_Workspace/kernel/src/driverspaceinit/init/initcommand.rs (line 14)
[struct] pub struct InitHandshake {
kernel_Workspace/kernel/src/driverspaceinit/init/initcommand.rs (line 21)
[function] pub fn send(&mut self, cmd: DsCmd,
kernel_Workspace/kernel/src/driverspaceinit/init/initcommand.rs (line 60)
[function] pub fn run(&mut self, params_va: u64) -> Result<(), DsError> {
kernel_Workspace/kernel/src/driverspaceinit/init/service.rs (line 86)
[function] pub fn vgpu_call(op: u32, m: &DsMsg, r: &mut DsMsg, ring_lvl: u8) -> i32 {
kernel_Workspace/kernel/src/driverspaceinit/init/service.rs (line 155)
[function] pub fn pci_call(op: u32, m: &DsMsg, r: &mut DsMsg, ring: u8) -> i32 {
kernel_Workspace/kernel/src/driverspaceinit/init/service.rs (line 247)
[function] pub fn video_call(op: u32, m: &DsMsg, r: &mut DsMsg) -> i32 {
kernel_Workspace/kernel/src/driverspaceinit/init/service.rs (line 430)
[function] pub fn poll() {
kernel_Workspace/kernel/src/driverspaceinit/init/service.rs (line 459)
[function] pub fn post_event(cmd: DsCmd, a0: u64, a1: u64) {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 24)
[enum] pub enum BufferTransferMode {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 32)
[enum] pub enum BufferError {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 56)
[struct] pub struct BufferFlags: u32 {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 66)
[struct] pub struct IpcBufferDescriptor {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 74)
[struct] pub struct IpcBufferSlot {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 89)
[function] pub fn new(id: u64, owner: u64) -> Self {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 106)
[function] pub fn is_free(&self) -> bool {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 110)
[function] pub fn mark_in_transfer(&self) -> bool {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 119)
[function] pub fn release(&self) {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 124)
[struct] pub struct IpcBufferPool {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 137)
[function] pub fn allocate_slot(&self, owner_task_id: u64) -> Result<u64,
BufferError> {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 150)
[function] pub fn register_buffer(

```

```

kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 192)
[function] pub fn get_slot(&self, slot_id: u64) ->
Option<core::sync::atomic::AtomicPtr<>>> {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 196)
[function] pub fn with_slot<F, R>(&self, slot_id: u64, f: F) -> Option<R>
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 206)
[struct] pub struct TransferContext<'a> {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 216)
[function] pub fn determine_mode(&self) -> BufferTransferMode {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 235)
[function] pub fn execute_transfer(&mut self) -> Result<(), BufferError> {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 498)
[function] pub fn revoke_ipc_buffer_capabilities(cnode: &mut CNode, buffer_slot_id:
u64) {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 508)
[function] pub fn cleanup_task_ipc_buffers(pool: &IpcBufferPool, task_id: u64) {
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 551)
[function] pub fn sys_ipc_buffer_register(
kernel_Workspace/kernel/src/ds_ipc/buffer.rs (line 582)
[function] pub fn sys_ipc_buffer_unregister(
kernel_Workspace/kernel/src/ds_ipc/endpoint.rs (line 8)
[struct] pub struct Endpoint {
kernel_Workspace/kernel/src/ds_ipc/endpoint.rs (line 23)
[function] pub fn new(badge: u64) -> Self {
kernel_Workspace/kernel/src/ds_ipc/endpoint.rs (line 34)
[function] pub fn enqueue_sender(&self, task: *mut Task) {
kernel_Workspace/kernel/src/ds_ipc/endpoint.rs (line 39)
[function] pub fn enqueue_receiver(&self, task: *mut Task) {
kernel_Workspace/kernel/src/ds_ipc/endpoint.rs (line 44)
[function] pub fn dequeue_sender(&self) -> Option<*mut Task> {
kernel_Workspace/kernel/src/ds_ipc/endpoint.rs (line 49)
[function] pub fn dequeue_receiver(&self) -> Option<*mut Task> {
kernel_Workspace/kernel/src/ds_ipc/msg.rs (line 5)
[struct] pub struct MessageInfo {
kernel_Workspace/kernel/src/ds_ipc/msg.rs (line 27)
[struct] pub struct MessageRegisters {
kernel_Workspace/kernel/src/ds_ipc/msg.rs (line 39)
[struct] pub struct IpcMessage {
kernel_Workspace/kernel/src/ds_ipc/syscall.rs (line 8)
[enum] pub enum IpcSyscall {
kernel_Workspace/kernel/src/ds_ipc/syscall.rs (line 15)
[function] pub fn sys_ipc_call(
kernel_Workspace/kernel/src/ds_ipc/syscall.rs (line 43)
[function] pub fn sys_ipc_recv(ep_cap_idx: u64) -> Result<IpcMessage, DSError> {
kernel_Workspace/kernel/src/fs/disc.rs (line 4)
[enum] pub enum DSError {
kernel_Workspace/kernel/src/fs/disc.rs (line 17)
[trait] pub trait BlockDevice {

```

```

kernel_Workspace/kernel/src/fs/disc.rs (line 33)
[enum] pub enum PartitionTableKind {
kernel_Workspace/kernel/src/fs/disc.rs (line 40)
[struct] pub struct Partition {
kernel_Workspace/kernel/src/fs/disc.rs (line 49)
[function] pub fn end_lba(&self) -> Option<u64> {
kernel_Workspace/kernel/src/fs/driver/ata_pio.rs (line 26)
[struct] pub struct AtaPio {
kernel_Workspace/kernel/src/fs/driver/ata_pio.rs (line 66)
[function] pub fn is_present(&self) -> bool {
kernel_Workspace/kernel/src/fs/driver/ata_pio.rs (line 129)
[function] pub fn identify(&self) -> Result<[u16; 256], DriverError> {
kernel_Workspace/kernel/src/fs/driver/ata_pio.rs (line 253)
[function] pub fn probe() -> usize {
kernel_Workspace/kernel/src/fs/driver/block.rs (line 2)
[enum] pub enum DriverError {
kernel_Workspace/kernel/src/fs/driver/block.rs (line 11)
[trait] pub trait BlockDevice {
kernel_Workspace/kernel/src/fs/driver/lock.rs (line 3)
[struct] pub struct IrqGuard {
kernel_Workspace/kernel/src/fs/driver/lock.rs (line 8)
[function] pub fn lock() -> Self {
kernel_Workspace/kernel/src/fs/driver/mod.rs (line 7)
[function] pub fn init() {
kernel_Workspace/kernel/src/fs/driver/mod.rs (line 17)
[function] pub fn self_test() -> bool {
kernel_Workspace/kernel/src/fs/driver/registry.rs (line 7)
[function] pub fn register(dev: &'static dyn BlockDevice) -> bool {
kernel_Workspace/kernel/src/fs/driver/registry.rs (line 22)
[function] pub fn get(index: usize) -> Option<&'static dyn BlockDevice> {
kernel_Workspace/kernel/src/fs/driver/registry.rs (line 28)
[function] pub fn first() -> Option<&'static dyn BlockDevice> {
kernel_Workspace/kernel/src/fs/driver/registry.rs (line 32)
[function] pub fn count() -> usize {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 6)
[enum] pub enum ExtError {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 15)
[struct] pub struct Superblock {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 27)
[struct] pub struct RawInode {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 33)
[struct] pub struct DirEntry {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 39)
[struct] pub struct Ext4 {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 65)
[function] pub fn mount(disk: &'static dyn BlockDevice) -> Result<Self, ExtError> {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 129)
[function] pub fn read_blk(&self, blk: u64) -> Result<Vec<u8>, ExtError> {

```

```

kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 156)
[function] pub fn read_inode(&self, ino: u32) -> Result<RawInode, ExtError> {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 227)
[function] pub fn read_file(&self, inode: &RawInode,
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 262)
[function] pub fn read_dir(&self, inode: &RawInode) -> Result<Vec<DirEntry>,
ExtError> {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 297)
[function] pub fn resolve(&self, path: &str) -> Result<RawInode, ExtError> {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 332)
[function] pub fn read_path(&self, path: &str, buf: &mut [u8]) -> Result<usize,
ExtError> {
kernel_Workspace/kernel/src/fs/ext4/mod.rs (line 337)
[function] pub fn list_path(&self, path: &str) -> Result<Vec<DirEntry>, ExtError> {
kernel_Workspace/kernel/src/fs/fat32/mod.rs (line 6)
[enum] pub enum FatError {
kernel_Workspace/kernel/src/fs/fat32/mod.rs (line 14)
[struct] pub struct Fat32 {
kernel_Workspace/kernel/src/fs/fat32/mod.rs (line 23)
[struct] pub struct FatEntry {
kernel_Workspace/kernel/src/fs/fat32/mod.rs (line 39)
[function] pub fn mount(disk: &'static dyn BlockDevice) -> Result<Self, FatError> {
kernel_Workspace/kernel/src/fs/fat32/mod.rs (line 142)
[function] pub fn read_dir_cluster(&self, start: u32) -> Result<Vec<FatEntry>,
FatError> {
kernel_Workspace/kernel/src/fs/fat32/mod.rs (line 226)
[function] pub fn resolve(&self, path: &str) -> Result<FatEntry, FatError> {
kernel_Workspace/kernel/src/fs/fat32/mod.rs (line 264)
[function] pub fn read_file(&self, entry: &FatEntry,
kernel_Workspace/kernel/src/fs/fat32/mod.rs (line 290)
[function] pub fn read_path(&self, path: &str, buf: &mut [u8]) -> Result<usize,
FatError> {
kernel_Workspace/kernel/src/fs/fat32/mod.rs (line 295)
[function] pub fn list_path(&self, path: &str) -> Result<Vec<FatEntry>, FatError> {
kernel_Workspace/kernel/src/fs/mbr.rs (line 9)
[struct] pub struct MbrEntry {
kernel_Workspace/kernel/src/fs/mbr.rs (line 16)
[struct] pub struct Partition {
kernel_Workspace/kernel/src/fs/mbr.rs (line 57)
[function] pub fn parse_mbr(buf: &[u8]) -> Option<[MbrEntry; 4]> {
kernel_Workspace/kernel/src/fs/mbr.rs (line 80)
[function] pub fn probe_disk(d: &'static dyn BlockDevice) -> usize {
kernel_Workspace/kernel/src/fs/mbr.rs (line 138)
[function] pub fn init() {
kernel_Workspace/kernel/src/fs/mod.rs (line 9)
[function] pub fn init() {
kernel_Workspace/kernel/src/fs/mod.rs (line 14)
[function] pub fn self_test() -> TestResult {
kernel_Workspace/kernel/src/fs/mod.rs (line 57)
[function] pub fn ensure_formatted(dev: &dyn BlockDevice) -> Result<()> {

```

```

kernel_Workspace/kernel/src/fs/mod.rs (line 113)
[function] pub fn root_device() -> Option<'static dyn driver::block::BlockDevice> {
kernel_Workspace/kernel/src/fs/tangfs/btree.rs (line 9)
[struct] pub struct BTreeNode {
kernel_Workspace/kernel/src/fs/tangfs/btree.rs (line 21)
[function] pub fn lookup_dir_entry(fs: &TangFs, dir_ino: u64, name: &str) ->
Result<u64> {
kernel_Workspace/kernel/src/fs/tangfs/btree.rs (line 91)
[function] pub fn read_dir_entries(fs: &TangFs, dir_ino: u64) ->
Result<Vec<DirEntry>> {
kernel_Workspace/kernel/src/fs/tangfs/dir.rs (line 2)
[struct] pub struct DirEntry {
kernel_Workspace/kernel/src/fs/tangfs/extern.rs (line 4)
[function] pub fn allocate_block(fs: &TangFs) -> Result<u64> {
kernel_Workspace/kernel/src/fs/tangfs/extern.rs (line 35)
[function] pub fn free_block(fs: &TangFs, block: u64) -> Result<()> {
kernel_Workspace/kernel/src/fs/tangfs/file.rs (line 4)
[function] pub fn read_file(fs: &TangFs, inode: &Inode, mut offset: u64, buf: &mut
[u8]) -> Result<usize> {
kernel_Workspace/kernel/src/fs/tangfs/file.rs (line 50)
[function] pub fn write_file(fs: &TangFs, inode: &mut Inode, mut offset: u64, data:
&[u8]) -> Result<usize> {
kernel_Workspace/kernel/src/fs/tangfs/inode.rs (line 7)
[struct] pub struct Inode {
kernel_Workspace/kernel/src/fs/tangfs/inode.rs (line 28)
[struct] pub struct Extent {
kernel_Workspace/kernel/src/fs/tangfs/inode.rs (line 35)
[struct] pub struct InodeHandle {
kernel_Workspace/kernel/src/fs/tangfs/journal.rs (line 5)
[struct] pub struct Journal {
kernel_Workspace/kernel/src/fs/tangfs/journal.rs (line 13)
[function] pub fn open(device: &'static dyn crate::fs::driver::BlockDevice, sb:
&super::Superblock) -> Result<Self> {
kernel_Workspace/kernel/src/fs/tangfs/journal.rs (line 22)
[function] pub fn write_block(&mut self, block: u64, data: &[u8]) -> Result<()> {
kernel_Workspace/kernel/src/fs/tangfs/journal.rs (line 39)
[function] pub fn replay(&self) -> Result<()> {
kernel_Workspace/kernel/src/fs/tangfs/mod.rs (line 18)
[struct] pub struct TangFs {
kernel_Workspace/kernel/src/fs/tangfs/mod.rs (line 26)
[function] pub fn mount(device: &'static dyn crate::fs::driver::BlockDevice) ->
Result<Self> {
kernel_Workspace/kernel/src/fs/tangfs/mod.rs (line 49)
[function] pub fn read_block(&self, block: u64) -> Result<Vec<u8>> {
kernel_Workspace/kernel/src/fs/tangfs/mod.rs (line 63)
[function] pub fn write_block(&self, block: u64, data: &[u8]) -> Result<()> {
kernel_Workspace/kernel/src/fs/tangfs/superblock.rs (line 5)
[struct] pub struct Superblock {
kernel_Workspace/kernel/src/fs/tangfs/superblock.rs (line 26)
[function] pub fn read(device: &dyn BlockDevice) -> Result<Self> {

```

```

kernel_Workspace/kernel/src/fs/tangfs/superblock.rs (line 45)
[function] pub fn write(&self, device: &dyn BlockDevice) -> Result<()> {
kernel_Workspace/kernel/src/fs/tangfs/superblock.rs (line 77)
[function] pub fn generate_uuid() -> [u8; 16] {
kernel_Workspace/kernel/src/fs/tfs.rs (line 19)
[enum] pub enum FsError {
kernel_Workspace/kernel/src/fs/tfs.rs (line 33)
[struct] pub struct Superblock {
kernel_Workspace/kernel/src/fs/tfs.rs (line 48)
[function] pub fn format(dev: &dyn BlockDevice) -> Result<()> {
kernel_Workspace/kernel/src/fs/tfs.rs (line 70)
[function] pub fn read_superblock(dev: &dyn BlockDevice) -> Result<Superblock> {
kernel_Workspace/kernel/src/fs/tfs.rs (line 85)
[function] pub fn write_superblock(dev: &dyn BlockDevice, sb: &Superblock) ->
Result<()> {
kernel_Workspace/kernel/src/fs/tfs.rs (line 156)
[function] pub fn list_dir(dev: &dyn BlockDevice, dir: u32, out: &mut impl Write) ->
Result<()> {
kernel_Workspace/kernel/src/fs/tfs.rs (line 186)
[function] pub fn write_file(dev: &dyn BlockDevice, dir: u32, name: &str, data:
&[u8]) -> Result<()> {
kernel_Workspace/kernel/src/fs/tfs.rs (line 246)
[function] pub fn remove(dev: &dyn BlockDevice, dir: u32, name: &str) -> Result<()> {
kernel_Workspace/kernel/src/fs/tfs.rs (line 274)
[function] pub fn mkdir(dev: &dyn BlockDevice, dir: u32, name: &str) -> Result<()> {
kernel_Workspace/kernel/src/fs/tfs.rs (line 325)
[function] pub fn find_dir(dev: &dyn BlockDevice, dir: u32, name: &str) ->
Result<u32> {
kernel_Workspace/kernel/src/fs/tfs.rs (line 333)
[function] pub fn entries(dev: &dyn BlockDevice, dir: u32) ->
Result<alloc::vec::Vec<(alloc::string::String, u32, u8)>> {
kernel_Workspace/kernel/src/fs/tfs.rs (line 352)
[function] pub fn read_file(dev: &dyn BlockDevice, dir: u32, name: &str) ->
Result<alloc::vec::Vec<u8>> {
kernel_Workspace/kernel/src/fs/vfs.rs (line 6)
[struct] pub struct FsEntry {
kernel_Workspace/kernel/src/fs/vfs.rs (line 12)
[enum] pub enum Mounted {
kernel_Workspace/kernel/src/fs/vfs.rs (line 19)
[function] pub fn read_path(&self, path: &str, buf: &mut [u8]) -> Option<usize> {
kernel_Workspace/kernel/src/fs/vfs.rs (line 27)
[function] pub fn list_path(&self, path: &str) -> Option<Vec<FsEntry>> {
kernel_Workspace/kernel/src/fs/vfs.rs (line 56)
[function] pub fn mount_all() {
kernel_Workspace/kernel/src/fs/vfs.rs (line 74)
[function] pub fn root() -> Option<&'static Mounted> {
kernel_Workspace/kernel/src/gdt.rs (line 52)
[function] pub fn init() {
kernel_Workspace/kernel/src/gfx/console.rs (line 32)
[function] pub fn cols() -> usize {

```

```

kernel_Workspace/kernel/src/gfx/console.rs (line 36)
[function] pub fn rows() -> usize {
kernel_Workspace/kernel/src/gfx/console.rs (line 40)
[function] pub fn fb_info() -> (u32, u32, u32, u64) {
kernel_Workspace/kernel/src/gfx/console.rs (line 75)
[function] pub(crate) fn framebuffer_info_pub() -> Option<(u32, u32, u32, u64,
bool)> {
kernel_Workspace/kernel/src/gfx/console.rs (line 79)
[function] pub fn set_enabled(enabled: bool) {
kernel_Workspace/kernel/src/gfx/console.rs (line 145)
[function] pub fn test_fill(r: u32, g: u32, b: u32) -> bool {
kernel_Workspace/kernel/src/gfx/console.rs (line 164)
[function] pub fn resync_background() -> bool {
kernel_Workspace/kernel/src/gfx/console.rs (line 188)
[function] pub fn init(fb_addr: u64, width: u32, height: u32, stride: u32, format:
PixelFormat) -> bool {
kernel_Workspace/kernel/src/gfx/console.rs (line 313)
[function] pub fn refresh() {
kernel_Workspace/kernel/src/gfx/framebuffer.rs (line 8)
[enum] pub enum PixelFormat {
kernel_Workspace/kernel/src/gfx/framebuffer.rs (line 21)
[struct] pub struct Framebuffer {
kernel_Workspace/kernel/src/gfx/framebuffer.rs (line 69)
[function] pub fn get(&self, x: usize, y: usize) -> u32 {
kernel_Workspace/kernel/src/gfx/framebuffer.rs (line 85)
[function] pub fn set(&mut self, x: usize, y: usize, c: u32) {
kernel_Workspace/kernel/src/gfx/framebuffer.rs (line 107)
[function] pub fn offset(&self, x: usize, y: usize) -> usize {
kernel_Workspace/kernel/src/gfx/framebuffer.rs (line 111)
[function] pub fn add(&mut self, x: usize, y: usize, r: u32, g: u32, b: u32) {
kernel_Workspace/kernel/src/gfx/framebuffer.rs (line 153)
[function] pub fn rgb(r: u32, g: u32, b: u32) -> u32 {
kernel_Workspace/kernel/src/gfx/galaxy.rs (line 96)
[function] pub fn render(fb: &mut Framebuffer, t: u32) {
kernel_Workspace/kernel/src/gfx/mod.rs (line 16)
[function] pub fn init() -> bool {
kernel_Workspace/kernel/src/gfx/mod.rs (line 25)
[function] pub fn init_mode(fb_phys: u64, width: u32, height: u32, stride: u32) ->
bool {
kernel_Workspace/kernel/src/gfx/mod.rs (line 30)
[function] pub fn set_resolution(mode: vga::VideoMode) -> bool {
kernel_Workspace/kernel/src/gfx/mod.rs (line 47)
[function] pub fn set_resolution_w_h(width: u32, height: u32) -> bool {
kernel_Workspace/kernel/src/gfx/mod.rs (line 73)
[function] pub fn current_resolution() -> (u32, u32) {
kernel_Workspace/kernel/src/gfx/mod.rs (line 77)
[function] pub fn refresh() {
kernel_Workspace/kernel/src/gfx/mod.rs (line 81)
[function] pub fn self_test() -> Result<&'static str, &'static str> {

```

```

kernel_Workspace/kernel/src/gfx/panic_screen.rs (line 11)
[function] pub fn show(info: &PanicInfo) -> ! {
kernel_Workspace/kernel/src/gfx/vga.rs (line 4)
[enum] pub enum VideoMode {
kernel_Workspace/kernel/src/gfx/vga.rs (line 47)
[function] pub fn set_mode(mode: VideoMode) {
kernel_Workspace/kernel/src/gfx/vga.rs (line 105)
[function] pub fn bochs_version() -> Option<u16> {
kernel_Workspace/kernel/src/gfx/vga.rs (line 114)
[function] pub fn bochs_disable() {
kernel_Workspace/kernel/src/gfx/vga.rs (line 120)
[function] pub fn bochs_set_mode(width: u32, height: u32, bpp: u32) -> bool {
kernel_Workspace/kernel/src/gfx/vga.rs (line 139)
[function] pub fn bochs_lfb_base() -> Option<u64> {
kernel_Workspace/kernel/src/hdmi/aut.rs (line 18)
[function] pub fn authorize(ring: u8, op: u32) -> bool {
kernel_Workspace/kernel/src/hdmi/aut.rs (line 46)
[function] pub fn tick_reset() {
kernel_Workspace/kernel/src/hdmi/bridge.rs (line 24)
[function] pub fn hdmi_call(op: u32, m: &DsMsg, r: &mut DsMsg, ring: u8, owner: u32)
-> i32 {
kernel_Workspace/kernel/src/hdmi/init.rs (line 6)
[function] pub fn init() -> bool {
kernel_Workspace/kernel/src/hdmi/init.rs (line 11)
[function] pub fn ready() -> bool {
kernel_Workspace/kernel/src/interrupts.rs (line 50)
[enum] pub enum InterruptIndex {
kernel_Workspace/kernel/src/interrupts.rs (line 83)
[function] pub fn init_idt() {
kernel_Workspace/kernel/src/main.rs (line 141)
[function] pub fn init() {
kernel_Workspace/kernel/src/main.rs (line 145)
[function] pub fn hlt_loop() -> ! {
kernel_Workspace/kernel/src/main.rs (line 149)
[function] pub fn init_permissions() {
kernel_Workspace/kernel/src/main.rs (line 162)
[function] pub fn kernel_main(boot_info: &'static bootloader::BootInfo) -> ! {
kernel_Workspace/kernel/src/main.rs (line 221)
[function] pub fn kernel_main_riscv() -> ! {
kernel_Workspace/kernel/src/mm/api.rs (line 5)
[function] pub fn kmalloc(size: usize) -> Option<*mut u8> {
kernel_Workspace/kernel/src/mm/api.rs (line 11)
[function] pub fn kcalloc(size: usize) -> Option<*mut u8> {
kernel_Workspace/kernel/src/mm/api.rs (line 17)
[function] pub fn krealloc(ptr: *mut u8, size: usize) -> Option<*mut u8> {
kernel_Workspace/kernel/src/mm/api.rs (line 23)
[function] pub fn kfree(ptr: *mut u8) {
kernel_Workspace/kernel/src/mm/api.rs (line 27)
[function] pub fn kalloc_pages(pages: usize) -> Option<*mut u8> {

```

```

kernel_Workspace/kernel/src/mm/api.rs (line 33)
[function] pub fn kfree_pages(ptr: *mut u8, _pages: usize) {
kernel_Workspace/kernel/src/mm/api.rs (line 37)
[function] pub fn virt_to_phys(ptr: *mut u8) -> u64 {
kernel_Workspace/kernel/src/mm/api.rs (line 41)
[struct] pub struct KernelAlloc;
kernel_Workspace/kernel/src/mm/api.rs (line 72)
[function] pub fn self_test() -> Result<&'static str, &'static str> {
kernel_Workspace/kernel/src/mm/ffi.rs (line 5)
[struct] pub struct RawMemEntry {
kernel_Workspace/kernel/src/mm/ffi.rs (line 14)
[struct] pub struct MmBootParams {
kernel_Workspace/kernel/src/mm/ffi.rs (line 25)
[function] pub fn mm_init(params: *const MmBootParams) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 26)
[function] pub fn mm_ready() -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 27)
[function] pub fn mm_total_ram() -> u64;
kernel_Workspace/kernel/src/mm/ffi.rs (line 28)
[function] pub fn mm_free_ram() -> u64;
kernel_Workspace/kernel/src/mm/ffi.rs (line 29)
[function] pub fn mm_dump();
kernel_Workspace/kernel/src/mm/ffi.rs (line 31)
[function] pub fn arch_memory_ready() -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 32)
[function] pub fn arch_memory_reserve_range(base: u64, len: u64);
kernel_Workspace/kernel/src/mm/ffi.rs (line 33)
[function] pub fn arch_memory_boot_alloc(len: u64, align: u64, out: *mut u64) ->
bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 35)
[function] pub fn paging_init(boot_phys_offset: u64);
kernel_Workspace/kernel/src/mm/ffi.rs (line 36)
[function] pub fn paging_enable_nx();
kernel_Workspace/kernel/src/mm/ffi.rs (line 37)
[function] pub fn paging_read_cr3() -> u64;
kernel_Workspace/kernel/src/mm/ffi.rs (line 38)
[function] pub fn paging_is_mapped(virt: u64) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 39)
[function] pub fn paging_aspace_switch(aspacer: *mut c_void);
kernel_Workspace/kernel/src/mm/ffi.rs (line 40)
[function] pub fn paging_map_page(virt: u64, phys: u64, flags: u64) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 42)
[function] pub fn pmm_init() -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 43)
[function] pub fn pmm_alloc_frame(out: *mut u64) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 44)
[function] pub fn pmm_alloc_zero_frame(out: *mut u64) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 45)
[function] pub fn pmm_alloc_frames(count: usize, out: *mut u64) -> bool;

```

```
kernel_Workspace/kernel/src/mm/ffi.rs (line 46)
[function] pub fn pmm_alloc_frames_aligned(count: usize,
kernel_Workspace/kernel/src/mm/ffi.rs (line 49)
[function] pub fn pmm_free_frame(phys: u64) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 50)
[function] pub fn pmm_free_frames(phys: u64, count: usize) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 53)
[function] pub fn vmm_init() -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 54)
[function] pub fn vmm_alloc(bytes: usize, flags: u32, out: *mut u64) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 55)
[function] pub fn vmm_free(virt: u64, bytes: usize) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 56)
[function] pub fn vmm_map_device(phys: u64, len: usize, out: *mut u64) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 57)
[function] pub fn vmm_unmap_device(virt: u64, len: usize) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 59)
[function] pub fn kmalloc(size: usize) -> *mut c_void;
kernel_Workspace/kernel/src/mm/ffi.rs (line 60)
[function] pub fn kcalloc(size: usize) -> *mut c_void;
kernel_Workspace/kernel/src/mm/ffi.rs (line 61)
[function] pub fn kcalloc(count: usize, size: usize) -> *mut c_void;
kernel_Workspace/kernel/src/mm/ffi.rs (line 62)
[function] pub fn krealloc(ptr: *mut c_void, size: usize) -> *mut c_void;
kernel_Workspace/kernel/src/mm/ffi.rs (line 63)
[function] pub fn kmalloc_aligned(size: usize, align: usize) -> *mut c_void;
kernel_Workspace/kernel/src/mm/ffi.rs (line 64)
[function] pub fn kfree(ptr: *mut c_void);
kernel_Workspace/kernel/src/mm/ffi.rs (line 65)
[function] pub fn kalloc_pages(pages: usize) -> *mut c_void;
kernel_Workspace/kernel/src/mm/ffi.rs (line 66)
[function] pub fn kfree_pages(ptr: *mut c_void, pages: usize);
kernel_Workspace/kernel/src/mm/ffi.rs (line 67)
[function] pub fn kvirt_to_phys(ptr: *mut c_void) -> u64;
kernel_Workspace/kernel/src/mm/ffi.rs (line 69)
[function] pub fn contig_alloc(bytes: usize,
kernel_Workspace/kernel/src/mm/ffi.rs (line 73)
[function] pub fn contig_free(phys: u64, bytes: usize);
kernel_Workspace/kernel/src/mm/ffi.rs (line 74)
[function] pub fn dma_alloc_coherent(bytes: usize,
kernel_Workspace/kernel/src/mm/ffi.rs (line 78)
[function] pub fn dma_free_coherent(phys: u64, virt: *mut c_void, bytes: usize);
kernel_Workspace/kernel/src/mm/ffi.rs (line 81)
[function] pub fn aspace_subsystem_init() -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 82)
[function] pub fn aspace_create() -> *mut c_void;
kernel_Workspace/kernel/src/mm/ffi.rs (line 83)
[function] pub fn aspace_destroy(pa: *mut c_void);
```

```

kernel_Workspace/kernel/src/mm/ffi.rs (line 84)
[function] pub fn aspace_paging_handle(pa: *mut c_void) -> *mut c_void;
kernel_Workspace/kernel/src/mm/ffi.rs (line 85)
[function] pub fn aspace_map_anon(pa: *mut c_void,
kernel_Workspace/kernel/src/mm/ffi.rs (line 89)
[function] pub fn aspace_unmap(pa: *mut c_void, addr: u64, len: usize) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 90)
[function] pub fn aspace_protect(pa: *mut c_void,
kernel_Workspace/kernel/src/mm/ffi.rs (line 94)
[function] pub fn aspace_brk(pa: *mut c_void, new_brk: u64) -> u64;
kernel_Workspace/kernel/src/mm/ffi.rs (line 95)
[function] pub fn mmap(pa: *mut c_void,
kernel_Workspace/kernel/src/mm/ffi.rs (line 100)
[function] pub fn paging_aspace_map(aspace: *mut c_void,
kernel_Workspace/kernel/src/mm/ffi.rs (line 105)
[function] pub fn munmap(pa: *mut c_void, addr: u64, len: usize) -> bool;
kernel_Workspace/kernel/src/mm/ffi.rs (line 106)
[function] pub fn paging_aspace_cr3(aspace: *mut c_void) -> u64;
kernel_Workspace/kernel/src/mm/init.rs (line 9)
[struct] pub struct Driverspace {
kernel_Workspace/kernel/src/mm/init.rs (line 24)
[function] pub fn prepare() -> Result<(), DsError> {
kernel_Workspace/kernel/src/mm/init.rs (line 91)
[function] pub fn self_test() -> Result<(), DsError> {
kernel_Workspace/kernel/src/mm/init.rs (line 134)
[function] pub fn ready() -> bool {
kernel_Workspace/kernel/src/mm/init.rs (line 138)
[function] pub fn k2d_view() -> Option<RingView> {
kernel_Workspace/kernel/src/mm/init.rs (line 142)
[function] pub fn d2k_view() -> Option<RingView> {
kernel_Workspace/kernel/src/mm/init.rs (line 146)
[function] pub fn scratch_view() -> Option<*mut u8> {
kernel_Workspace/kernel/src/mm/init.rs (line 150)
[function] pub fn map_into_ds(va: u64, phys: u64, len: usize, prot:
space::ProtFlags) -> bool {
kernel_Workspace/kernel/src/mm/init.rs (line 159)
[function] pub fn unmap_from_ds(va: u64, len: usize) -> bool {
kernel_Workspace/kernel/src/mm/mm_bridge.rs (line 14)
[struct] pub struct RawMemEntry {
kernel_Workspace/kernel/src/mm/mod.rs (line 20)
[function] pub fn init(params: &ffi::MmBootParams) -> bool {
kernel_Workspace/kernel/src/mm/mod.rs (line 25)
[function] pub fn ready() -> bool {
kernel_Workspace/kernel/src/mm/mod.rs (line 30)
[function] pub fn dump() {
kernel_Workspace/kernel/src/mm/mod.rs (line 35)
[function] pub fn init_riscv() -> bool {

```

```

kernel_Workspace/kernel/src/mm/mod.rs (line 57)
[function] pub fn init_from_boot_info(boot_info: &'static bootloader::BootInfo) ->
bool {
kernel_Workspace/kernel/src/mm/mod.rs (line 110)
[function] pub fn self_test() -> Result<&'static str, &'static str> {
kernel_Workspace/kernel/src/mm/phys.rs (line 3)
[function] pub fn alloc_frame() -> Option<u64> {
kernel_Workspace/kernel/src/mm/phys.rs (line 13)
[function] pub fn alloc_zero_frame() -> Option<u64> {
kernel_Workspace/kernel/src/mm/phys.rs (line 23)
[function] pub fn alloc_frames(count: usize) -> Option<u64> {
kernel_Workspace/kernel/src/mm/phys.rs (line 33)
[function] pub fn alloc_frames_aligned(count: usize, align: usize) -> Option<u64> {
kernel_Workspace/kernel/src/mm/phys.rs (line 43)
[function] pub fn free_frame(phys: u64) -> bool {
kernel_Workspace/kernel/src/mm/phys.rs (line 47)
[function] pub fn free_frames(phys: u64, count: usize) -> bool {
kernel_Workspace/kernel/src/mm/phys.rs (line 51)
[function] pub fn reserve(base: u64, len: u64) {
kernel_Workspace/kernel/src/mm/phys.rs (line 55)
[function] pub fn total_bytes() -> u64 {
kernel_Workspace/kernel/src/mm/phys.rs (line 59)
[function] pub fn free_bytes() -> u64 {
kernel_Workspace/kernel/src/mm/phys.rs (line 65)
[function] pub fn phys_to_virt(phys: u64) -> *mut u8 {
kernel_Workspace/kernel/src/mm/phys.rs (line 69)
[function] pub fn self_test() -> Result<&'static str, &'static str> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 29)
[function] pub fn init() {
kernel_Workspace/kernel/src/mm/riscv.rs (line 47)
[function] pub fn reserve(base: u64, len: u64) {
kernel_Workspace/kernel/src/mm/riscv.rs (line 91)
[function] pub fn alloc_frame() -> Option<u64> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 95)
[function] pub fn alloc_zero_frame() -> Option<u64> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 103)
[function] pub fn alloc_frames(count: usize) -> Option<u64> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 108)
[function] pub fn alloc_frames_aligned(count: usize, align_frames: usize) ->
Option<u64> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 121)
[function] pub fn free_frame(pa: u64) -> bool {
kernel_Workspace/kernel/src/mm/riscv.rs (line 125)
[function] pub fn free_frames(pa: u64, count: usize) -> bool {
kernel_Workspace/kernel/src/mm/riscv.rs (line 152)
[function] pub fn total_bytes() -> u64 {
kernel_Workspace/kernel/src/mm/riscv.rs (line 156)
[function] pub fn free_bytes() -> u64 {

```

```

kernel_Workspace/kernel/src/mm/riscv.rs (line 165)
[function] pub fn phys_to_virt(phys: u64) -> *mut u8 {
kernel_Workspace/kernel/src/mm/riscv.rs (line 195)
[function] pub fn init() {
kernel_Workspace/kernel/src/mm/riscv.rs (line 206)
[function] pub fn pool_bytes() -> u64 {
kernel_Workspace/kernel/src/mm/riscv.rs (line 210)
[function] pub fn heap_free_bytes() -> u64 {
kernel_Workspace/kernel/src/mm/riscv.rs (line 334)
[function] pub fn kmalloc(size: usize) -> Option<*mut u8> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 342)
[function] pub fn kmalloc_aligned(size: usize, align: usize) -> Option<*mut u8> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 350)
[function] pub fn kcalloc(size: usize) -> Option<*mut u8> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 356)
[function] pub fn kcalloc(count: usize, size: usize) -> Option<*mut u8> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 360)
[function] pub fn kfree(ptr: *mut u8) {
kernel_Workspace/kernel/src/mm/riscv.rs (line 378)
[function] pub fn krealloc(ptr: *mut u8, size: usize) -> Option<*mut u8> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 395)
[function] pub fn kalloc_pages(pages: usize) -> Option<*mut u8> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 399)
[function] pub fn kfree_pages(ptr: *mut u8, _pages: usize) {
kernel_Workspace/kernel/src/mm/riscv.rs (line 403)
[function] pub fn kvirt_to_phys(ptr: *mut u8) -> u64 {
kernel_Workspace/kernel/src/mm/riscv.rs (line 420)
[struct] pub struct VmmFlags(u32);
kernel_Workspace/kernel/src/mm/riscv.rs (line 470)
[function] pub fn init() {
kernel_Workspace/kernel/src/mm/riscv.rs (line 504)
[function] pub fn alloc(bytes: usize, _flags: VmmFlags) -> Option<u64> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 513)
[function] pub fn free(va: u64, bytes: usize) -> bool {
kernel_Workspace/kernel/src/mm/riscv.rs (line 525)
[function] pub fn map_device(phys: u64, len: usize) -> Option<u64> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 536)
[function] pub fn unmap_device(va: u64, len: usize) -> bool {
kernel_Workspace/kernel/src/mm/riscv.rs (line 570)
[struct] pub struct ProtFlags(u32);
kernel_Workspace/kernel/src/mm/riscv.rs (line 601)
[struct] pub struct MapFlags(u32);
kernel_Workspace/kernel/src/mm/riscv.rs (line 663)
[struct] pub struct AddressSpace {
kernel_Workspace/kernel/src/mm/riscv.rs (line 674)
[function] pub fn new() -> Option<Self> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 687)
[function] pub fn handle(&self) -> *mut c_void {

```

```

kernel_Workspace/kernel/src/mm/riscv.rs (line 691)
[function] pub fn cr3(&self) -> u64 {
kernel_Workspace/kernel/src/mm/riscv.rs (line 732)
[function] pub fn translate(&self, va: u64) -> Option<u64> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 758)
[function] pub fn map_phys(&self, virt: u64, phys: u64, len: usize, prot: ProtFlags)
-> bool {
kernel_Workspace/kernel/src/mm/riscv.rs (line 773)
[function] pub fn map_anon(&self, hint: u64, len: usize, prot: ProtFlags) ->
Option<u64> {
kernel_Workspace/kernel/src/mm/riscv.rs (line 807)
[function] pub fn mmap(
kernel_Workspace/kernel/src/mm/riscv.rs (line 821)
[function] pub fn munmap(&self, addr: u64, len: usize) -> bool {
kernel_Workspace/kernel/src/mm/riscv.rs (line 840)
[function] pub fn protect(&self, addr: u64, len: usize, prot: ProtFlags) -> bool {
kernel_Workspace/kernel/src/mm/riscv.rs (line 878)
[function] pub fn brk(&self, new_brk: u64) -> u64 {
kernel_Workspace/kernel/src/mm/riscv.rs (line 886)
[function] pub fn switch(&self) {
kernel_Workspace/kernel/src/mm/riscv.rs (line 931)
[function] pub fn init() -> bool {
kernel_Workspace/kernel/src/mm/space.rs (line 7)
[struct] pub struct ProtFlags(u32);
kernel_Workspace/kernel/src/mm/space.rs (line 39)
[struct] pub struct MapFlags(u32);
kernel_Workspace/kernel/src/mm/space.rs (line 64)
[struct] pub struct AddressSpace {
kernel_Workspace/kernel/src/mm/space.rs (line 69)
[function] pub fn new() -> Option<Self> {
kernel_Workspace/kernel/src/mm/space.rs (line 79)
[function] pub fn handle(&self) -> *mut c_void {
kernel_Workspace/kernel/src/mm/space.rs (line 83)
[function] pub fn cr3(&self) -> u64 {
kernel_Workspace/kernel/src/mm/space.rs (line 86)
[function] pub fn map_phys(&self, virt: u64, phys: u64, len: usize, prot: ProtFlags)
-> bool {
kernel_Workspace/kernel/src/mm/space.rs (line 90)
[function] pub fn map_anon(&self, hint: u64, len: usize, prot: ProtFlags) ->
Option<u64> {
kernel_Workspace/kernel/src/mm/space.rs (line 96)
[function] pub fn mmap(&self, addr: u64, len: usize, prot: ProtFlags, flags:
MapFlags) -> Option<u64> {
kernel_Workspace/kernel/src/mm/space.rs (line 102)
[function] pub fn munmap(&self, addr: u64, len: usize) -> bool {
kernel_Workspace/kernel/src/mm/space.rs (line 106)
[function] pub fn protect(&self, addr: u64, len: usize, prot: ProtFlags) -> bool {
kernel_Workspace/kernel/src/mm/space.rs (line 110)
[function] pub fn brk(&self, new_brk: u64) -> u64 {

```

```

kernel_Workspace/kernel/src/mm/space.rs (line 114)
[function] pub fn switch(&self) {
kernel_Workspace/kernel/src/mm/space.rs (line 126)
[function] pub fn self_test() -> Result<'static str, &'static str> {
kernel_Workspace/kernel/src/mm/virt.rs (line 6)
[struct] pub struct VmmFlags(u32);
kernel_Workspace/kernel/src/mm/virt.rs (line 35)
[function] pub fn alloc(bytes: usize, flags: VmmFlags) -> Option<u64> {
kernel_Workspace/kernel/src/mm/virt.rs (line 45)
[function] pub fn free(virt: u64, bytes: usize) -> bool {
kernel_Workspace/kernel/src/mm/virt.rs (line 49)
[function] pub fn map_device(phys: u64, len: usize) -> Option<u64> {
kernel_Workspace/kernel/src/mm/virt.rs (line 59)
[function] pub fn unmap_device(virt: u64, len: usize) -> bool {
kernel_Workspace/kernel/src/mm/virt.rs (line 63)
[function] pub fn self_test() -> Result<'static str, &'static str> {
kernel_Workspace/kernel/src/nic/arp.rs (line 13)
[struct] pub struct ArpPacket {
kernel_Workspace/kernel/src/nic/arp.rs (line 22)
[function] pub fn parse(bytes: &[u8]) -> Result<Self, PacketError> {
kernel_Workspace/kernel/src/nic/arp.rs (line 62)
[function] pub fn write_to(&self, out: &mut [u8]) -> Result<usize, PacketError> {
kernel_Workspace/kernel/src/nic/arp.rs (line 112)
[struct] pub struct ArpCache<const N: usize> {
kernel_Workspace/kernel/src/nic/arp.rs (line 121)
[function] pub fn lookup(&mut self, ip: Ipv4Address, now_ms: u64) ->
Option<MacAddress> {
kernel_Workspace/kernel/src/nic/arp.rs (line 134)
[function] pub fn insert(&mut self, ip: Ipv4Address, mac: MacAddress, expires_at_ms:
u64) {
kernel_Workspace/kernel/src/nic/checksum.rs (line 2)
[function] pub fn ones_complement_sum(bytes: &[u8]) -> u32 {
kernel_Workspace/kernel/src/nic/checksum.rs (line 18)
[function] pub fn fold(sum: u32) -> u16 {
kernel_Workspace/kernel/src/nic/checksum.rs (line 27)
[function] pub fn checksum(bytes: &[u8]) -> u16 {
kernel_Workspace/kernel/src/nic/checksum.rs (line 32)
[function] pub fn is_valid(bytes: &[u8]) -> bool {
kernel_Workspace/kernel/src/nic/command.rs (line 7)
[struct] pub struct NetworkCommandRunner<const ARP_ENTRIES: usize> {
kernel_Workspace/kernel/src/nic/command.rs (line 24)
[function] pub fn start_ping(
kernel_Workspace/kernel/src/nic/command.rs (line 49)
[function] pub fn poll(
kernel_Workspace/kernel/src/nic/command.rs (line 69)
[function] pub fn parse_ipv4(input: &str) -> Option<Ipv4Address> {
kernel_Workspace/kernel/src/nic/device.rs (line 4)
[struct] pub struct PollResult {

```

```

kernel_Workspace/kernel/src/nic/device.rs (line 11)
[struct] pub struct TxFrame<'a> {
kernel_Workspace/kernel/src/nic/device.rs (line 23)
[struct] pub struct RxFrame<'a> {
kernel_Workspace/kernel/src/nic/device.rs (line 28)
[trait] pub trait NetworkDevice {
kernel_Workspace/kernel/src/nic/driver.rs (line 5)
[struct] pub struct DriverRegistry<'a> {
kernel_Workspace/kernel/src/nic/driver.rs (line 11)
[function] pub fn new() -> Self {
kernel_Workspace/kernel/src/nic/driver.rs (line 18)
[function] pub fn register(&mut self, device: &'a mut dyn NetworkDevice) -> bool {
kernel_Workspace/kernel/src/nic/driver.rs (line 27)
[function] pub fn device(&mut self, index: usize) -> Option<&mut (dyn NetworkDevice
+ 'a)> {
kernel_Workspace/kernel/src/nic/driver.rs (line 31)
[function] pub fn len(&self) -> usize {
kernel_Workspace/kernel/src/nic/driver.rs (line 35)
[function] pub fn is_empty(&self) -> bool {
kernel_Workspace/kernel/src/nic/error.rs (line 2)
[enum] pub enum NetworkError {
kernel_Workspace/kernel/src/nic/error.rs (line 18)
[enum] pub enum PacketError {
kernel_Workspace/kernel/src/nic/ethernet.rs (line 11)
[struct] pub struct EthernetHeader {
kernel_Workspace/kernel/src/nic/ethernet.rs (line 18)
[struct] pub struct EthernetFrame<'a> {
kernel_Workspace/kernel/src/nic/ethernet.rs (line 24)
[function] pub fn parse(bytes: &'a [u8]) -> Result<Self, PacketError> {
kernel_Workspace/kernel/src/nic/ethernet.rs (line 46)
[function] pub fn is_for(&self, local: MacAddress) -> bool {
kernel_Workspace/kernel/src/nic/ethernet.rs (line 51)
[function] pub fn build(
kernel_Workspace/kernel/src/nic/ethernet.rs (line 66)
[function] pub fn pad_to_minimum(frame: &mut [u8], logical_len: usize) ->
Result<usize, PacketError> {
kernel_Workspace/kernel/src/nic/ethernet.rs (line 78)
[function] pub fn self_test() -> Result<&'static str, &'static str> {
kernel_Workspace/kernel/src/nic/icmp.rs (line 8)
[struct] pub struct IcmpPacket<'a> {
kernel_Workspace/kernel/src/nic/icmp.rs (line 17)
[function] pub fn parse(bytes: &'a [u8]) -> Result<Self, PacketError> {
kernel_Workspace/kernel/src/nic/icmp.rs (line 37)
[function] pub fn is_reply_for(&self, identifier: u16, sequence: u16) -> bool {
kernel_Workspace/kernel/src/nic/icmp.rs (line 44)
[function] pub fn write_echo_request(
kernel_Workspace/kernel/src/nic/icmp.rs (line 66)
[function] pub fn write_echo_reply(out: &mut [u8], request: IcmpPacket<'>) ->
Result<usize, PacketError> {

```

```

kernel_Workspace/kernel/src/nic/ipv4.rs (line 9)
[struct] pub struct Ipv4Packet<'a> {
kernel_Workspace/kernel/src/nic/ipv4.rs (line 19)
[function] pub fn parse(bytes: &'a [u8]) -> Result<Self, PacketError> {
kernel_Workspace/kernel/src/nic/ipv4.rs (line 64)
[struct] pub struct Ipv4Header {
kernel_Workspace/kernel/src/nic/ipv4.rs (line 73)
[function] pub fn write<'a>{
kernel_Workspace/kernel/src/nic/packet.rs (line 8)
[struct] pub struct IcmpEchoRequest<'a> {
kernel_Workspace/kernel/src/nic/packet.rs (line 19)
[function] pub fn build_icmp_echo(
kernel_Workspace/kernel/src/nic/packet.rs (line 43)
[function] pub fn self_test() -> Result<&'static str, &'static str> {
kernel_Workspace/kernel/src/nic/ping.rs (line 9)
[enum] pub enum PingResult {
kernel_Workspace/kernel/src/nic/ping.rs (line 16)
[struct] pub struct PingClient<const ARP_ENTRIES: usize> {
kernel_Workspace/kernel/src/nic/ping.rs (line 37)
[function] pub fn start(
kernel_Workspace/kernel/src/nic/ping.rs (line 51)
[function] pub fn poll(
kernel_Workspace/kernel/src/nic/protocols.rs (line 3)
[struct] pub struct MacAddress(pub [u8; 6]);
kernel_Workspace/kernel/src/nic/protocols.rs (line 9)
[function] pub fn is_broadcast(&self) -> bool {
kernel_Workspace/kernel/src/nic/protocols.rs (line 16)
[enum] pub enum EtherType {
kernel_Workspace/kernel/src/nic/protocols.rs (line 24)
[struct] pub struct EthernetHeader {
kernel_Workspace/kernel/src/nic/protocols.rs (line 32)
[struct] pub struct Ipv4Header {
kernel_Workspace/kernel/src/nic/protocols.rs (line 47)
[struct] pub struct UdpHeader {
kernel_Workspace/kernel/src/nic/runtime.rs (line 30)
[function] pub fn init() -> Result<(), NetworkError> {
kernel_Workspace/kernel/src/nic/runtime.rs (line 46)
[function] pub fn start_ping(destination: Ipv4Address, now_ms: u64) ->
Result<PingResult, NetworkError> {
kernel_Workspace/kernel/src/nic/runtime.rs (line 54)
[function] pub fn poll(now_ms: u64) -> Result<Option<PingResult>, NetworkError> {
kernel_Workspace/kernel/src/nic/runtime.rs (line 63)
[function] pub fn is_ready() -> bool {
kernel_Workspace/kernel/src/nic/stack.rs (line 11)
[struct] pub struct NetworkConfig {
kernel_Workspace/kernel/src/nic/stack.rs (line 31)
[enum] pub enum StackEvent {
kernel_Workspace/kernel/src/nic/stack.rs (line 54)
[struct] pub struct PingRequest<'a> {

```

```

kernel_Workspace/kernel/src/nic/stack.rs (line 62)
[struct] pub struct NetworkStack<const ARP_ENTRIES: usize> {
kernel_Workspace/kernel/src/nic/stack.rs (line 83)
[function] pub fn next_hop_mac(&mut self, destination: Ipv4Address, now_ms: u64) ->
Option<MacAddress> {
kernel_Workspace/kernel/src/nic/stack.rs (line 87)
[function] pub fn build_arp_request(
kernel_Workspace/kernel/src/nic/stack.rs (line 103)
[function] pub fn build_ping(
kernel_Workspace/kernel/src/nic/stack.rs (line 132)
[function] pub fn process_rx(
kernel_Workspace/kernel/src/nic/types.rs (line 5)
[struct] pub struct MacAddress(pub [u8; 6]);
kernel_Workspace/kernel/src/nic/types.rs (line 49)
[struct] pub struct Ipv4Address(pub [u8; 4]);
kernel_Workspace/kernel/src/nic/virtio/descriptor.rs (line 7)
[struct] pub struct VirtqDescriptor {
kernel_Workspace/kernel/src/nic/virtio/device.rs (line 5)
[struct] pub struct Capabilities {
kernel_Workspace/kernel/src/nic/virtio/device.rs (line 11)
[trait] pub trait NetworkDevice {
kernel_Workspace/kernel/src/nic/virtio/net.rs (line 48)
[struct] pub struct VirtioNetHeader {
kernel_Workspace/kernel/src/nic/virtio/net.rs (line 169)
[struct] pub struct VirtioNetDevice {
kernel_Workspace/kernel/src/nic/virtio/net.rs (line 196)
[function] pub fn mmio_base(&self) -> usize {
kernel_Workspace/kernel/src/nic/virtio/pci_legacy.rs (line 159)
[struct] pub struct VirtioPciLegacyNetDevice {
kernel_Workspace/kernel/src/nic/virtio/queue.rs (line 8)
[struct] pub struct Descriptor {
kernel_Workspace/kernel/src/nic/virtio/queue.rs (line 26)
[struct] pub struct DescriptorId(pub u16);
kernel_Workspace/kernel/src/nic/virtio/queue.rs (line 28)
[struct] pub struct DescriptorPool<const N: usize> {
kernel_Workspace/kernel/src/nic/virtio/queue.rs (line 59)
[function] pub fn allocate(&mut self) -> Result<DescriptorId, NetworkError> {
kernel_Workspace/kernel/src/nic/virtio/queue.rs (line 75)
[function] pub fn configure(
kernel_Workspace/kernel/src/nic/virtio/queue.rs (line 101)
[function] pub fn descriptor(&self, id: DescriptorId) -> Result<&Descriptor,
NetworkError> {
kernel_Workspace/kernel/src/nic/virtio/queue.rs (line 107)
[function] pub fn release_chain(&mut self, head: DescriptorId) -> Result<(),
NetworkError> {
kernel_Workspace/kernel/src/nic/virtio/queue.rs (line 141)
[struct] pub struct QueueMemory {
kernel_Workspace/kernel/src/nic/virtio/queue.rs (line 148)
[function] pub fn self_test() -> Result<&'static str, &'static str> {

```

```

kernel_Workspace/kernel/src/nic/virtio/transport.rs (line 4)
[struct] pub struct QueueSetup {
kernel_Workspace/kernel/src/nic/virtio/transport.rs (line 11)
[trait] pub trait VirtioTransport {
kernel_Workspace/kernel/src/nimcore.rs (line 14)
[function] pub fn banner(buf: &mut [u8]) -> usize {
kernel_Workspace/kernel/src/nimcore.rs (line 18)
[function] pub fn shell_register(name: &str, h: ShellHandler) -> bool {
kernel_Workspace/kernel/src/nimcore.rs (line 22)
[function] pub fn shell_run(line: &str) -> i32 {
kernel_Workspace/kernel/src/nimcore.rs (line 26)
[function] pub fn key_push(c: u8) {
kernel_Workspace/kernel/src/nimcore.rs (line 30)
[function] pub fn key_pop() -> Option<u8> {
kernel_Workspace/kernel/src/nimcore.rs (line 35)
[function] pub fn parse_u64(s: &str, base: u8) -> Option<u64> {
kernel_Workspace/kernel/src/pci.rs (line 11)
[struct] pub struct PciAddress {
kernel_Workspace/kernel/src/pci.rs (line 18)
[struct] pub struct PciDevice {
kernel_Workspace/kernel/src/pci.rs (line 36)
[function] pub fn config_read_u32(addr: PciAddress, offset: u8) -> u32 {
kernel_Workspace/kernel/src/pci.rs (line 45)
[function] pub fn config_write_u32(addr: PciAddress, offset: u8, value: u32) {
kernel_Workspace/kernel/src/pci.rs (line 54)
[function] pub fn config_read_u16(addr: PciAddress, offset: u8) -> u16 {
kernel_Workspace/kernel/src/pci.rs (line 60)
[function] pub fn config_read_u8(addr: PciAddress, offset: u8) -> u8 {
kernel_Workspace/kernel/src/pci.rs (line 88)
[function] pub fn enumerate() -> Vec<PciDevice> {
kernel_Workspace/kernel/src/pci.rs (line 121)
[enum] pub enum NickKind {
kernel_Workspace/kernel/src/pci.rs (line 126)
[function] pub fn find_nic(devices: &[PciDevice]) -> Option<(PciDevice, NickKind)> {
kernel_Workspace/kernel/src/pci.rs (line 142)
[function] pub fn bar(addr: PciAddress, bar_index: u8) -> u32 {
kernel_Workspace/kernel/src/pci.rs (line 147)
[function] pub fn io_base_from_bar(bar_value: u32) -> Option<u16> {
kernel_Workspace/kernel/src/pci.rs (line 155)
[function] pub fn mem_base_from_bar(bar_value: u32) -> Option<u32> {
kernel_Workspace/kernel/src/pci.rs (line 163)
[function] pub fn enable_bus_mastering(addr: PciAddress) {
kernel_Workspace/kernel/src/pci.rs (line 172)
[function] pub fn init() {
kernel_Workspace/kernel/src/pci.rs (line 186)
[function] pub fn self_test() -> TestResult {
kernel_Workspace/kernel/src/policy/bridge.rs (line 9)
[function] pub fn check(ring: u8, cmd: u32, arg: u64) -> bool {

```

```

kernel_Workspace/kernel/src/policy/mod.rs (line 34)
[function] pub fn evaluate(ring: u8, class: u8, op: u8, _arg: u64) -> u8 {
kernel_Workspace/kernel/src/policy/mod.rs (line 59)
[struct] pub struct PolicyEntry {
kernel_Workspace/kernel/src/policy/mod.rs (line 82)
[function] pub fn policy_log(ring: u8, cls: u8, op: u8, dec: u8) {
kernel_Workspace/kernel/src/policy/mod.rs (line 94)
[function] pub fn denies() -> u64 {
kernel_Workspace/kernel/src/policy/mod.rs (line 98)
[function] pub fn total() -> u64 {
kernel_Workspace/kernel/src/policy/mod.rs (line 103)
[function] pub fn entry(idx: usize) -> Option<PolicyEntry> {
kernel_Workspace/kernel/src/policy/mod.rs (line 112)
[function] pub fn ring_of(world: u32) -> u8 {
kernel_Workspace/kernel/src/policy/mod.rs (line 125)
[function] pub fn required_caps(class: u8) -> &'static [Capability] {
kernel_Workspace/kernel/src/policy/mod.rs (line 136)
[function] pub fn hook(world: u32, cap: Capability) -> bool {
kernel_Workspace/kernel/src/policy/mod.rs (line 149)
[function] pub fn install() {
kernel_Workspace/kernel/src/policy/mod.rs (line 153)
[function] pub fn decide(world: u32, cmd: u32, arg: u64) -> Result<(), &'static str>
{
kernel_Workspace/kernel/src/policy/mod.rs (line 185)
[function] pub fn decide_current(cmd: u32, arg: u64) -> Result<(), &'static str> {
kernel_Workspace/kernel/src/process/fd.rs (line 9)
[enum] pub enum FdKind {
kernel_Workspace/kernel/src/process/fd.rs (line 16)
[struct] pub struct Fd {
kernel_Workspace/kernel/src/process/fd.rs (line 54)
[function] pub fn open_for(pid: u32, path: &str) -> i32 {
kernel_Workspace/kernel/src/process/fd.rs (line 105)
[function] pub fn read_for(pid: u32, fd: i32, out: &mut [u8]) -> i32 {
kernel_Workspace/kernel/src/process/fd.rs (line 157)
[function] pub fn write_for(pid: u32, fd: i32, data: &[u8]) -> i32 {
kernel_Workspace/kernel/src/process/fd.rs (line 187)
[function] pub fn close_for(pid: u32, fd: i32) -> i32 {
kernel_Workspace/kernel/src/process/proc.rs (line 3)
[struct] pub struct Process {
kernel_Workspace/kernel/src/serial.rs (line 9)
[function] pub fn init() {
kernel_Workspace/kernel/src/serial.rs (line 32)
[function] pub fn write_byte(byte: u8) {
kernel_Workspace/kernel/src/serial.rs (line 47)
[function] pub fn init() {
kernel_Workspace/kernel/src/serial.rs (line 54)
[function] pub fn write_byte(byte: u8) {
kernel_Workspace/kernel/src/serial.rs (line 60)
[function] pub fn init() {

```

```

kernel_Workspace/kernel/src/serial.rs (line 67)
[function] pub fn write_byte(byte: u8) {
kernel_Workspace/kernel/src/serial.rs (line 74)
[function] pub fn write_str(s: &str) {
kernel_Workspace/kernel/src/serial.rs (line 80)
[struct] pub struct SerialWriter;
kernel_Workspace/kernel/src/serial.rs (line 89)
[function] pub fn print_args(args: fmt::Arguments) {
kernel_Workspace/kernel/src/terminal/mod.rs (line 116)
[function] pub fn push_scancode(code: u8) {
kernel_Workspace/kernel/src/terminal/mod.rs (line 164)
[function] pub fn pop_keycode() -> Option<u32> {
kernel_Workspace/kernel/src/terminal/mod.rs (line 177)
[function] pub fn set_keycode_capture(on: bool) {
kernel_Workspace/kernel/src/terminal/mod.rs (line 564)
[function] pub fn init() {
kernel_Workspace/kernel/src/terminal/mod.rs (line 569)
[function] pub fn run() -> ! {
kernel_Workspace/kernel/src/terminal/mod.rs (line 583)
[function] pub fn self_test() -> crate::testing::TestResult {
kernel_Workspace/kernel/src/testing.rs (line 6)
[struct] pub struct Test {
kernel_Workspace/kernel/src/testing.rs (line 14)
[function] pub fn self_test() -> $crate::testing::TestResult {
kernel_Workspace/kernel/src/testing.rs (line 20)
[function] pub fn run_test(test: &Test) -> bool {
kernel_Workspace/kernel/src/testing.rs (line 41)
[function] pub fn run_all(tests: &[Test]) {
kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.rs (line 15)
[struct] pub struct CpuCtx {
kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.rs (line 24)
[function] pub fn tr_restore_ctx(ctx: *mut CpuCtx) -> !;
kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.rs (line 27)
[struct] pub struct World {
kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.rs (line 40)
[function] pub fn init() {
kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.rs (line 46)
[function] pub fn add_world(ring: u8, ttbr0: u64, entry: u64,
kernel_Workspace/kernel/src/trampoline_rings/arch/aarch64/trampoline_rings.rs (line 90)
[function] pub fn start() -> ! {
kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.rs (line 16)
[struct] pub struct CpuCtx {
kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.rs (line 26)
[function] pub fn tr_restore_ctx(ctx: *mut CpuCtx) -> !;
kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.rs (line 29)
[struct] pub struct World {
kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.rs (line 44)
[function] pub fn init() {

```

```

kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.rs (line 51)
[function] pub fn add_world(ring: u8, satp: u64, entry: u64,
kernel_Workspace/kernel/src/trampoline_rings/arch/risc-v/trampoline_rings.rs (line 95)
[function] pub fn start() -> ! {
kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.rs (line 20)
[struct] pub struct CpuCtx {
kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.rs (line 30)
[function] pub fn tr_restore_ctx(ctx: *mut CpuCtx) -> !;
kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.rs (line 65)
[struct] pub struct World {
kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.rs (line 87)
[function] pub fn init() {
kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.rs (line 129)
[function] pub fn add_world(ring: u8, cr3: u64, entry: u64,
kernel_Workspace/kernel/src/trampoline_rings/arch/x86_64/trampoline_rings.rs (line 174)
[function] pub fn start() -> ! {
kernel_Workspace/kernel/src/userspace/process/proc.rs (line 5)
[struct] pub struct IpcMsg {
kernel_Workspace/kernel/src/userspace/process/proc.rs (line 13)
[struct] pub struct Process {
kernel_Workspace/kernel/src/userspace/process/proc.rs (line 29)
[function] pub fn register(world: usize, parent: u32) -> Option<u32> {
kernel_Workspace/kernel/src/userspace/process/proc.rs (line 55)
[function] pub fn by_pid(pid: u32) -> Option<&'static mut Process> {
kernel_Workspace/kernel/src/userspace/process/proc.rs (line 69)
[function] pub fn by_world(world: usize) -> Option<&'static mut Process> {
kernel_Workspace/kernel/src/userspace/process/proc.rs (line 83)
[function] pub fn send(dst: u32, msg: IpcMsg) -> bool {
kernel_Workspace/kernel/src/userspace/process/proc.rs (line 101)
[function] pub fn recv(pid: u32) -> Option<IpcMsg> {
kernel_Workspace/kernel/src/userspace/process/runc1.rs (line 27)
[function] pub fn run(path: &str) -> i32 {
kernel_Workspace/kernel/src/userspace/process/spawn.rs (line 6)
[function] pub fn spawn_init() -> Result<(), &'static str> {
kernel_Workspace/kernel/src/userspace/process/syscall.rs (line 121)
[function] pub fn handle(world: usize, c: &mut tr::CpuCtx) {
kernel_Workspace/kernel/src/vga_buffer.rs (line 6)
[enum] pub enum Color {
kernel_Workspace/kernel/src/vga_buffer.rs (line 57)
[struct] pub struct Writer {
kernel_Workspace/kernel/src/vga_buffer.rs (line 64)
[function] pub fn write_byte(&mut self, byte: u8) {
kernel_Workspace/kernel/src/vga_buffer.rs (line 104)
[function] pub fn write_string(&mut self, s: &str) {
kernel_Workspace/kernel/src/vga_buffer.rs (line 113)
[function] pub fn set_color(&mut self, foreground: Color) {
kernel_Workspace/kernel/src/vga_buffer.rs (line 117)
[function] pub fn set_column(&mut self, col: usize) {

```

```
kernel_workspace/kernel/src/vga_buffer.rs (line 121)
[function] pub fn clear_to_end(&mut self) {
kernel_workspace/kernel/src/vga_buffer.rs (line 133)
[function] pub fn write_byte_colored(&mut self, byte: u8, fg: Color) {
kernel_workspace/kernel/src/vga_buffer.rs (line 140)
[function] pub fn column(&self) -> usize {
kernel_workspace/kernel/src/vga_buffer.rs (line 144)
[function] pub fn clear_screen(&mut self) {
kernel_workspace/kernel/src/vga_buffer.rs (line 176)
[function] pub fn text_cell(row: usize, col: usize) -> (u8, u8) {
kernel_workspace/kernel/src/vga_buffer.rs (line 183)
[function] pub(super) fn write(args: fmt::Arguments) {
kernel_workspace/kernel/src/vga_buffer.rs (line 188)
[function] pub(super) fn write_colored(color: Color, args: fmt::Arguments) {
kernel_workspace/kernel/src/vga_buffer.rs (line 197)
[function] pub(super) fn self_test() -> crate::testing::TestResult {
kernel_workspace/kernel/src/vga_buffer.rs (line 222)
[function] pub fn _print(args: fmt::Arguments) {
kernel_workspace/kernel/src/vga_buffer.rs (line 229)
[function] pub fn _print_colored(_color: Color, args: fmt::Arguments) {
server_workspace/c/c-bridge/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/compute/dag-exec/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/compute/jit-compiler/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/compute/wasm-runtime/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/core/core-alloc/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/core/core-crypto/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/core/core-ffi/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/core/core-io/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/core/core-math/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/core/core-sync/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/core/core-telemetry/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/domain/auth-core/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/domain/catalog-core/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/domain/graph-core/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
```

```

server_workspace/domain/identity-core/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/domain/ledger-core/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/domain/orders-core/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/domain/ts-core/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/net/dns-resolver/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/net/proto-grpc/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/net/proto-rpc/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/net/stack-http3/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/net/stack-quick/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/net/stack-tcp/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/net/stack-udp/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/storage/store-btree/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/storage/store-disk/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/storage/store-lsm/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/storage/store-page/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
server_workspace/storage/store-wal/src/lib.rs (line 1)
[function] pub fn add(left: u64, right: u64) -> u64 {
trangorgellibc/src/abi/errno.rs (line 5)
[enum] pub enum Errno {
trangorgellibc/src/abi/ktable.rs (line 6)
[struct] pub struct KernelTable {
trangorgellibc/src/abi/ktable.rs (line 25)
[function] pub fn validate(&self) -> bool {
trangorgellibc/src/abi/ktable.rs (line 30)
[function] pub fn print(&self, s: &str) {
trangorgellibc/src/abi/syscall.rs (line 4)
[enum] pub enum Syscall {
trangorgellibc/src/abi/types.rs (line 4)
[struct] pub struct SystemInfo {
trangorgellibc/src/abi/types.rs (line 14)
[struct] pub struct Handle(pub u64);
trangorgellibc/src/abi/types.rs (line 19)
[function] pub fn is_valid(self) -> bool {

```

trangorgellibc/src/abi/types.rs (line 26)

```
[struct] pub struct Stat {
```

trangorgellibc/src/lib.rs (line 69)

```
[function] pub fn log(s: &str) {
```

trangorgellibc/src/lib.rs (line 75)

```
[function] pub fn print(s: &str) {
```

trangorgellibc/src/lib.rs (line 79)

```
[function] pub fn yield_cpu() { sc0(SYS_YIELD); }
```

trangorgellibc/src/lib.rs (line 81)

```
[function] pub fn exit(code: i32) -> ! {
```

trangorgellibc/src/lib.rs (line 86)

```
[function] pub fn getpid() -> u32 { sc0(SYS_GETPID) as u32 }
```

trangorgellibc/src/lib.rs (line 88)

```
[function] pub fn spawn(path: &str) -> i32 {
```

trangorgellibc/src/lib.rs (line 92)

```
[function] pub fn key() -> Option<u8> {
```

trangorgellibc/src/lib.rs (line 96)

```
[function] pub fn ipc_send(pid: u32, a0: u64, a1: u64) -> bool {
```

trangorgellibc/src/lib.rs (line 100)

```
[struct] pub struct Mail { pub from: u32, pub a0: u64, pub a1: u64 }
```

trangorgellibc/src/lib.rs (line 102)

```
[function] pub fn ipc_recv() -> Option<Mail> {
```

trangorgellibc/src/lib.rs (line 119)

```
[function] pub fn open(path: &str) -> i32 {
```

trangorgellibc/src/lib.rs (line 124)

```
[function] pub fn read(fd: i32, buf: &mut [u8]) -> i32 {
```

trangorgellibc/src/lib.rs (line 128)

```
[function] pub fn write(fd: i32, buf: &[u8]) -> i32 {
```

trangorgellibc/src/lib.rs (line 132)

```
[function] pub fn close(fd: i32) -> i32 {
```

trangorgellibc/src/lib.rs (line 137)

```
[function] pub fn wait() -> Option<(u32, i32)> {
```

trangorgellibc/src/lib.rs (line 148)

```
[function] pub fn readdir(idx: u64, name: &mut [u8]) -> Option<u8> {
```

trangorgellibc/src/lib.rs (line 154)

```
[function] pub fn runc1(path: &str) -> i32 {
```

trangorgellibc/src/lib.rs (line 170)

```
[function] pub fn ui_open() -> Option<(u32, u32, u32)> {
```

trangorgellibc/src/lib.rs (line 187)

```
[function] pub fn ui_pixel(stride: u32, x: i32, y: i32, color: u32, w: u32, h: u32) {
```

trangorgellibc/src/lib.rs (line 199)

```
[function] pub fn ui_clear(stride: u32, w: u32, h: u32, color: u32) {
```

trangorgellibc/src/lib.rs (line 215)

```
[function] pub fn ui_text(stride: u32, x: i32, y: i32, s: &str, color: u32, w: u32, h: u32) {
```

trangorgellibc/src/lib.rs (line 242)

```
[function] pub fn malloc(n: usize) -> *mut u8 {
```

trangorgellibc/src/lib.rs (line 253)

```
[function] pub fn put_u32(v: u32) {
```

```

triang-lang/src/ast.rs (line 2)
[struct] pub struct Program {
triang-lang/src/ast.rs (line 7)
[struct] pub struct Function {
triang-lang/src/ast.rs (line 15)
[enum] pub enum Param {
triang-lang/src/ast.rs (line 21)
[enum] pub enum Type {
triang-lang/src/ast.rs (line 27)
[struct] pub struct Layout {
triang-lang/src/ast.rs (line 34)
[enum] pub enum Stmt {
triang-lang/src/ast.rs (line 55)
[struct] pub struct OpCall {
triang-lang/src/ast.rs (line 62)
[enum] pub enum Target {
triang-lang/src/ast.rs (line 69)
[enum] pub enum Expr {
triang-lang/src/ast.rs (line 76)
[struct] pub struct Cond {
triang-lang/src/ast.rs (line 83)
[enum] pub enum CmpOp {
triang-lang/src/codegen/asm.rs (line 44)
[function] pub fn emit(ir: &[Ir]) -> String {
triang-lang/src/codegen/c.rs (line 67)
[function] pub fn emit(ir: &[Ir]) -> String {
triang-lang/src/codegen/mod.rs (line 6)
[enum] pub enum Emit {
triang-lang/src/codegen/target.rs (line 2)
[enum] pub enum Target {
triang-lang/src/codegen/target.rs (line 9)
[function] pub fn reg(&self, name: &str) -> &'static str {
triang-lang/src/codegen/target.rs (line 19)
[function] pub fn ret_reg(&self) -> &'static str {
triang-lang/src/ir.rs (line 5)
[enum] pub enum Val {
triang-lang/src/ir.rs (line 11)
[enum] pub enum BinOp {
triang-lang/src/ir.rs (line 22)
[enum] pub enum Ir {
triang-lang/src/ir.rs (line 43)
[struct] pub struct Lower {
triang-lang/src/ir.rs (line 50)
[function] pub fn new() -> Self {
triang-lang/src/ir.rs (line 63)
[function] pub fn lower_program(mut self, program: &Program) -> Vec<Ir> {
triang-lang/src/iso.rs (line 1)
[struct] pub struct IsoFile {

```

```

triang-lang/src/iso.rs (line 58)
[function] pub fn build(volume: &str, files: &[IsoFile]) -> Vec<u8> {
triang-lang/src/lexer.rs (line 5)
[struct] pub struct LexError {
triang-lang/src/lexer.rs (line 11)
[struct] pub struct Lexer {
triang-lang/src/lexer.rs (line 19)
[function] pub fn new(src: &str) -> Self {
triang-lang/src/lexer.rs (line 50)
[function] pub fn tokenize(&mut self) -> Result<Vec<Token>, LexError> {
triang-lang/src/parser.rs (line 5)
[struct] pub struct ParseError {
triang-lang/src/parser.rs (line 11)
[struct] pub struct Parser {
triang-lang/src/parser.rs (line 17)
[function] pub fn new(tokens: Vec<Token>) -> Self {
triang-lang/src/parser.rs (line 71)
[function] pub fn parse_program(&mut self) -> Result<Program, ParseError> {
triang-lang/src/project.rs (line 4)
[struct] pub struct Project {
triang-lang/src/project.rs (line 12)
[function] pub fn load(path: &str) -> Result<Project, String> {
triang-lang/src/sema.rs (line 5)
[enum] pub enum Symbol {
triang-lang/src/sema.rs (line 11)
[struct] pub struct SemaError {
triang-lang/src/sema.rs (line 15)
[struct] pub struct Sema {
triang-lang/src/sema.rs (line 21)
[function] pub fn new() -> Self {
triang-lang/src/sema.rs (line 32)
[function] pub fn check(&mut self, program: &Program) -> Result<(), SemaError> {
triang-lang/src/token.rs (line 4)
[enum] pub enum TokenKind {
triang-lang/src/token.rs (line 41)
[struct] pub struct Token {

```